VU Medical Zone

ADMIN:

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ZOO 503 MIDTERM PPT

LECTURE 1 TO 90

Paleontology as a science

Paleontology in the modern world

Paleontology as a science

What is Paleontology?

- The branch of science concerned with fossil animals and plants.
- the study of fossils to determine organisms' evolution & ecology.

Paleontology as a science

What is Paleontology?

- Originates from Greek “palaios-old”, “ontos-creature" “logos-study"
- Paleontology lies on the border between Biology and Geology.

Paleontology as a science

Use of Paleontology?

1. Origin of life
2. Curiosity about different worlds
3. Climate and biodiversity change
4. The shape of evolution

Paleontology as a science

Use of Paleontology?

5. Extinction.

6. Dating rocks.

END

Paleontology as a science

What is science?

Paleontology as a science

What is science?

- Science is a systematic enterprise that builds and organizes knowledge in the form of testable explanations and predictions about the universe.

Paleontology as a science

What is science?

- From Latin scientia, meaning "knowledge"
- Science is supposed to be about reality, about hard facts, calculations and proof.

Paleontology as a science

What is science?

- In mathematics, and many areas of physics, this might be true.
- In natural sciences, there have been two main approaches

1. Induction
2. deduction

Paleontology as a science

Induction

- Sir Francis Bacon
- The enquirer might hope to see common patterns among the observations, and these common patterns would point to an explanation, or law of nature.

Paleontology as a science

Deduction

- Aristotle
- A series of observations point to an inevitable Outcome
- "All *men are mortal. Socrates is a man. Therefore Socrates is mortal."*
- Mathematics and in detective work.

Paleontology as a science

Hypotheticodeductive method

- Karl Popper
- "Proof is impossible"
- Scientists set up the hypotheses
- hypothesis testing;
- They seek to refute, or disprove, hypotheses rather than to prove them.

Paleontology as a science

Paradigm shifts

- Thomas Kuhn
- science shuttles between so-called times of normal science and times of scientific revolution.
- Scientific revolutions, or paradigm shifts, are when a whole new idea invades an area of science.

Paleontology as a science

An Example of Paradigm shift

- The paper by Luis Alvarez and colleagues (1980)
- The Earth had been hit by a meteorite 65 million years ago,
- Caused the extinction of the dinosaurs and other groups.

Paleontology as a science

A good Hypothesis

1. hypotheses should be sensible and testable
2. Speculation vs. "Informed deduction"
3. Should be based on observable facts, not the fantasy.

Paleontology as a science

Paleontology rendition in Popular culture

- Jurassic Park and Walking with Dinosaurs
- The slow evolution of reconstructions of ancient life over the centuries reflects the growth of paleontology as a discipline.

Paleontology as a science

Steps to understanding Paleontology

Paleontology as a science

STEPS TO UNDERSTANDING

1. Earliest fossil finds
2. Fossils as magical stones.
3. Fossils as fossils
4. The idea of extinction
5. The vastness of geological time

Paleontology as a science

Earliest fossil finds

- Prehistoric
- peoples picked up and used them as Ornaments
- Early Greeks, Xenophanes (576–480 bce) and Herodotus(484–426)

Paleontology as a science

Fossils as magical stones

- Roman and medieval times
- Fossil sharks' teeth were known as *glossopetrae* worn as amulets
- The idea of *vis plastica*
- The work of Johann Beringer

Paleontology as a science

Fossils as fossils

- The genius of Leonardo da Vinci
- Nicolaus Steno, true nature of *glossopetrae*
- Robert Hooke, descriptions of Fossils using a crude microscope
- Magical explanations of fossils were debunked

Paleontology as a science

The idea of extinction

- Robert Hooke was one of the first to hint at the idea of extinction
- William Hunter (1768): "American incognitum"
- Reality of extinction: French natural scientist Georges Cuvier
- Cuvier : father of comparative anatomy

Paleontology as a science

The vastness of geological time

- Sedimentary rocks and their contained fossils document the history of long spans of time.
- Naming of the geological periods and eras : 1820s and 1830s
- Stratigraphy is an understanding of geologic time

Paleontology as a science

FOSSILS AND
EVOLUTION

Paleontology as a science

FOSSILS AND EVOLUTION

1. Progressionism and Evolution
2. Darwinian evolution

Paleontology as a science

Progressionism and Evolutio

- Limited Knowledge of the fossil record in the 1820s and 1830s
- Paleontologists debated whether there was a progression from simple organisms in the most ancient rocks to more complex forms later.

Paleontology as a science

Progressionism and Evolution

- Charles Lyell (1797–1875), was an antiprogressionist
- Progressionism (Orthogenesis) was linked to the idea of evolution.
- The work of Comte de Buffon (1707–1788) and Jean-Baptiste Lamarck (1744–1829)

Paleontology as a science

Progressionism and Evolution

- Lamarck s' idea of "Great Chain of Being" or the *Scala naturae*.
- "All organisms were linked in time by a unidirectional ladder, from simplest at the bottom to most complex at the top"

Paleontology as a science

Darwinian evolution

- Charles Darwin (1809–1882)
- Developed the theory of evolution by natural selection in the 1830s
- Abandoning the usual belief that species were fixed and unchanging.

Paleontology as a science

Darwinian evolution

- Idea of evolution by common descent, that all species today had evolved from other species in the past.
- The problem: how the variation within species could be harnessed to produce evolutionary change?

Paleontology as a science

Darwinian evolution

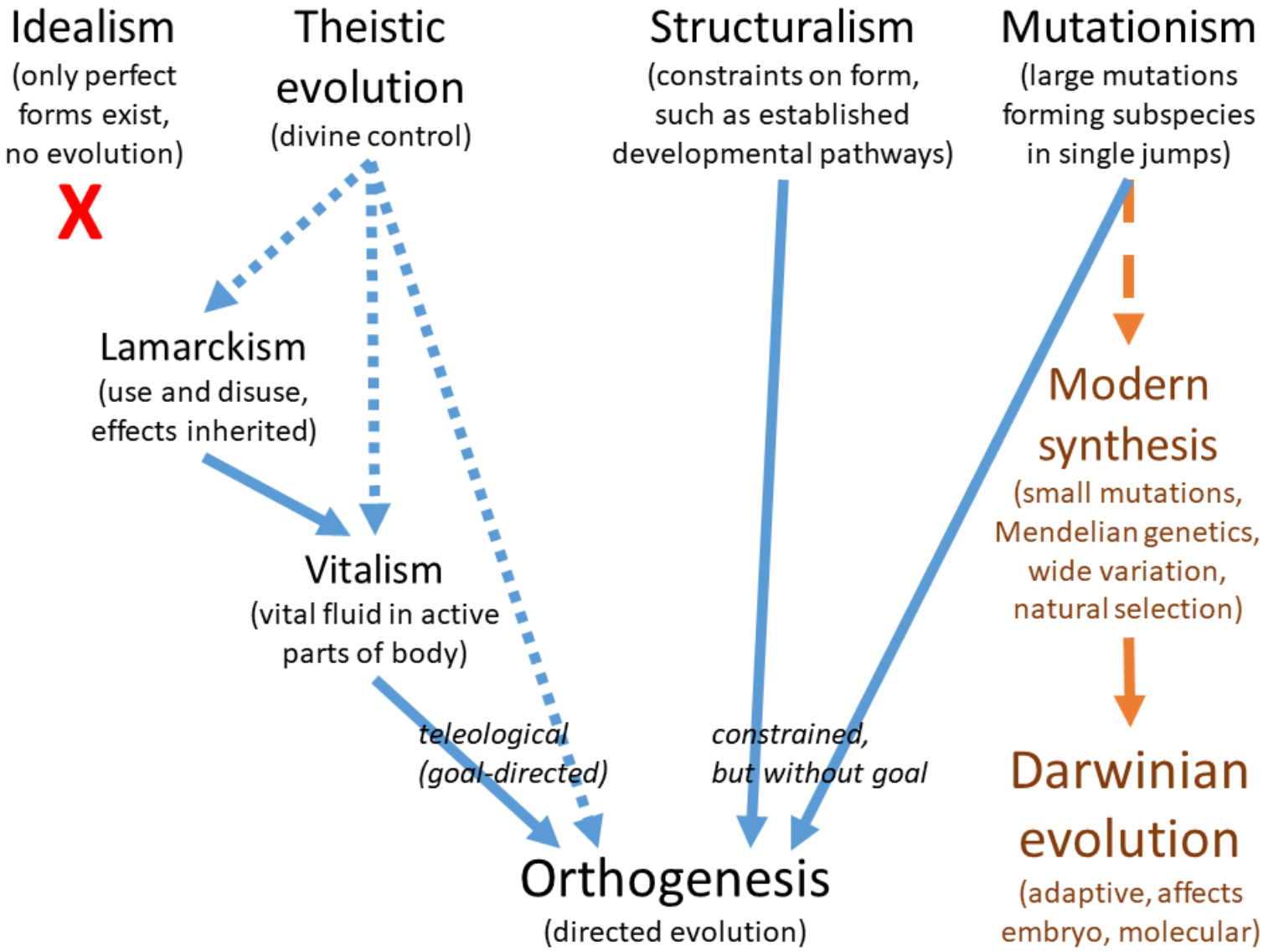
- Ideas of Thomas Malthus (1766–1834)
- Survival of the fittest through adaptations
- Book "On the Origin of Species (1859)"

Paleontology as a science

Darwinian evolution

- "Modern synthesis"
- Modern genetics early in the 20th century
- Amalgamation of Genetics with "natural history" (systematics, ecology, paleontology)
- Establishment of Darwinian evolution by natural selection

Paleontology as a science



https://en.wikipedia.org/wiki/Orthogenesis#/media/File:Alternatives_to_Darwinism.png

Paleontology as a science

PALEONTOLOY
TODAY

Paleontology as a science

PALEONTOLOGY TODAY

1. Dinosaurs and fossil humans
2. Evidence of earliest life
3. Macroevolution
4. Paleontological research

Paleontology as a science

Dinosaurs and fossil humans

- 19th century paleontology
- "Archaeopteryx", true "missing link" predicted by Darwin only 2 years before.
- Incomplete remains of Neandertal man in 1856

Paleontology as a science



Archaeopteryx

- *Archaeopteryx* lithographica,
- Specimen displayed at the Museum für Naturkunde in Berlin

https://en.wikipedia.org/wiki/File:Archaeopteryx_lithographica_(Berlin_specimen).jpg

Paleontology as a science

Evidence of earliest life

- Extraordinary progress in understanding the earliest stages in the evolution of life
- Cambrian fossils
- Precambrian fossils
- In 1947, the first soft-bodied Ediacaran fossils were found in Australia

Paleontology as a science

Macroevolution

- Rates of evolution,
- The nature of speciation
- The timing and extent of mass extinctions
- The diversification of life
- other topics that involve long time scales

Paleontology as a science

END

Paleontological research

- Done by paid professionals
- By amateur enthusiasts
- Amateurs are not paid to work as paleontologists
- Collaborations between different sciences is the key

Fossils in Time and Space

Fossils in time and space

Fossils in Time and Space

Fossils in Time and Space

- The Earth is old, and the distribution of continents and oceans has changed radically over time.
- Past: History of life into a relatively short span of time

Fossils in Time and Space

Life on Earth

- Evolving for up to 4 billion years
- Complex story of fossil groups coming and going
- Continents moving from place to place

Fossils in Time and Space

How to Study fossils in time and space

- Development of geographic and temporal frameworks
- Accurate and reliable enough to chart the distributions of fossil organisms through time and space.

Fossils in Time and Space

How to Study fossils in time and space

- Paleogeographers and Stratigraphers are equipped with a range of computer-based high-tech methods.

Fossils in Time and Space

How to Study fossils in time and space

- Models describing the distributions of the continents, oceans and their biotas throughout geological time

Fossils in Time and Space

Use of Fossils informatio

- Fossils information is also useful in Geology
- Allows the tectonic history of mountain ranges to be reconstructed

Fossils in Time and Space

Use of Fossils information

- Helps in identifying the levels of thermal maturity of rocks
- The gas and oil windows in hydrocarbon exploration

END

Fossils in time and space

Frameworks

Fossils in time and space

Rock stratigraphy

- A rock stratigraphy is the essential framework that paleontologists use to accurately locate fossil collections in both temporal and spatial frameworks.

Fossils in time and space

Leonardo's legacy

- The origin of modern stratigraphy: Leonardo da Vinci and his drawings
- da Vinci portrayed a clear sequence of laterally continuous, horizontal strata showing the concept of superposition

Fossils in time and space

Steno's Work

- Nicolaus Steno: established the simple fact that older rocks are overlain by younger rocks if the sequence has not been inverted
- law of superposition of strata

Fossils in time and space

Steno's Work

- The principle of original horizontality: "Strata either perpendicular to the horizon or inclined to the horizon were at one time parallel to the horizon";

Fossils in time and space

Steno's Work

- The principle of lateral continuity: "Material forming any stratum were continuous over the surface of the Earth unless some other solid bodies stood in the way"

Fossils in time and space

Steno's Work

- The principle of cross-cutting relationships:
- "If a body or discontinuity cuts across a stratum, it must have formed after that stratum.

Fossils in time and space

Types of Stratigraphy

1. Lithology (Lithostratigraphy)
2. Fossils (Biostratigraphy)
3. Tectonic units, such as thrust sheets (Tectonostratigraphy)
4. Magnetic polarity (Magnetostratigraphy),

Fossils in time and space

Types of Stratigraphy

5. Chemical compositions (Chemostratigraphy)
6. Discontinuities (Allostratigraphy),
7. Seismic data (Seismic stratigraphy)
8. Depositional trends (Cyclo- and sequence stratigraphies)

END

Fossils in time and space

Lithostratigraphy

Fossils in time and space

Early geologists

- Early geologists thought the Earth was very young
- Scottish scientist James Hutton (1726–1797) chellenged
- “no vestige of a beginning,– no prospect of an end”.
- “ruins of an earlier world”

Fossils in time and space

Gaia hypothesis and Modern Concept

- Gaia hypothesis: Earth as a living organism in equilibrium with its biosphere.
- Modern Concept: Earth as a dynamic and changing system,

Fossils in time and space

Lithostratigraphy

- All aspects of stratigraphy start from the rocks themselves
- The order and succession of rocks is called lithostratigraphy

Fossils in time and space

Lithostratigraphy

- Building blocks for any study of biological and geological change through time.
- Successions of rock are often divided by gaps or unconformities.

Fossils in time and space

Lithostratigraphy

- A stratigraphy is required to monitor biological and geological changes through time and underpins the whole basis of Earth history.

Fossils in time and space



The Permian through Jurassic strata of the Colorado Plateau area of southeastern Utah demonstrate the principles of stratigraphy

https://en.wikipedia.org/wiki/Stratigraphy#/media/File:SEUtahStrat.JPG

Fossils in time and space

Stratigraphic unit

- A stratigraphic unit is a volume of rock of identifiable origin and relative age range that is defined by the distinctive and dominant, paleontologic features
- It is fundamental Unit of Lithostratigraphy

Fossils in time and space

Lithostratigraphic units

1. A bed, is a lithologically distinct layer within a member or formation and is the smallest recognizable stratigraphic unit.
2. A member is a more local lithologic development, usually part of a formation

Fossils in time and space

Lithostratigraphic units

3. The formation, a rock unit that can be mapped and recognized across country
4. A succession of contiguous formations with common characteristics is called a group

Fossils in time and space

Lithostratigraphic units

5. Groups themselves may comprise a supergroup.

END

Fossils in time and space

Use of fossils: Discovery of Biostratigraphy (Part 1)

Fossils in time and space

Early Palaeontologists

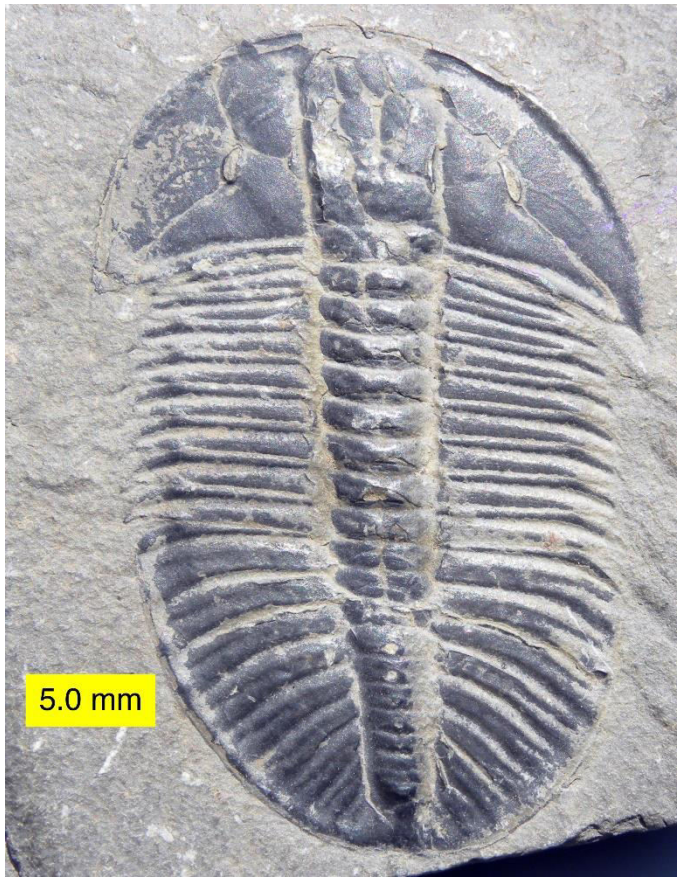
- Work of William Smith in Britain
- "Assemblages of fossils in different layers"
- Georges Cuvier (Anatomist) and Alexandre Brongniart in France (Mollusck expert)

Fossils in time and space

Early Palaeontologists

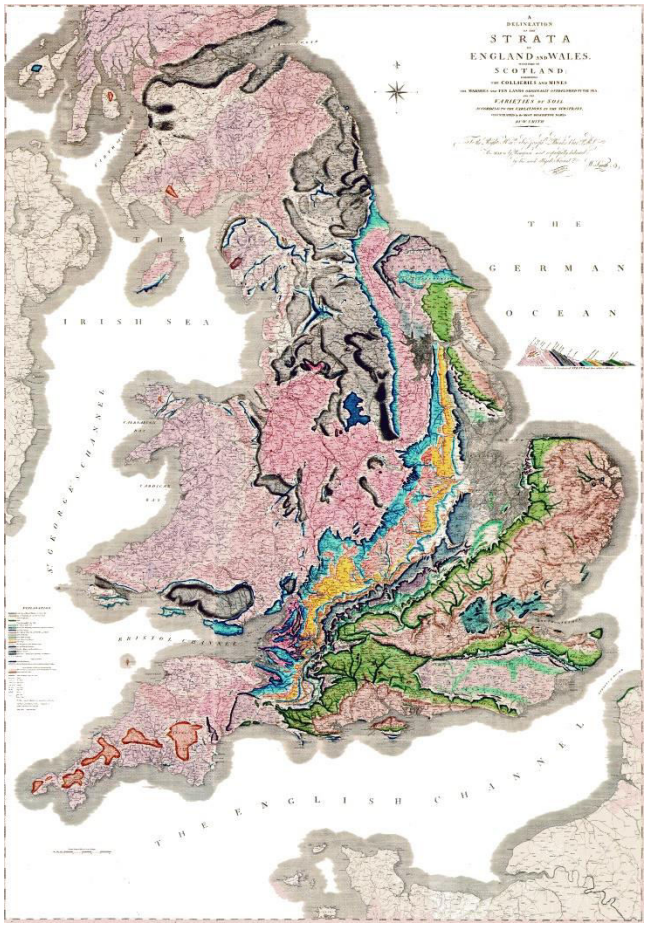
- William Smith: Trilobite-Dominated assemblages in England and Wales area.
- Cuvier and Brongniart: Comparison of different strata in Paris basin and supposed Biological catastrophs in between.

Fossils in time and space



A Trilobite Fossil

https://en.wikipedia.org/wiki/Trilobite#/media/File:Cambrian_Trilobite_Olenoides_Mt._Stephen.jpg



Smith's famous 1815 geological map

https://en.wikipedia.org/wiki/William_Smith_(geologist)#/media/File:Geological_map_Britain_William_Smith_1815.jpg

Fossils in time and space

The Work of John Phillips

- Formally defined the three great eras
- Paleozoic ("ancient life")
- Mesozoic ("middle life")
- Cenozoic ("recent life")
- separated by extinction events

Fossils in time and space

The Work of John Phillips

- Precise biotic and morphological changes along phylogenetic lineages
- Accurate correlation using a wide variety of fossil organisms

Fossils in time and space

Biostratigraphy: the means of correlation

- Biostratigraphy is the establishment of fossil based successions and their use in stratigraphic correlation
- Biozones: The main operational units of a biostratigraphy.

Fossils in time and space

Biostratigraphy: the means of correlation

- Measurements of the stratigraphic ranges of fossils, or assemblages of fossils, form the basis for the definition of biozones

Fossils in time and space

Biostratigraphy: the means of correlation

- The known range of a zone fossil is the time between its first and last appearances in a specific rock section,
- First appearance datum (FAD)
- Last appearance datum (LAD)

Fossils in time and space

Biostratigraphy: the means of correlation

- Establishment of biozones
- Quantitative techniques to understand the relationships between rock thickness and time, and to make links from locality to locality

END

Fossils in time and space

Use of fossils: Discovery of Biostratigraphy (Part 2)

Fossils in time and space

Biostratigraphy: the means of correlation

- The fossil record is rarely complete.
- Only a small percentage of potential fossils are ever preserved.
- Stratigraphic ranges can also be influenced by the Signor–Lipps effect

Fossils in time and space

Biostratigraphy: the means of correlation

- Signor–Lipps effect
- "Since the fossil record of organisms is never complete, neither the first nor the last organism in a given taxon will be recorded as a fossil"

Fossils in time and space

Biostratigraphy: the means of correlation

- Many different animal and plants are used in biostratigraphic correlation
- Graptolites (1 myr) and Ammonites (25 kyr) are the best known and most reliable zone macrofossils

Fossils in time and space

Biostratigraphy: the means of correlation

- Microfossil groups: conodonts, dinoflagellates, foraminiferans and plant spores used in petroleum exploration
- Comprise of small samples, drill cores and chippings

Fossils in time and space

Biostratigraphy: the means of correlation

- Microfossil groups
- many groups are widespread and rapidly evolving.
- Drawback: Techniques used to extract them from rocks are specialized, like acid digestion and thin sections.

Fossils in time and space

Chronostratigraphy

- Dividing up geological time
- Chronostratigraphy or global standard stratigraphy
- Most fundamental of all stratigraphic concepts

Fossils in time and space

Chronostratigraphy

- "The branch of geology concerned with establishing the absolute ages of strata."

Fossils in time and space

Sequence stratigraphy

- "Sequence Stratigraphy is a branch of geology that attempts to subdivide and link sedimentary deposits into unconformity bound units on a variety of scales."

Fossils in time and space

Cyclostratigraphy:

- Finding the rhythm
- Cyclostratigraphy is the study of astronomically forced climate cycles within sedimentary successions

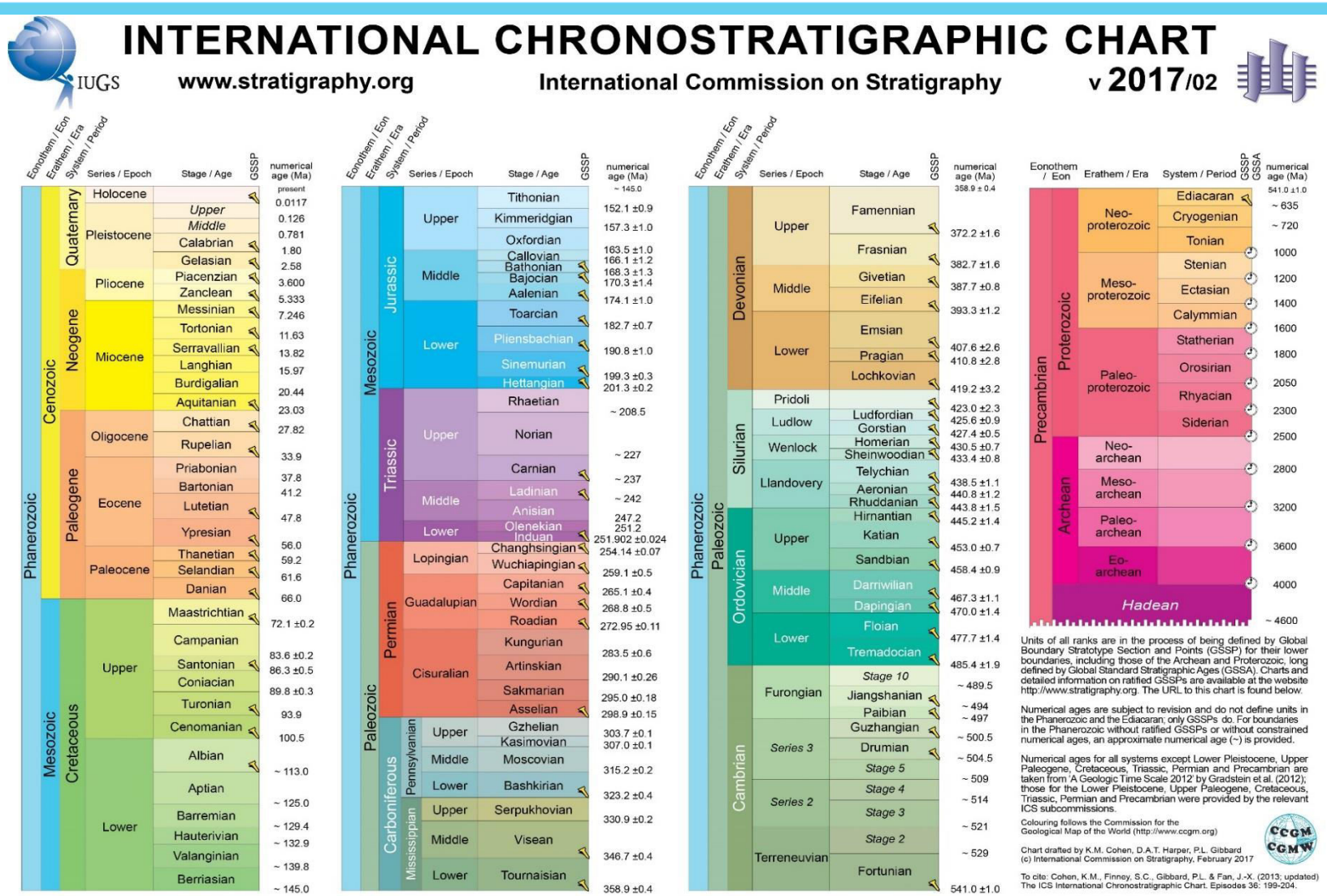
Fossils in time and space

Geologic time scale

- System of chronological dating that relates geological strata (stratigraphy) to time
- Used to describe the timing and relationships of events that have occurred during Earth's history.

END

Fossils in time and space



http://www.stratigraphy.org/index.php/ics-chart-timescale

Fossils in time and space

Paleobiogeography

Fossils in time and space

Paleobiogeography

- All living organisms have a defined geographic range
- The ranges may be large or small, and controlled by a variety of factors including climate and latitude.

Fossils in time and space

Early Biogeographists

- Charles Darwin (Galápagos islands) and Alfred Russel Wallace (East Indies)
- Recognized the reality of biogeographic provinces.

Fossils in time and space

Early Biogeographists

- Philip Sclater and Alfred Russel Wallace
- The Earth today can be divided into six main provinces
- Nearctic, Palearctic, Neotropical, Ethiopian, Oriental and Australasian.

Fossils in time and space



Six Biogeographic Realms according to the WWF

https://en.wikipedia.org/wiki/Biogeographic_realm#/media/File:Ecozones.svg

Fossils in time and space

Early Biogeographists

- Early 1900s, the German scientist Alfred Wegener suggested that the continents moved across the Earth's surface on a liquid core, suggesting that continents could in fact drift.

Fossils in time and space

Computerized paleogeographic systems

- Our understanding of plate movements has been greatly advanced
- Paleomap Project
- taking the Earth far into the future as well as deep into the past

Fossils in time and space

Faunal and floral barriers

- Barriers of various types have partitioned biogeographic provinces through time.

Fossils in time and space

The work of Gaylord Simpson

- George G. Simpson Three passages
- Corridors (open at all times)
- Filters (allowed restricted
- Access)
- Sweepstake routes (opened only occasionally)

Fossils in time and space

Barriers

- Continental settings: mountain ranges, inland seas or even rain forests.
- Marine faunas: wide expanses of deep ocean, swift ocean currents or land.

Fossils in time and space

END

Isthmus of Panama

- Barrier for some organisms may provide a corridor for others.
- The emergence of the Isthmus of Panama (3 Mya)
- Connected North and South America
- Separated the Atlantic and Pacific oceans.

Fossils in time and space



The isthmus of Panama

https://eoimages.gsfc.nasa.gov/images/imagerecords/4000/4073/Panama_SRTM_lrg.jpg

Fossils in time and space

Island Paleobiogeography

Fossils in time and space

Island biogeography

- Modern oceans are littered with islands.
- The biogeography of modern islands is complex and it is hard to apply models based on modern islands to ancient examples

Fossils in time and space

Island biogeography

- Island biogeography is the study of the species composition and species richness on islands.

Fossils in time and space

Island biogeography

- Island biogeography is a study aimed at establishing and explaining the factors that affect species diversity of a specific community.

Fossils in time and space

Island biogeography

- Most islands are isolated from the mainland, and they are important powerhouses of speciation

Fossils in time and space

Island biogeography

- Moving island complexes: the cross-latitude transfer of evolving animals and plants may have acted as "Noah's arks".
- Example: The transit of India from Gondwana to Asia

Fossils in time and space

Island biogeography

- Island biotas (faunas and floras) are often diverse, with many endemic species and commonly with evidence that these species came originally from one or more source continents.

Fossils in time and space

Island biogeography

- Galápagos, or Aldabra, have become important sites for biologists to watch "evolution in action".
- It is difficult to understand the role of such islands through geological time

Fossils in time and space

Island biogeography

- By their very nature, being short lived and located in tectonically active areas
- They are quickly lost and often destroyed.

END

Fossils in time and space

FOSSILS IN FOLD BELTS

Fossils in time and space

FOSSILS IN FOLD BELTS

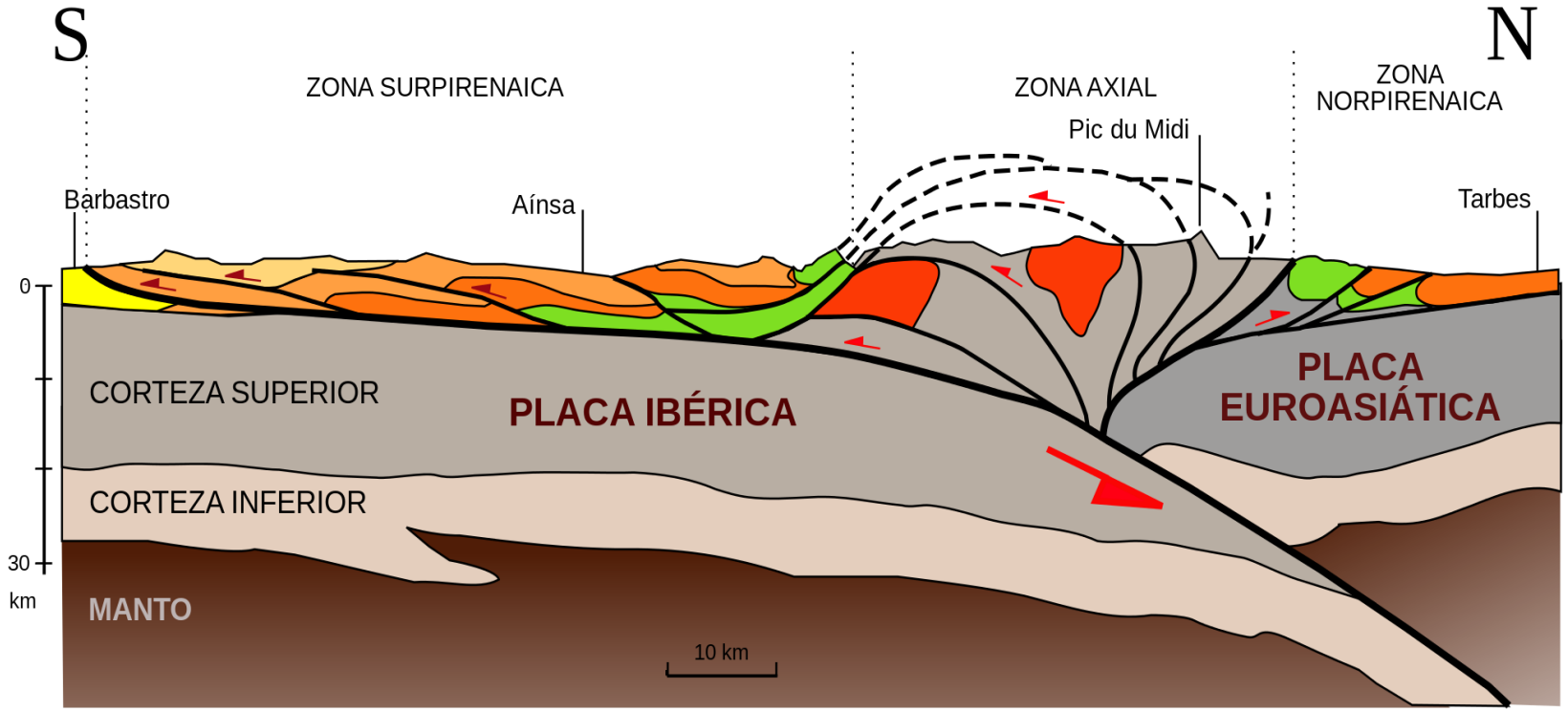
- "One bad fossil is worth a good working hypothesis"
- Rudolf Trümpy, an eminent Alpine geologist.

Fossils in time and space

FOLD BELTS

- A fold and thrust belt is a series of mountainous foothills adjacent to an orogenic belt, which forms due to contractional tectonics.

Fossils in time and space



Profile through the Pyrenees. In the south a fold and thrust belt exists as sediments are folded and stacked (thrusted) on top of the other. WF

https://en.wikipedia.org/wiki/Fold_and_thrust_belt#/media/File:Profile_through_the_Pyrenees_ES.svg

Fossils in time and space

FOSSILS IN FOLD BELTS

- Fossils from the deformed zones of mountain belts
- They are usually poorly preserved
- Metamorphosed and tectonized

Fossils in time and space

FOSSILS IN FOLD BELTS

- We can reassemble ancient mountain belts and trace the origins of their jumbled structure using paleontological data.

Fossils in time and space

FOSSILS IN FOLD BELTS

- Fossils in fold belts can be of great value to structural geologists in understanding the rates and timing of tectonic events.

Fossils in time and space

FOSSILS IN FOLD BELTS

- A fossil that was originally symmetric, but now squeezed, or stretched
- The precise evidence of the magnitude of the tectonic forces on it

Fossils in time and space

FOSSILS IN FOLD BELTS

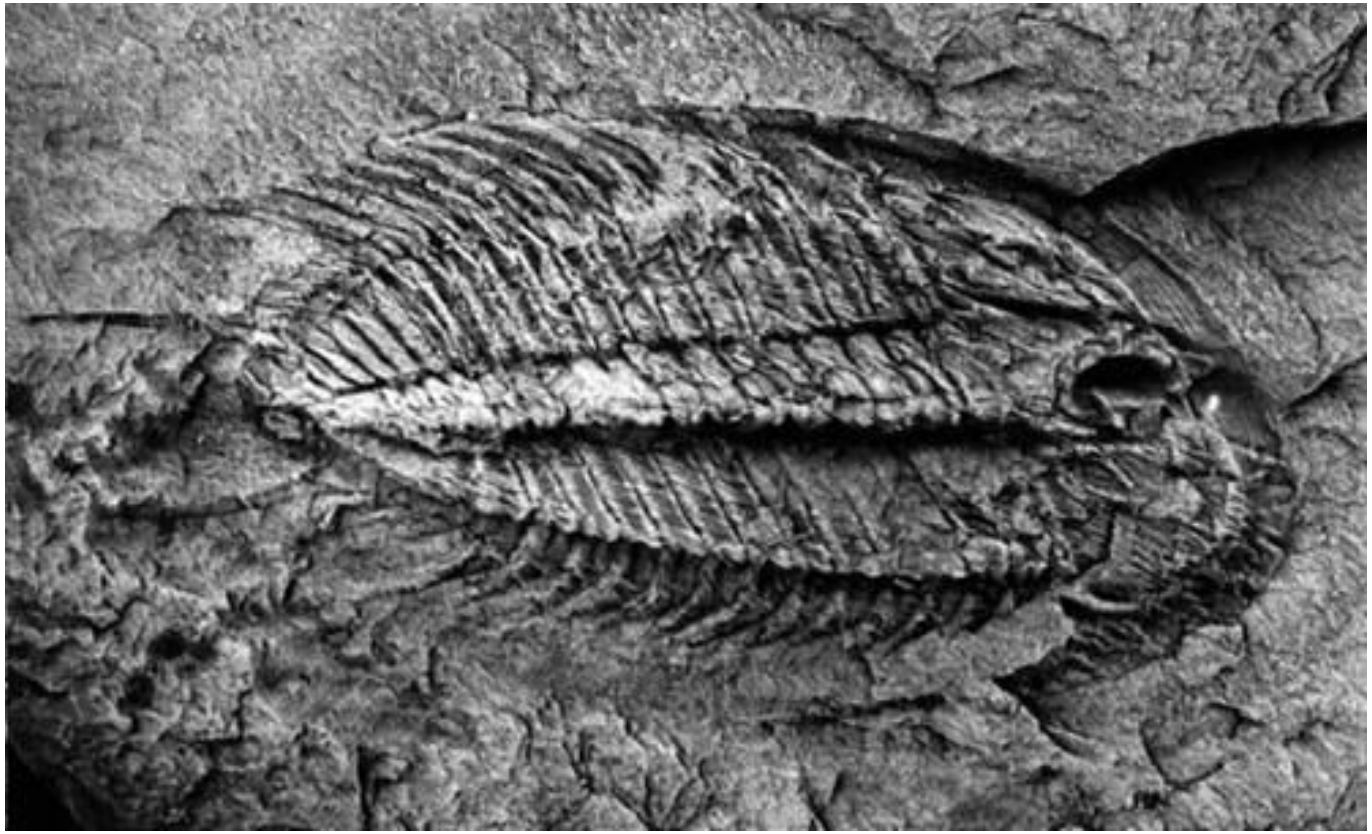
- Example is the "Delabole butterfly"
- In fact, the wide-hinged fossils are "Spiriferide brachiopods"

Fossils in time and space

FOSSILS IN FOLD BELTS

- Hughes and Jell (1992)
- Techniques to unstrain Cambrian trilobites from Kashmir
- distorted by earth movements during the uplift of the Himalayan mountain belt

Fossils in time and space



Strained Cambrian trilobites from Himalaya.

(Courtesy of Nigel Hughes.)

http://www.blackwellpublishing.com/paleobiology/figure.asp?chap=02&fig=Fig2-19&img=c02f019

Fossils in time and space

FOSSILS IN FOLD BELTS

- The investigation of thermal maturation is now a routine petroleum exploration technique.
- A number of groups of microfossils change color with changing paleotemperature.

END

Taphonomy and fossil record

Taphonomy and Fossil record

Taphonomy and fossil record

Taphonomy

- "Taphonomy is the paleontological study includes all the processes that occur after the death of an organism and before its fossilization in the rock"

Taphonomy and fossil record

Taphonomy

- Plants and animals with hard tissues are most frequently preserved in the fossil record.
- Soft tissues usually decay rapidly, but rapid burial or early mineralization may prevent decay

Taphonomy and fossil record

Taphonomy

- Physical and chemical processes during transport and compaction damage hard tissues
- Plants preserve as permineralized tissues, coalified compressions and cemented casts.

Taphonomy and fossil record

Taphonomy

- Many of the analytic approaches used by taphonomists are also used by forensic scientists.
- Both observe the state of decay of remains
- Measurement of the chemistry of the bone

Taphonomy and fossil record

Taphonomy

- Assessment of the rare earth elements (scandium, yttrium and the 15 lanthanides) to help pinpoint the time of death
- Archeologist and Paleontologists both use these techniques

Taphonomy and fossil record

Fossil Record

- Quality of the fossil record?
- Fidelity and quality of the fossil record.
- Can paleontologists trust their patchy fossil finds to understand large-scale patterns of evolution?

Taphonomy and fossil record

Fossil Record

- Paleontologists should be careful when they attempt to reconstruct a whole plant or animal, and try to understand its biomechanics, when they have just a few bones or bits of twigs.

Taphonomy and fossil record

END

Fossil Record

- Care is required in seeking to understand patterns of diversity change and evolution when many fossil species are missing.
- Diverse opinions and arguments in favour and against the fossil records

Taphonomy and fossil record

Fossil Preservation (Part 1)

Taphonomy and fossil record

Fossilization

- Sometimes a plant or an animal dies ends up as a fossil.
- There are several stages that normally occur in the transition from a dead body to a fossil

Taphonomy and fossil record

Transition to fossil

1. Decay of the soft tissues of the plant or animal.
2. Transport and breakage of hard tissues.
3. Burial and modification of the hard tissues.

Taphonomy and fossil record

Fossilization

- In rare cases, soft parts may be preserved, and these examples of exceptional preservation are crucially important in reconstructing past life.
- "Lagerstätten"

Taphonomy and fossil record

Two kinds of fossils

1. Body fossils: the partial or complete remains of plants or animals
2. Trace fossils: the remains of the activity of ancient organisms, such as burrows and tracks.

Taphonomy and fossil record

Decay

- When large animals feed on dead plant or animal tissues, the process is termed scavenging
- When microbes, such as fungi or bacteria, transform tissues of the dead organism, the process is termed decay

Taphonomy and fossil record

Factors controlling Decay

- Oxygen
- Microbial nutrients
- Temperature
- pH

- Example of Neolithic and younger "bog bodies" of northern Europe

Taphonomy and fossil record

Volatiles and Refractories

- Soft parts of animals are made from volatiles, forms of carbon that break down readily.
- Refractories are the organic carbons much less liable to break down, such as cellulose.

Taphonomy and fossil record

Mineralization

- Early mineralization of soft tissues depends on three factors:

1. Rate of burial
2. Organic content
3. Salinity

- Physical and chemical effects occurring after burial, are termed as Diagenesis

Taphonomy and fossil record

Conservation traps

- Obrution deposits: sedimentation rates are so rapid that carcasses are buried virtually instantly
- Amber: fossilised resin

Taphonomy and fossil record

Mineralization Types

- Mineralization of soft tissues occurs in three ways
1. Permineralization
2. Formation of mineral coats
3. Formation of tissue casts and shells

END

Taphonomy and fossil record

Fossil Preservation
(Part 2)

Taphonomy and fossil record

Breakage and transport

- Several processes of breakage
- Physical: disarticulation, fragmentation, abrasion
- Chemical: bioerosion, corrosion, dissolution

Taphonomy and fossil record

Breakage and transport

- Disarticulation: Skeletons that are made from several parts may become disarticulated, separated into their component parts.
- After scavenging or decay of connective tissue

Taphonomy and fossil record

Breakage and transport

- Fragmentation: Skeletons may also become fragmented
- Individual shells, bones or pieces of woody tissue break up into smaller pieces usually along lines of weakness

Taphonomy and fossil record

Breakage and transport

- Abrasion: Shells, bones and wood may be abraded by physical grinding and polishing against each other and against other sedimentary grains.

Taphonomy and fossil record

Breakage and transport

- Bioerosion:
- The removal of skeletal materials by boring organisms such as sponges, algae and bivalves
- Shells, bones and wood may undergo bioerosion

Taphonomy and fossil record

Breakage and transport

- Corrosion and Dissolution
- Before and after burial, skeletal materials are commonly corroded and dissolved by chemical action, i.e. by weak acidic waters

Taphonomy and fossil record

Burial and Modifications

- The remains are typically buried after scavenging, decay, breakage and transport.
- Sediment is washed or blown over the remains, and the specimen becomes deeply buried.

Taphonomy and fossil record

Burial and Modifications

- During and after burial, the specimen may undergo physical and chemical change
- Addition of $CaCO_3$ is called Carbonate concretion
- Coprolites are phosphtised.

Taphonomy and fossil record

Plant Preservation

- Through petrification or cellular permineralization

1. Coal balls
2. Coalified compression
3. Cementation
4. Direct preservation of hard parts

END

Taphonomy and fossil record

QUALITY OF THE FOSSIL RECORD

Taphonomy and fossil record

Incompleteness of the fossil record

- Charles Darwin "imperfection of the geological record"
- in 1972, David Raup explained all the factors that make the fossil record incomplete

Taphonomy and fossil record

Anatomic filters

- Organisms are likely to be preserved only if they have hard parts, a skeleton of some kind.
- Entirely soft-bodied organisms, such as worms and jellyfish, are only preserved in rare cases

Taphonomy and fossil record

Biological filters

- Behavior and population size matter
- Common organisms such as rats are more likely to be fossilized than rare ones such as pandas
- Shorter life span of rats

Taphonomy and fossil record

Ecological filters

- Where an organism lives matters.
- Animals that live in shallow seas, or plants that live around lakes and rivers, are more likely to be buried under sediment
- Flying animals

Taphonomy and fossil record

Sedimentary filters

- Some environments are typically sites of deposition, and organisms are more likely to be buried there.
- Mountainside or a beach VS a shallow lagoon or a lake

Taphonomy and fossil record

Preservation filters

- Once the organism is buried in sediment, the chemical conditions must be right for the hard parts to survive.
- The destruction of bones or shells by acidic waters. worn and damaged by physical movement.

Taphonomy and fossil record

Diagenetic filters

- After a rock has formed, the rock may be transformed by the passage of mineralizing waters
- Either enhancing the fossils, by replacing biomolecules with mineral molecules, or destroying the fossil

Taphonomy and fossil record

Metamorphic filters

- Over millions of years, the fossiliferous rock might be baked or subjected to high pressure.
- In these kinds of metamorphic fossils may survive these terrible indignities, or they may be destroyed.

Taphonomy and fossil record

Vertical movement filters

- Burial means the rock has been covered by younger rock
- Tectonic movements must subsequently raise the fossiliferous rock to the Earth's surface, or the fossil remains forever buried and unseen.

Taphonomy and fossil record

Human filters:

- Fossils are seen and collected by a human being
- The fossil has to be registered in a museum as part of collective human paleontological knowledge
- Individual amateur collections wasted

Taphonomy and fossil record

Sampling and reality

- The fossil record give us a good outline of the key events in the history of life.
- Understanding the history of life with the help of the fossils (small remnant of what once existed)is a big question.

END

Paleoecology and Paleoclimates

Paleoecology and
Paleoclimates

Paleoecology and Paleoclimates

Importance of Fossils

- Fossil organisms provide fundamental evidence of evolution
- They also allow the reconstruction of ancient animal and plant communities.

Paleoecology and Paleoclimates

Paleoecologists

- The functions of single fossil organisms (paleoautecology)
- The composition and structure of fossil communities (paleosynecology)

Paleoecology and Paleoclimates

The Paleoecology of fossil organisms

- Described in terms of life strategies and trophic modes together with their habitats
- All fossil organisms interacted with other fossil organisms

Paleoecology and Paleoclimates

The Paleoecology of fossil organisms

- Described in terms of life strategies and trophic modes together with their habitats
- All fossil organisms interacted with other fossil organisms

Paleoecology and Paleoclimates

The Paleoecology Analysis

- Populations and paleocommunities may be analyzed with a range of statistical techniques.

Paleoecology and Paleoclimates

Evolutionary Paleoecology

- Charts the changing structure and composition of Paleocommunities through time
- Ecological events can be ranked in importance.

Paleoecology and Paleoclimates

Evolutionary Paleoecology

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- Ecological events can be ranked in importance.

Paleoecology and Paleoclimates

Paleoclimates

- Described on the basis of climatically-sensitive biotas and sediments together with stable isotopes.
- Climate has been an important factor in driving evolutionary change at a number of different levels.

Paleoecology and Paleoclimates

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Paleoecology and Paleoclimates

Feedback loops and Gai Hypothesis

- Feedback loops between organisms and their environments indicate that the Gaia hypothesis is a useful model for some of geological time.

END

Paleoecology and Paleoclimates

Paleoecology (Part 1)

Paleoecology and Paleoclimates

Paleoecology

- Study of the life and times of fossil organisms, the lifestyles of individual animals and plants together with their relationships to each other and their surrounding environment.

Paleoecology and Paleoclimates

Speculation

- Element of speculation has prompted some paleontologists to exclude paleoecology from mainstream science

Paleoecology and Paleoclimates

Solution

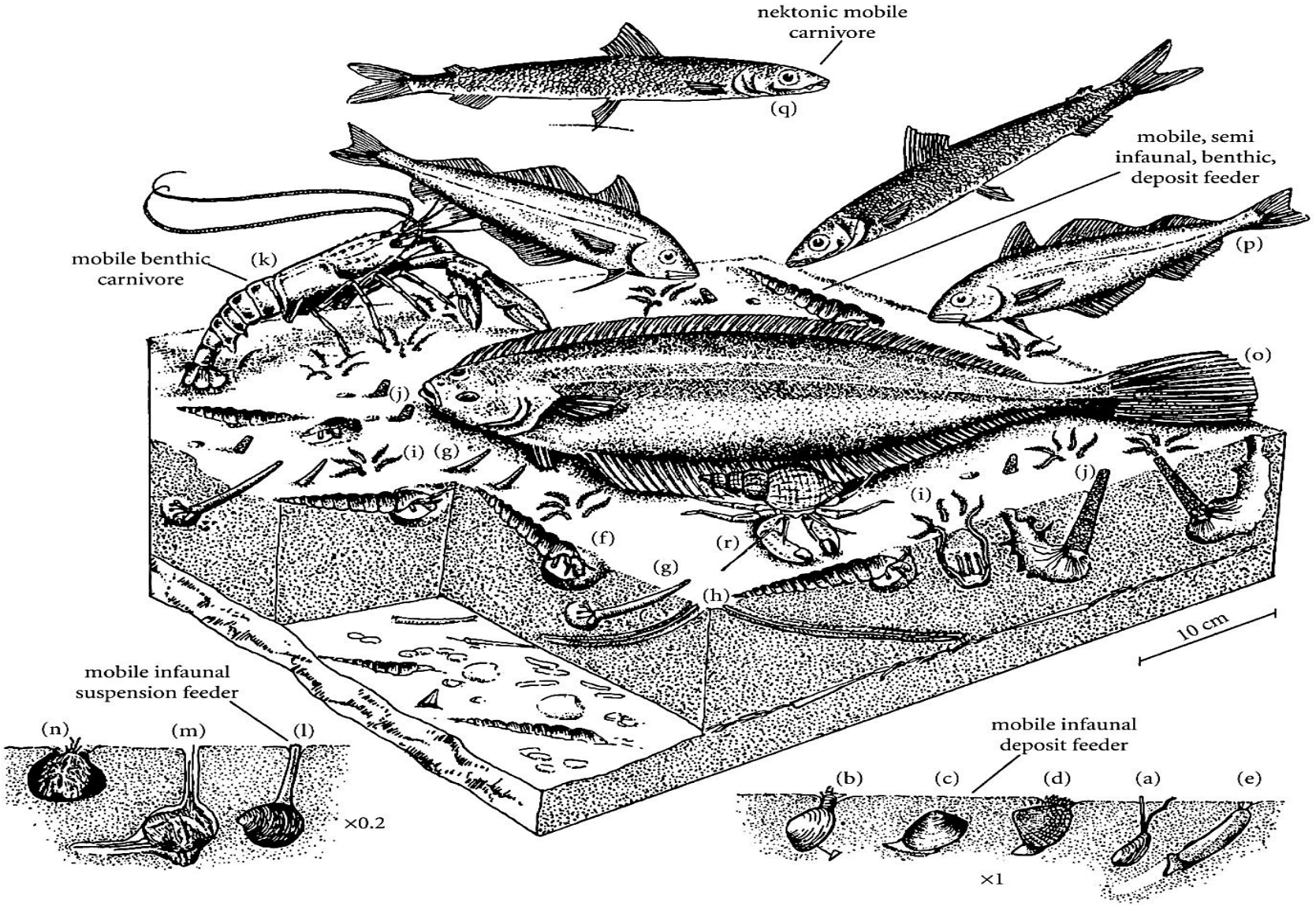
- Emerging numerical and statistical techniques can help us frame and test hypotheses
- Paleoecology is actually not very different from other sciences.

Paleoecology and Paleoclimates

Two main areas

- Paleoautecology is the study of the ecology of a single organism
- Paleosynecology looks at communities or associations of organisms.

Paleoecology and Paleoclimates



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Paleoecology and Paleoclimates

Fidelity

- The similarity of a death assemblage to its living counterpart, its fidelity
- Measurements of biomass and taxonomic composition

Paleoecology and Paleoclimates

Structure of Community

- Numerical abundance and diversity are not the best estimates
- Counts of stable adult populations are the most realistic monitors of community structure.

Paleoecology and Paleoclimates

Structure of Community

- A census of an extraordinarily preserved Lagerstätte deposit can help in the estimation of the Taphonomic loss.

Paleoecology and Paleoclimates

Structure of Community

- The degree of breakage with the attitude of fossils in sediments is useful to separate autochthonous (in place) from allochthonous (transported) assemblages.

Paleoecology and Paleoclimates

END

The -Coenoses

- The living assemblage, or biocoenosis, is transformed into a thanetocoenosis after death and decay.
- Taphocoenosis is the end product that is finally preserved

Paleoecology and Paleoclimates

Paleoecology
Part 2

Paleoecology and Paleoclimates

Population

- Populations are the building blocks of communities
- A population is a naturally occurring assemblage of plants and animals that live in the same place at the same time and regularly interbreed.

Paleoecology and Paleoclimates

Ecosystem

- "All the populations of species living in association"
- Keystone species help shape the ecosystem and that can trigger large-scale changes if they disappear.
- A classic keystone species is the elephant

Paleoecology and Paleoclimates

Incumbent Species

- Incumbent species can occupy the same ecological niche for many millions of years, adding stability to many ecosystems.

- Dinosaurs and mammals

Paleoecology and Paleoclimates

Habitat and Niche

- Fossil organisms can be classified in terms of
- Habitat, where they live.(their address)
- Niche, their lifestyle (their occupation).

Paleoecology and Paleoclimates

Marine Sediments

- The majority of fossil animals have been found in marine sediments
- Occupy a wide range of depths and conditions.
- Photic zone is the depth of water penetrated by light

Paleoecology and Paleoclimates

Marine Variety of Life

- Marine environments host a variety of lifestyles
- The upper surface waters are rich in floating plankton, and nektonic organisms swim at various levels in the water

Paleoecology and Paleoclimates

Marine Sediments

- Benthos are the beasts that
- live in or on the seabed
- Mobile nektobenthos, scuttle across the seafloor
- the fixed or sessile benthos are fixed

Paleoecology and Paleoclimates

Marine Sediments

- Infaunal organisms live beneath the sediment–water interface
- Epifaunal organisms live above it The sediment–water interface

Paleoecology and Paleoclimates

Guilds

- Guilds are groups of functionally similar organisms occurring together in a community.

Paleoecology and Paleoclimates

END

Megaguild

- Megaguilds are a range of adaptive strategies based on a combination of life position (e.g. shallow, active, infaunal burrower)

Paleoecology and Paleoclimates

Paleoclimates

Paleoecology and Paleoclimates

Paleoclimatology

- The study of changes in climate taken on the scale of the entire history of Earth.
- Uses the methods from the Earth and life sciences

Paleoecology and Paleoclimates

Paleoclimatology

- It obtains data from previously preserved things such as rocks, sediments, ice sheets, tree rings, corals, shells, and microfossils.

Paleoecology and Paleoclimates

The Climate change in life history

- Last 600 million years
- Earth has oscillated at least five times between icehouse and greenhouse conditions
- Spending most of the time in greenhouse climates

Paleoecology and Paleoclimates

Five climate zones

1. Humid Tropical (no winters and average temperatures above 18°C)
2. Dry subtropical (evaporation exceeds precipitation)

Paleoecology and Paleoclimates

Five climate zones

3. Warm temperate (mild winters)
4. Cool temperate (severe winters)
5. Polar (no summers and temperatures below 10°C).

Paleoecology and Paleoclimates

A big Question

- Can these zones be recognized through deep time and be used to develop models for both short- and long-term climate change?

Paleoecology and Paleoclimates

Identification of climate zones

- A range of geological and paleontological criteria has helped identify climatic zones through time.

Paleoecology and Paleoclimates

Identification of climate zones

- calcretes (soils rich in calcium carbonate) and evaporites (evaporated salts) help identify dry, arid climates

Paleoecology and Paleoclimates

Identification of climate zones

- Dropstones (stones that plummet from the bottoms of melting icebergs into seabed sediments) and Tillites (rocks and sand left behind by an advancing glacier) indicate polar conditions.

Paleoecology and Paleoclimates

Trends of climate change

- Short term trends
- Long term trends
- Consequences for evolution
- Biological Feedback and Gaia Hypothesis

END

Macroevolution & The Tree of Life

Macroevolution and the Tree of Life

Macroevolution & The Tree of Life

Evolution by natural selection

- Evolution by natural selection is a core scientific model that was set out by Darwin
- It has been confirmed again and again in every branch of Biology.

Macroevolution & The Tree of Life

Intelligent design

- Creationists attempts to promote their religious beliefs, such as "intelligent design" or belief in a flat Earth, are not testable and therefore are not science.

Macroevolution & The Tree of Life

Speciation

- Speciation often occurs by the establishment of a barrier, and the isolation of part of a previously interbreeding population.

Macroevolution & The Tree of Life

Evolution

- Evolution takes place both within
- Species lineages (phyletic gradualism)
- at the time of speciation (punctuated equilibrium)

Macroevolution & The Tree of Life

Tree of Life

- The evolution of life may be represented by a single branching phylogenetic tree.
- Cladistics is a method of reconstructing phylogeny based on the identification of shared derived characters (homologies).

Macroevolution & The Tree of Life

END

Evolution

- Molecular sequencing provides additional evidence for reconstructing and dating the tree of life.
- DNA has been extracted from fossils such as woolly mammoths, but not from truly ancient fossils

Macroevolution & The Tree of Life

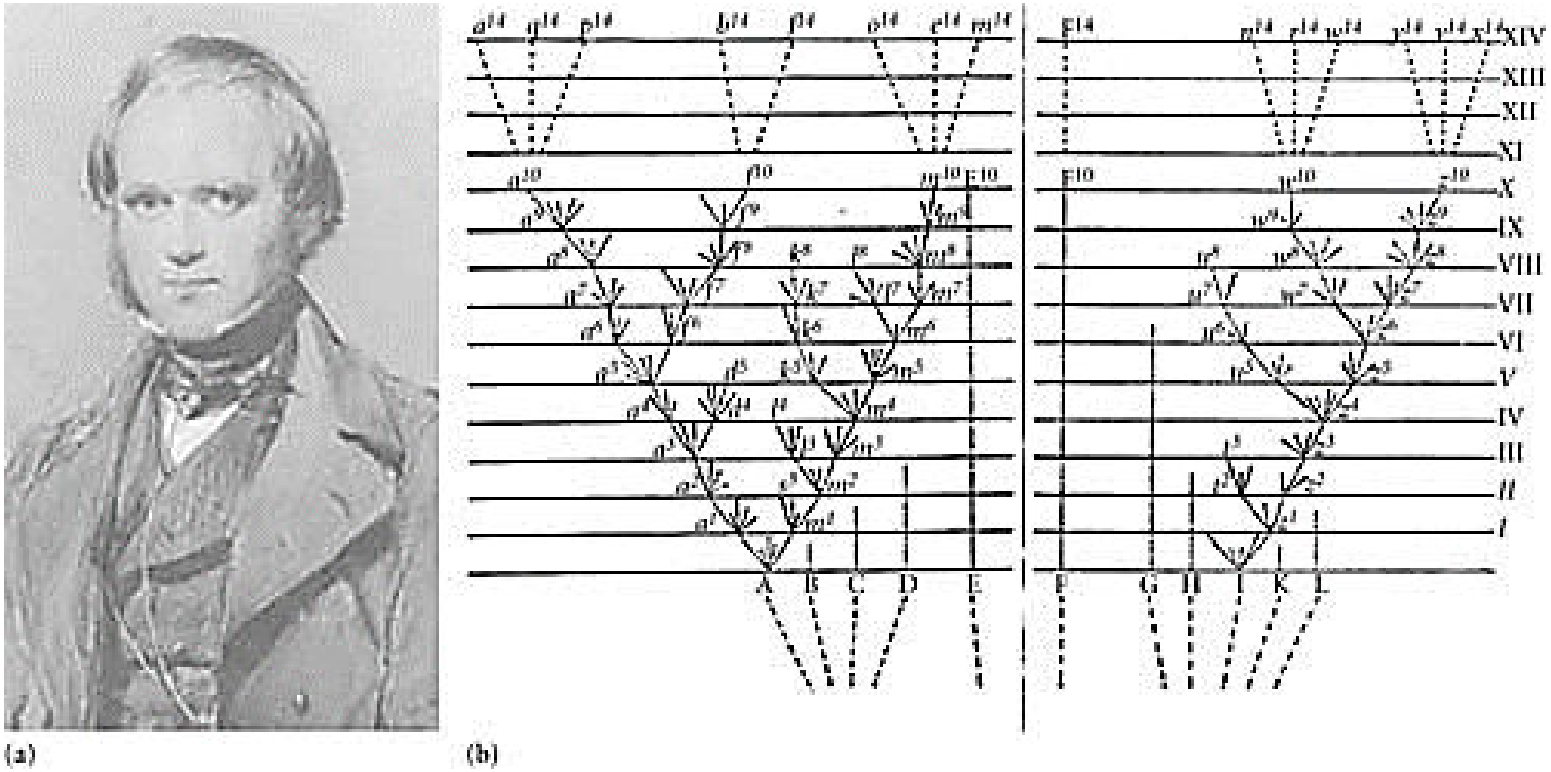
Evolution by Natural Selection

Macroevolution & The Tree of Life

Early Evolutionists

- Darwin laid the framework for Evolutionary Biology 150 years ago.
- No one has yet falsified Darwin's theory of evolution by natural selection
- On The Origins of Species (1859)

Macroevolution & The Tree of Life



(a) Charles Darwin. (b) Branching diagram of phylogeny, the only illustration in On the Origin of Species (1859). It shows how two species, A and I, branch and radiate through time. The units I–XIV are time intervals of variable length, and the lower case letters (a, b, c) represent new species.

Macroevolution & The Tree of Life

Tree of Life

- Darwin: life had diversified to millions of species by the continued splitting of species from a common stem
- Explained for the first time the meaning of the natural hierarchy of life.

Macroevolution & The Tree of Life

Macroevolution

- Paleontological aspects of evolution, such as the tree of life and studies of processes over thousands and millions of years, are sometimes called macroevolution ("big evolution")

Macroevolution & The Tree of Life

Microevolution

- Microevolution ("small evolution") are all the smaller-scale and shorter-term processes studied by biologists and geneticists in the laboratory or in the field.

Macroevolution & The Tree of Life

Darwin's Quest of the Truth

- The Voyage of *HMS Beagle*
- Observation of Diversity of life
- Evidence for relationship in time and space
- Awareness of Fossil record
- Malthus' work

Macroevolution & The Tree of Life

Principle of Natural Selection

- *"Only the organisms best adapted to their environment tend to survive and transmit their genetic characteristics in increasing numbers to succeeding generations while those less adapted tend to be eliminated."*

Macroevolution & The Tree of Life

Natural Selection

1. Nearly all species produce far more young than can survive to adulthood (Malthus'principle)
2. The young that survive tend to be those best adapted to survive.

Macroevolution & The Tree of Life

Natural Selection

3. Characters are inherited from parent to offspring, so the characters that ensure survival will tend to be passed on.
4. These survival characteristics will increase generation by generation.

END

Macroevolution & The Tree of Life

The Tree of Life

Macroevolution & The Tree of Life

Tree Thinking

- Darwin was the first to picture evolution as a great branching tree.
- "Which species of ape is closest to humans – the gorilla or chimp?"
- Putting together complete trees of all species

Macroevolution & The Tree of Life

Tree Thinking

- Scala Naturae: the ladder of life.
- Ladder go from lower to higher, simple to complex
- Tree of life: Splits and Branches
- Tree is a better analogy for life

Macroevolution & The Tree of Life

Cladistics

- In Cladistics, we reconstruct the Life's Hierarchy.
- The true patterns of Relationship among the organisms.
- Similarities and Differences among organisms

Macroevolution & The Tree of Life

Phylogenetically informative characters

- Good characters are phylogenetically informative, that is, indicative of the true phylogeny.
- These characters identify the clads

Macroevolution & The Tree of Life

Phylogenetically uninformative characters

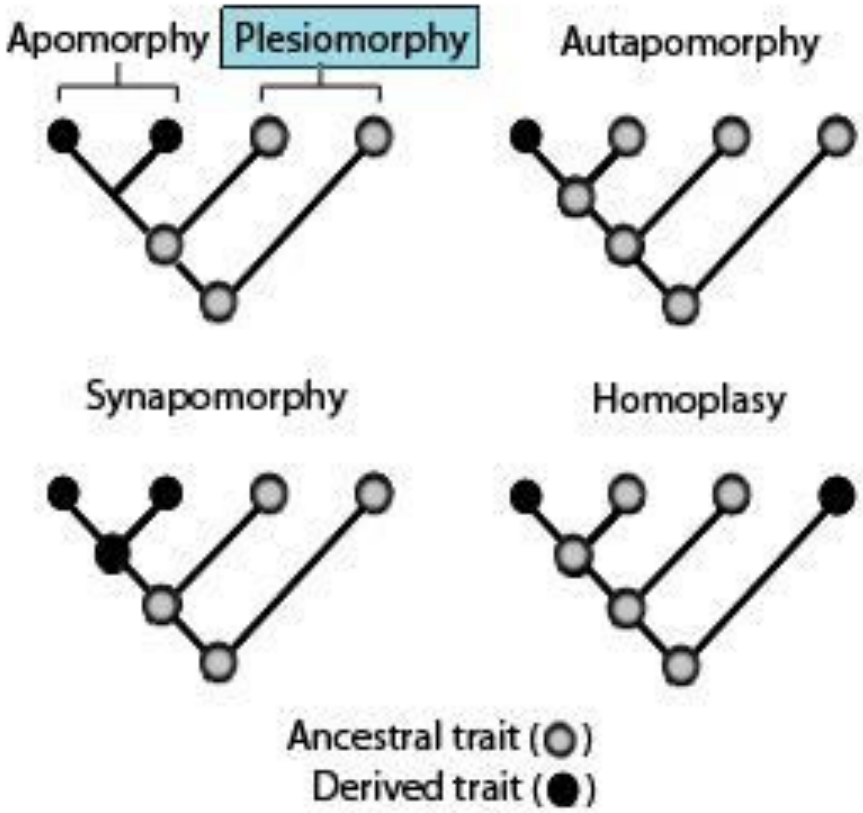
- Phylogenetically uninformative characters fall into two main categories:
- Convergences
- Plesiomorphies

Macroevolution & The Tree of Life

Convergence

- Convergence in evolution is when features, or organisms, evolve to look the same perhaps because they live the same way.

Macroevolution & The Tree of Life



Plesiomorphies

- Plesiomorphies are characters that are shared by the organisms of interest, but also by other groups.

https://upload.wikimedia.org/wikipedia/commons/1/1b/Plesiomorphy.jpg

Macroevolution & The Tree of Life

Monophyletic Groups

- These are groups that had a single origin and include all the descendants of that common ancestor.
- Psittaciformes, the parrots, long been identified as real and distinct from all others by naturalists

Macroevolution & The Tree of Life

Non Natural Groups

- Clades are distinguished from two kinds of non-natural groups:

- Paraphyletic groups
- Polyphyletic groups

END

Macroevolution & The Tree of Life

Cladistics

Macroevolution & The Tree of Life

Early Scientists

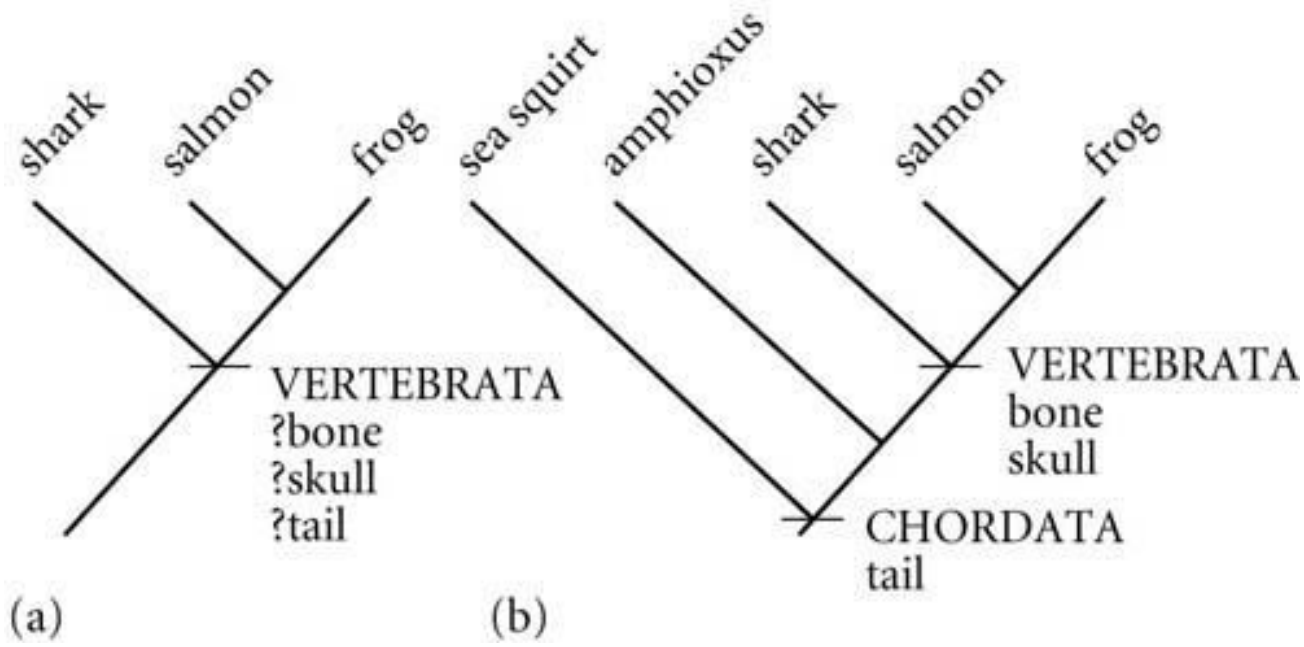
- Willi Hennig (1913–1976), an eminent German entomologist, stressed the need to develop a new, more objective method in systematics, which has come to be called cladistics.

Macroevolution & The Tree of Life

Cladistics

- The fundamental aim of cladistics is to identify clades, and so to discover, or reconstruct, the tree of life.
- Patterns of relationships are shown as branching diagrams, or cladograms

Macroevolution & The Tree of Life



Reconstructing the phylogeny of vertebrates by cladistic methods. (a) Are the defining features of vertebrates the possession of bone, a skull and a tail? (b) The tail is found in a wider group, termed the Chordata, but the skull and bone define the Vertebrata.

Macroevolution & The Tree of Life

Nodes

- The most closely related species are joined most closely.
- The branches in the cladogram join at branching points, or nodes, each of which marks the base of a clade

Macroevolution & The Tree of Life

Apomorphies

- Hennig called the phylogenetically informative characters "Apomorphies" or derived characters
- Apomorphies are features that arose once only in evolution.

Macroevolution & The Tree of Life

Synapomorphies

- "Apomorphies" shared by two or more species are termed synapomorphies
- Synapomorphies of Order Psittaciformes include the deep, hooked beak and the unusual foot.

Macroevolution & The Tree of Life

Older distinctions

- "Homology" and "Analogy"
- Vertebrate Limbs
- Wing of bird, paddle of whale arm of human are the same. Homologs.
- Peddles of whales and seals

Macroevolution & The Tree of Life

Molecular Revolution

- Birth of Molecular Biology 1950s and 1960s
- Comparisons of molecules allow analysts to do two things:

1. To draw up trees of relationships
2. To estimate time.

Macroevolution & The Tree of Life

Dendrograms

- Trees of relationships can be based on a comparison of the amount of difference between protein sequences, and a best-fitting dendrogram, or branching diagram, is drawn.

Macroevolution & The Tree of Life

Molecular Clocks

- Time estimation comes from the concept of the molecular clock.
- The amount of difference in the fine structure of a protein is proportional to the time since they last shared a common ancestor.

Macroevolution & The Tree of Life

Nucleic Acids

- The nucleic acid sequences must be aligned, that is matched, so that the code of a particular gene in one species is lined up with the same sequence in another species.

END

Fossils form and function

Fossil form and Function

Fossils form and function

Fossil species

- Fossil species are identified according to their external form; this is termed the morphological species concept.
- Variations in form include normal levels of individual variation

Fossils form and function

Fossil species

- The other types of variations in the forms of the fossils include
- Geographic distribution
- Sexual dimorphism
- Different growth stages
- ecophenotypic variation

Fossils form and function

Allometry

- Fossil species may show allometry, or changes in relative proportions during growth;
- Specific organs may show positive (grow faster) or negative (grow slower) allometry.

Fossils form and function

Development and Phylogeny

- The development of an organism may give some evidence about phylogeny.
- Changes in developmental rates and timing (heterochrony) may affect evolution.

Fossils form and function

The New "Evo-Devo" Perspective

- shows how certain developmental genes control fundamental aspects of form
- Symmetry
- Front–back orientation
- Segmentation
- Limb form

Fossils form and function

Inferring function from ancient organisms

- Is difficult
- Various methods of doing this:
- Comparison with modern analogs
- Biomechanical testing
- Circumstantial evidence

Fossils form and function

Modern analogs

- Modern analogs may provide exact parallels with some fossil organisms
- More often they provide only principles or rules.

Fossils form and function

Biomechanical models

- May be used to assess how the design of an ancient organism matches the hypothetical forces acting on it

Fossils form and function

Biomechanical models

- May be used to assess how the design of an ancient organism matches the hypothetical forces acting on it
- Models of locomotion are easy to produce

Fossils form and function

Biomechanical models

- Circumstantial evidence, such as
1. The enclosing rocks
2. Associated fossils
3. Trace fossils and
4. Close study of the fossils themselves

can add considerable information on fossil function.

END

Fossils form and function

Growth and Form

Fossils form and function

Evolution and Form

- Darwin gave us an evolutionary view of form.
- He was astonished by
- The variety of external appearances
- By their wonderful adaptations to life.

Fossils form and function

Form

- The form, or external appearance, of any microbe, plant or animal is shaped by evolution.
- Wings for flight
- Longs beak and tongue of humming bird

Fossils form and function

Sexual Selection

- Sexual selection is the set of evolutionary processes that depend on interactions between the sexes.
- Example: The astonishing tails and colors of male birds

Fossils form and function

Adaptations

- Plants and animals have adaptations that function in the context of both natural and sexual selection
- *An adaptation is an aspect of form that performs a physical or behavioral function.*

Fossils form and function

Importance of Form

- The form of fossils is important for three reasons

1. Form is the only evidence we have in fossils for identifying species and wider relationships

Fossils form and function

Importance of Form

2. Form can tell us about behavior and ecology

3. The study of changes in form through time informs us about evolution

Fossils form and function

Recognizing ancient species

- Paleontologists use the morphological species concept
- Statistical observations define the mean or average characteristics in a specie

END

Fossils form and function

Growth and Form
(Continued)

Fossils form and function

Variations Within the Species

- Within a species, there may be a range of morphologies
- Individual variation
- The stuff of natural selection, as Darwin stressed.

Fossils form and function

Variations Within the Species

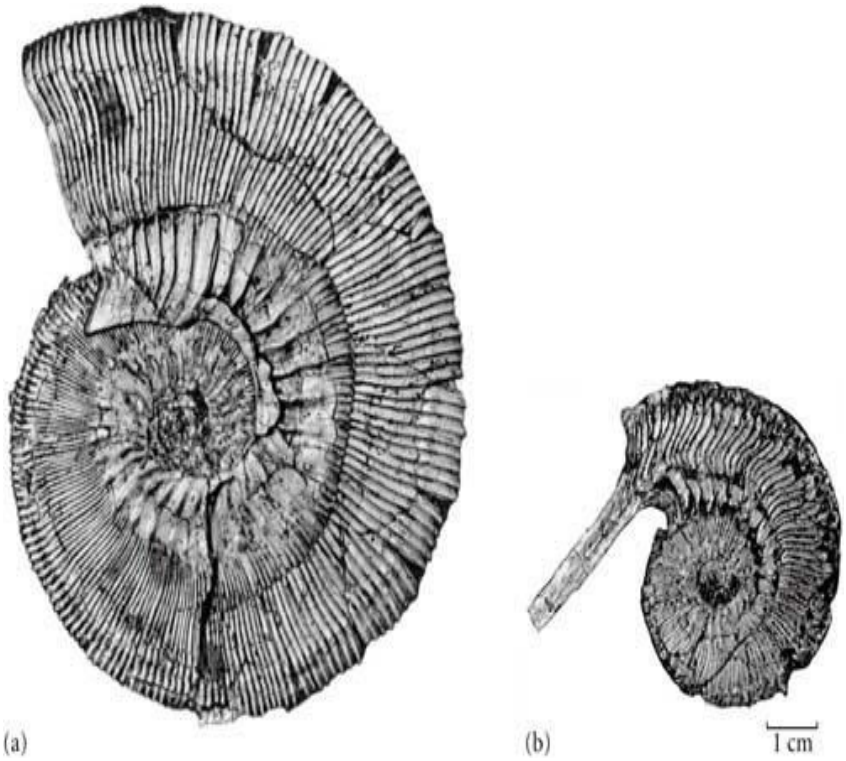
- Geographic variation and physical differences between populations or subspecies in different parts of the overall species range.

Fossils form and function

Variations Within the Species

- Sexual dimorphism, in which males and females may show different sizes, and different specialized features (horns, antlers, tail feathers)

Fossils form and function



Sexual dimorphism in ammonites, the Jurassic Kosmoceras. The larger shell (a) was probably the female, the smaller (b) the male. (Courtesy of Jim Kennedy and Peter Skelton.)

http://www.blackwellpublishing.com/paleobiology/figure.asp?chap=06&fig=Fig6-2&img=c06f002

Fossils form and function

Variations Within the Species

- Growth stages, where there may be quite different larval and adult stages, or where body form alters during growth

Fossils form and function

Variations Within the Species

- Ecophenotypic effects,
- Where local ecological conditions affect the form of an organism during its lifetime

Fossils form and function

Allometry

- Changes in form during growth are common.
- Think of human growth: babies have relatively large heads and eyes, and small limbs.

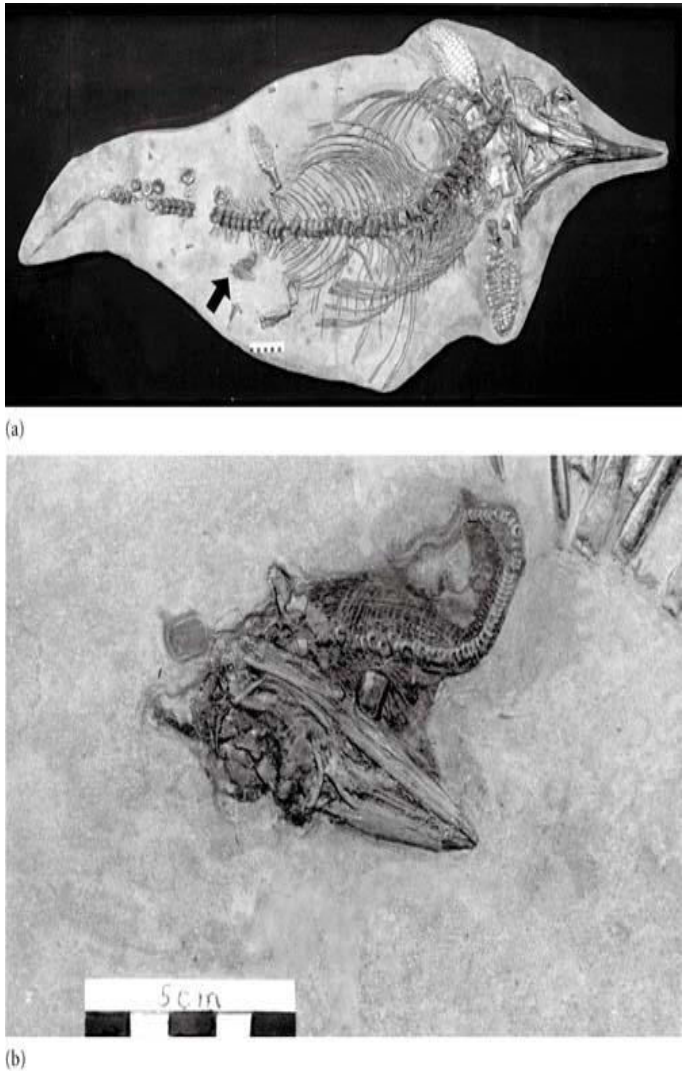
Fossils form and function

Allometry

- If measurements of the variable parts are scaled against a standard measure of the animal, it is evident that the proportions change as the animal grows older

Fossils form and function

Adult female Ichthyosaurus (a) from the Lower Jurassic of Somerset, England, showing an embryo that has just been born (arrow), and detail of the curled embryo (b). (Courtesy of Makoto Manabe.)



http://www.blackwellpublishing.com/paleobiology/figure.asp?chap=06&fig=Fig6-3&img=c06f003

Fossils form and function

Allometric Growth

- The ratio of eye diameter to body length diminishes as the animal approaches adulthood.
- This is an example of allometric ("different measure") growth.

Fossils form and function

Isometric Growth

- If there is no change in proportions during growth, the feature is said to show isometric ("same measure") growth.

Fossils form and function

Biological Scaling Principle

- Some organs and functions relate to the mass of an animal (a three-dimensional measure)
- others relate to body length or body outline (one- and two-dimensional measures).

END

Fossils form and function

Evolution and Development

Fossils form and function

Ontogeny and Phylogeny

- Biologists have long sought a link between ontogeny (development) and phylogeny (evolutionary history).

Fossils form and function

Biogenetic Law

- In 1866, Ernst Haeckel, a German evolutionist, announced his Biogenetic Law, that "ontogeny recapitulates phylogeny".

Fossils form and function

Biogenetic Law

- Ernst Haeckel's idea was that the sequence of embryonic stages mimicked the past evolutionary history of an animal.

Fossils form and function

Von Baer's Law

- A parallel between the sequence of development, and a cladogram.
- In human development embryo passes through the major nodes of the cladogram of vertebrates.

Fossils form and function

Von Baer's Law

- The synapomorphies of vertebrates appear first, then those of tetrapods, then those of amniotes, then those of mammals, of primates, and of the species *Homo sapiens* last.

Fossils form and function

Development and Phylogeny

- Three other aspects of development throw light on phylogeny
- Atavisms
- Vestigial structures
- Patterns of Development

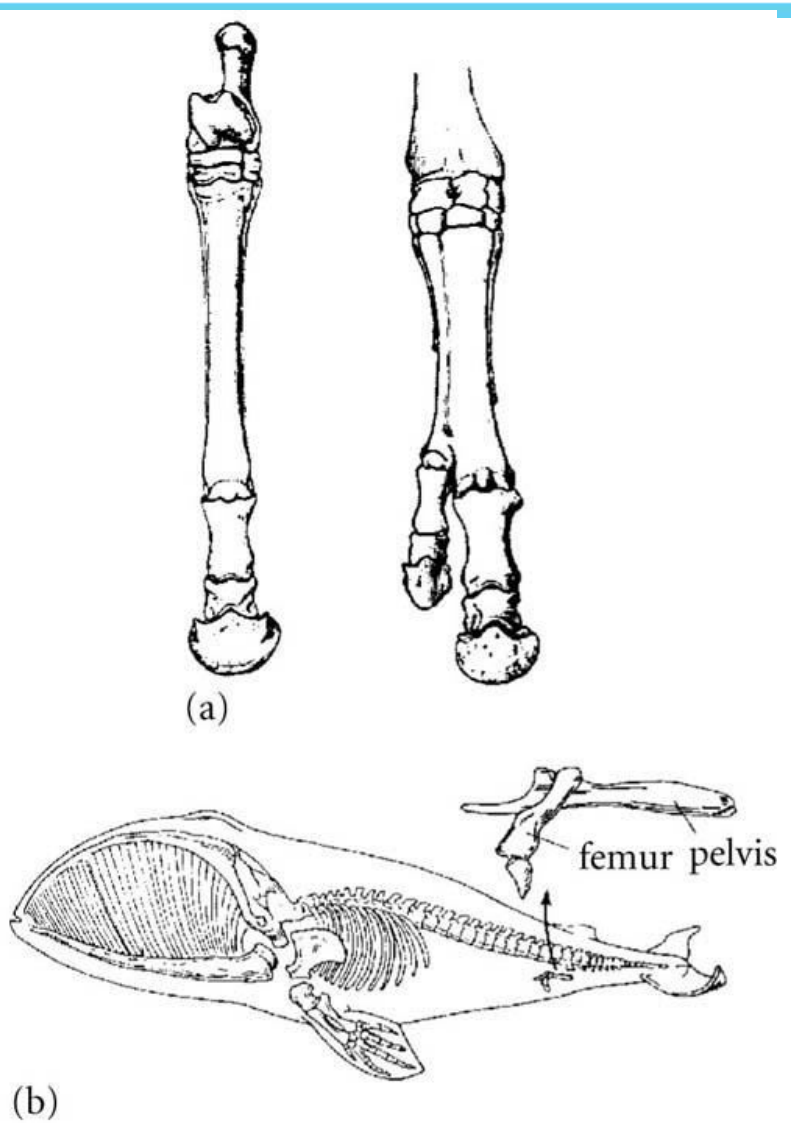
Fossils form and function

Atavisms

- Certain developmental abnormalities called atavisms, or throwbacks, show former stages of evolution, such as human babies with small tails or excessive hair

Fossils form and function

Hints of ancestry in modern animals. (a) Extra toes in a horse; normal horse leg (left), extra toes (right). (b) The vestigial hip girdle and hindlimb of a whale; the rudimentary limb is the rudiment of a hindlimb that functioned 50 Ma.



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Fossils form and function

Developmental Genes

- Widely shared among organisms
- Determine aspects of form
- Symmetry
- Anteroposterior orientation
- Limb differentiation

Fossils form and function

END

Developmental Genes

- Since the 1980s a major new research field "evodevo" (short for evolution–development)
- Most famous developmental genes are the homeobox genes

Mass extinctions and biodiversity loss

Introduction

Mass extinctions and biodiversity loss

Mass Extinctions

- During mass extinctions, 20–90% of species were wiped out; these include a broad range of organisms, and the events appear to have happened rapidly.

Mass extinctions and biodiversity loss

Mass Extinctions

- It is difficult to study mass extinctions in the Precambrian
- There have been a Neoproterozoic event between the Ediacaran and Early Cambrian faunas.

Mass extinctions and biodiversity loss

Mass Extinctions

- The "big five" Phanerozoic mass extinctions occurred in the
  - End-Ordovician
  - Late Devonian
  - End-Permian
  - End-Triassic
  - End-Cretaceous

Mass extinctions and biodiversity loss

Mass Extinctions

- The end-Permian mass extinction was the largest of all time
- Probably caused massive volcanic eruptions, leading to acid rain and global anoxia.

Mass extinctions and biodiversity loss

Mass Extinctions

- The end-Cretaceous mass extinction has been most studied, and it was probably caused by a major impact on the Earth.

Mass extinctions and biodiversity loss

Mass Extinctions

- Smaller-scale extinction events include the loss of mammals at the end of the Pleistocene, perhaps the result of climate change and human hunting.

Mass extinctions and biodiversity loss

Recovery from **Mass Extinctions**

- Takes a long time
- First, some unusual disaster taxa that cope well in harsh conditions
- Gives way to the longer-lived taxa that rebuild normal ecosystems

Mass extinctions and biodiversity loss

Major Concern Today

- Calculated species loss as high as during any mass extinction of the past.
- The severity of the current extinction episode is still debated.

END

Mass extinctions and biodiversity loss

Mass Extinctions

Mass extinctions and biodiversity loss

Extinction

- Extinction, is now a key theme in discussions about the future
- The dodo is perhaps the most iconic of icons
- An icon of human carelessness rather than of avian extinction.

Mass extinctions and biodiversity loss

An image of a dodo from another era. Lewis Carroll introduced the dodo as a kindly and wise old gentleman in Alice Through the Looking Glass, although at the time most people probably regarded the dodo as rather foolish. Driven to extinction in the 17th century by overhunting, the dodo is now an image of human thoughtlessness.



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Mass extinctions and biodiversity loss

Mass Extinctions

- The most spectacular extinctions are known as mass extinctions
- Times when a large cross-section of species died out rather rapidly.

Mass extinctions and biodiversity loss

Extinctions Events

- There may have been only five or six mass extinctions throughout the known history of life, although there were many extinction events,

Mass extinctions and biodiversity loss

Study of Mass Extinctions

- From the 1980s
- Wide interdisciplinary links
- Stratigraphy
- Geochemistry
- Climate modeling
- Ecology
- Conservation
- Astronomy

Mass extinctions and biodiversity loss

Study of Mass Extinctions

- The study of mass extinctions involves careful hypothesis testing at all levels, from the broadest scale to the narrowest

Mass extinctions and biodiversity loss

Back ground

Extinctions

- Extinction happens all the time
- Species have a natural duration of few million years and so they live for a time and then disappear

Mass extinctions and biodiversity loss

Features of Mass Extinction

- More than 30% of plants and animals of the time.
- The extinct organisms spanned a broad range of ecologies

Mass extinctions and biodiversity loss

Features of Mass Extinction

- The extinctions were worldwide covering most continents and ocean basins.
- The extinctions all happened within a relatively short time

Mass extinctions and biodiversity loss

Features of Mass Extinction

- The level of extinction stands out as considerably higher than the background extinction level.

END

Mass extinctions and biodiversity loss

Mass Extinctions (Continued)

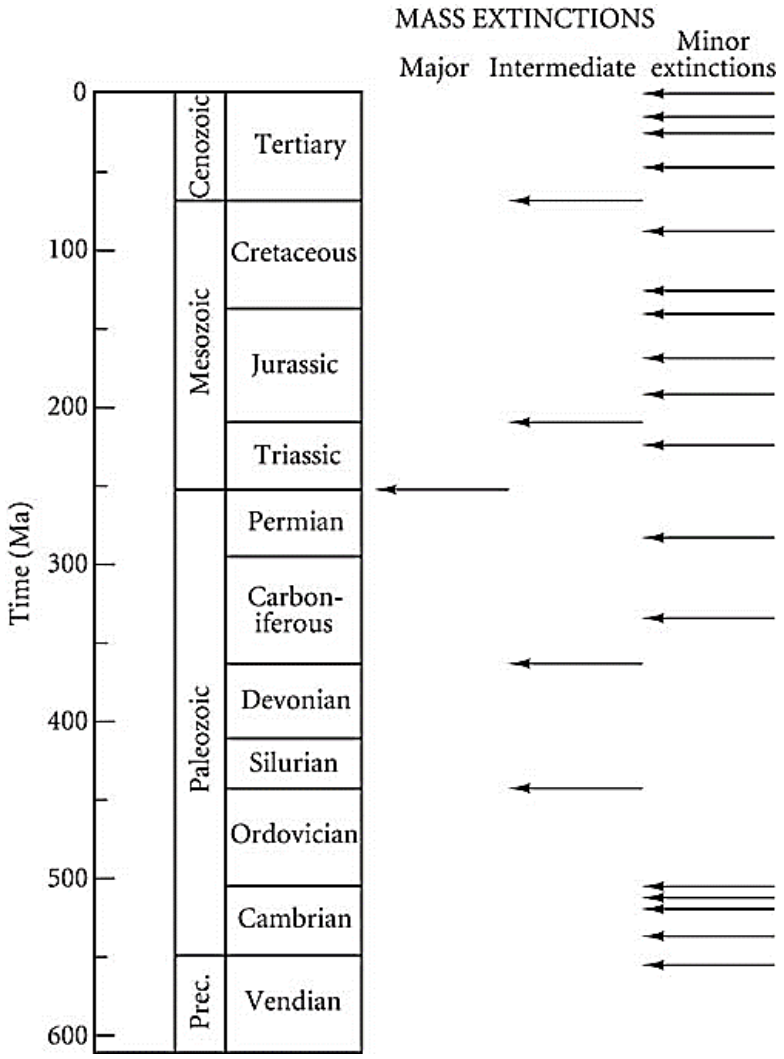
Mass extinctions and biodiversity loss

Mass Extinctions

- Paleontologists talk about the "big five" mass extinctions of the last 540 myr
- The Phanerozoic, and the current extinction crisis is sometimes called the "sixth extinction".

Mass extinctions and biodiversity loss

Mass extinctions through the past 600 myr. Major: 50% of families and up to 96% of species Intermediate: global in extent, and involved losses of 20% of families and 75–85% of species. Minor mass extinctions: causing losses of 10% of families and up to 50% of species



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Mass extinctions and biodiversity loss

Patterns and Timings

- Good-quality fossil records indicate a variety of patterns of extinction
- The rock record can be misleading
- Gradual extinctions might look catastrophic

Mass extinctions and biodiversity loss

Signor–Lipps effect

- Paleontologists will never find the very last fossil of a species.
- Signor and Lipps showed how this backward smearing now termed the Signor–Lipps effect

Mass extinctions and biodiversity loss

Selectivity and mass extinctions

- The second defining character of mass extinctions was that they should be ecologically catholic
- There should be little evidence of selectivity.

Mass extinctions and biodiversity loss

Periodicity of mass extinctions

- Fundamental debate has been whether each event had its own unique causes, or whether a unifying principle linking all mass extinctions might be found

Mass extinctions and biodiversity loss

Periodicity of mass extinctions

- The idea of periodicity of impacts was reawakened by Rohde and Muller (2005) who argued for a 62 myr periodicity in mass extinction

END

Mass extinctions and biodiversity loss

The "Big Five"
Mass
Extinction Events

Mass extinctions and biodiversity loss

The "big five" or the "big three"?

- There is some debate about whether there were five or three mass extinctions
- Key points about three of the five events

Mass extinctions and biodiversity loss

End-Ordovician mass extinction

- 445 Ma
- Substantial turnovers occurred among marine faunas
- Reef-building animals, brachiopods, echinoderms, ostracodes trilobites died out

Mass extinctions and biodiversity loss

End-Ordovician **mass extinction**

- Major Climatic changes
- Tropical-type reefs and their rich faunas lived around the shores of North America that then lay around the equator
- Southern glaciation and lowering sea levels globally

Mass extinctions and biodiversity loss

Late Devonian mass extinction

- A succession of extinction pulses 380 to 360 Ma
- The cephalopods were decimated
- Causes: A major cooling phase associated with anoxia or an extraterrestrial impact

Mass extinctions and biodiversity loss

The end-Triassic mass extinction

- The fourth of the big five mass extinctions
- A marine mass extinction event at, or close to, the Triassic-Jurassic boundary, 200 Ma
- Anoxia and global warming due to Volcanic eruptions

Mass extinctions and biodiversity loss

Rest of Mass Extinctions

- The third and fifth of the "big five" were the
- Permo-Triassic (PT)
- Cretaceous-Tertiary (KT) events

Mass extinctions and biodiversity loss

The Permo-Triassic event

- The end-Permian, or Permo-Triassic, extinction was the most devastating of all time
- Less understood than KT event

Mass extinctions and biodiversity loss

The Permo-Triassic event

- The peak of eruptions by the Siberian Traps was dated at 251 Ma, matching precisely the date of the PT boundary
- Environmental changes

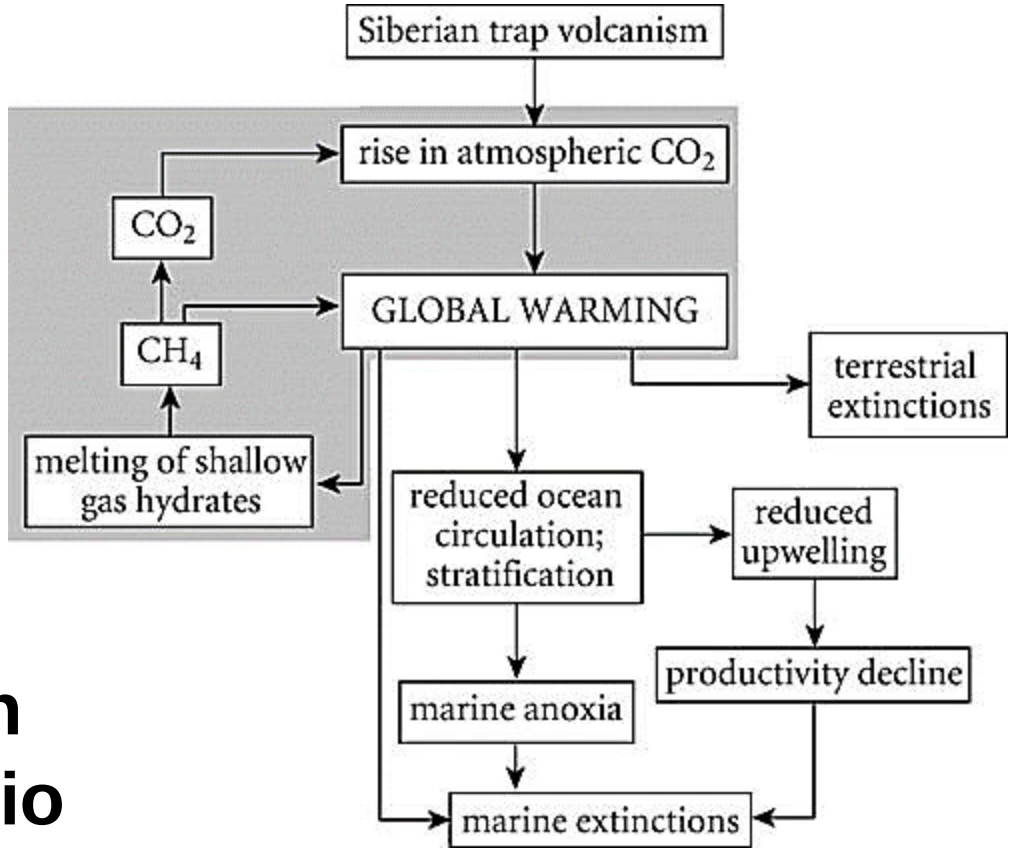
Mass extinctions and biodiversity loss

The Permo-Triassic event

- studies of stable isotopes (oxygen, carbon) in those rock sections revealed a common story of environmental turmoil

Mass extinctions and biodiversity loss

The possible chain of events following the eruption of the Siberian Traps



Shaded Grey: Runway Green House Scenario

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Mass extinctions and biodiversity loss

The Cretaceous-Tertiary event

Mass extinctions and biodiversity loss

The KT event

- Intense scrutiny since 1980
- Reasonable theories (global climate change, change in plants, impact, plate tectonic movements, sealevel change)

Mass extinctions and biodiversity loss

The KT event

- June 1980, paper in Science by Luis Alvarez and colleagues
- 10 km meteorite (asteroid) had hit the Earth
- The impact threw up a great cloud of dust that encircled the globe

Mass extinctions and biodiversity loss

The KT event

- An unusual clay band right at the KT boundary, within a succession of marine limestones
- Iridium spike: where the iridium content shot up from normal background levels

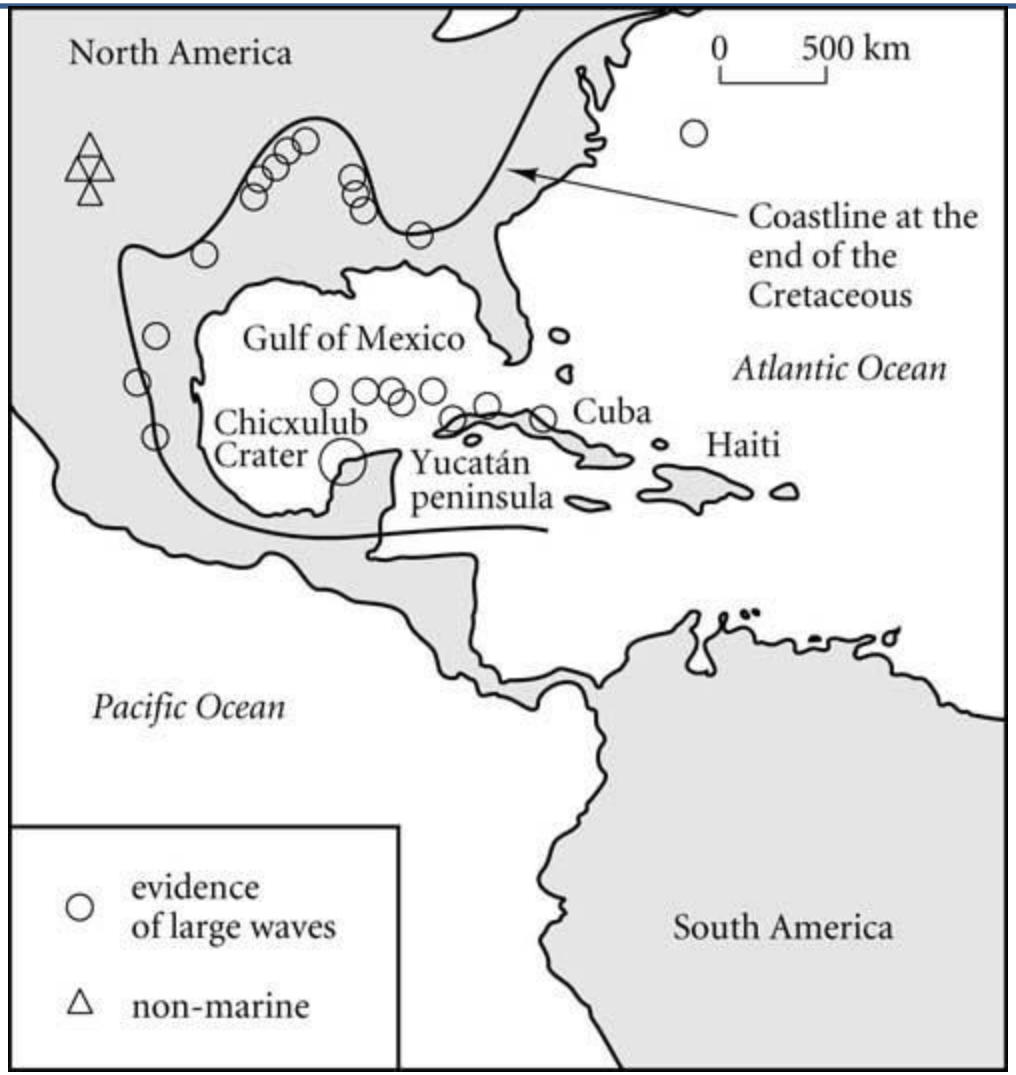
Mass extinctions and biodiversity loss

The KT event

- Alvarez Predictions: 150KM crater, 10 KM Diameter of Meteorite
- 1991, Crater was identified in Chicxulub in Mexico

Mass extinctions and biodiversity loss

The KT impact site identified. Location of the Chicxulub Crater on the Yucatán peninsula



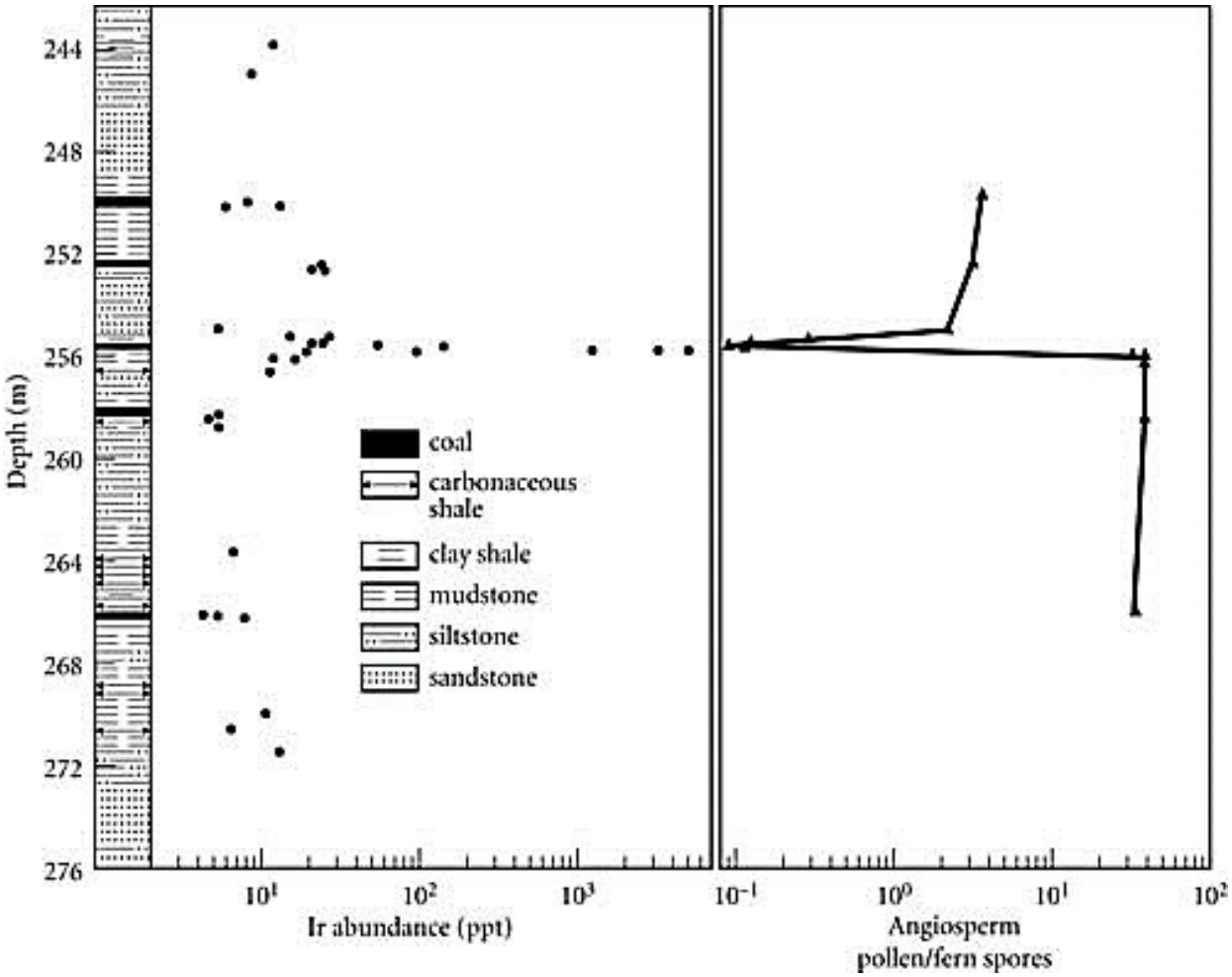
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Mass extinctions and biodiversity loss

The KT event

- Catastrophic Extinction in plants, lower level of pollens, sudden loss of angiosperms and replacement with ferns
- Progressive return to Microflora

Mass extinctions and biodiversity loss



The Iridium (Ir) spike and Fern spike

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Mass extinctions and biodiversity loss

Another Alternative

- The gradualist model
- Extinctions occurred over long intervals of time as a result of climatic changes

Mass extinctions and biodiversity loss

Another Alternative

- On land, subtropical lush habitats with dinosaurs gave way to strongly seasonal, temperate, conifer-dominated habitats with mammals

Mass extinctions and biodiversity loss

Third school of Thought

- Volcanic activity
- The Deccan Traps in India represent a vast outpouring of lava that occurred over the 2–3 myr spanning the KT boundary

END

The Origin of Life

Introduction

The Origin of Life

RNA world

- By fusion of organic molecules in the first billion years after the formation of the Earth
- Self-replicating RNA "RNA world"

The Origin of Life

The Origin of Life

- Photosynthesis by a group of bacteria, called cyanobacteria, generated molecular oxygen (O_2)
- The atmosphere became oxygenated

The Origin of Life

Three great domains

- By gene sequencing of modern organisms,
- Bacteria, Archaea and Eucarya
- The first two are prokaryotes, the last eukaryotes

The Origin of Life

The Earliest Fossils

- Bacteria in rocks up to 3.2 Ga, indicated by stromatolites
- Structures built by alternating algal mats and sediment layers

The Origin of Life

Cellular Organism

- Cellular fossils 3.5 Ga are highly controversial
- The first widely accepted cellular fossils date from 2.5 Ga

The Origin of Life

Origin of Eukarya

- Lipids Biomarkers, evidence for cyanobacteria and eukaryotes 2.7 Ga
- The oldest eukaryotes, 1.9 Ga

The Origin of Life

Origin of Sexes

- Red algae from 1.2 Ga show that sex had originated
- They show mitosis, but also meiosis, which is unique to sexual reproduction

The Origin of Life

Origin of Multicellularity

- Together with sex came multicellularity
- The possession of many, often specialized, cells, first seen in 1.2 Ga red algae

END

The Origin of Life

Scientific Methods

The Origin of Life

Scientific models

- Some of them now rejected by the evidence
- Others still available as potentially valid hypotheses:

1. Spontaneous generation
2. Inorganic model

The Origin of Life

Scientific models

3. Extraterrestrial origins
4. Biochemical model
5. Hydrothermal model

The Origin of Life

Spontaneous generation

- Medieval scholars
- Organisms sprang into life directly from nonliving matter
- Pasteur, 1861
- Meat in airtight containers: maggots did not appear

The Origin of Life

The inorganic model

- Complex organic molecules arose gradually on a pre-existing, non-organic replication platform
- Silicate crystals
- Graham Cairns-Smith of Glasgow University, 2007
- not conclusive

The Origin of Life

Extraterrestrial Model

- The building blocks for life were seeded on Earth from outer space
- Simple organic molecules in meteorites called carbonaceous chondrites
- Panspermia
- 1996, David McKay

The Origin of Life

Biochemical Model

- 1920s, independently
- Russian biochemist, A. I. Oparin
- British evolutionary biologist, J. B. S. Haldane

The Origin of Life

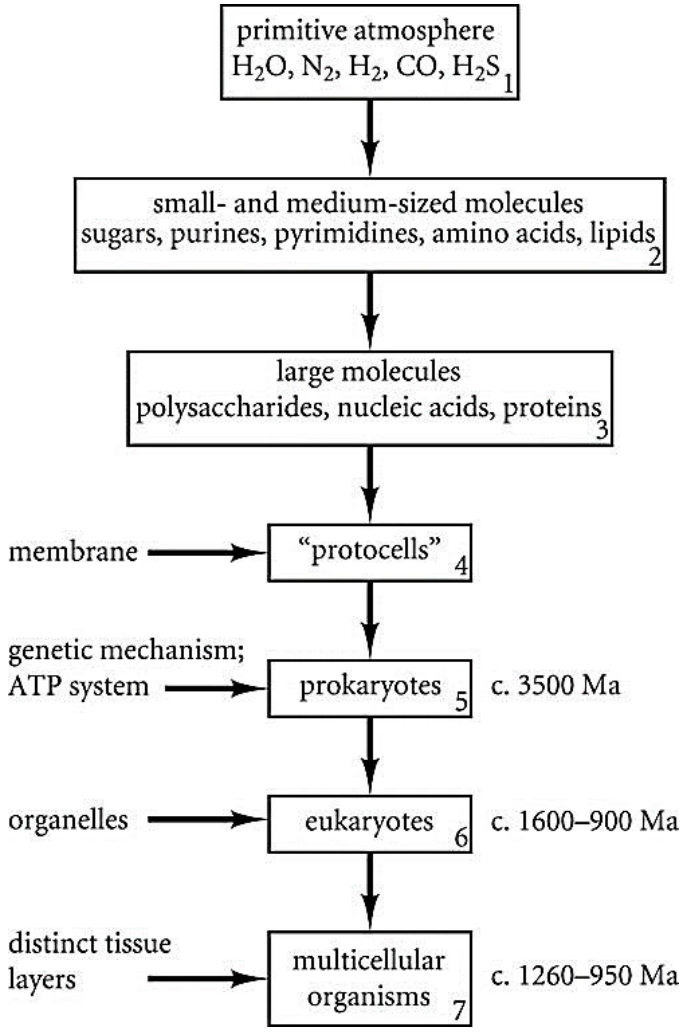
Biochemical Model

- Life could have arisen through a series of organic chemical reactions that produced ever more complex biochemical structures
- Biochemical theory of Origin of Life

The Origin of Life

The biochemical theory for the origin of life, as proposed by Oparin and Haldane in the 1920s.

Biochemists have achieved steps 1–3 in the laboratory, but scientists have so far failed to create life.



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The Origin of Life

Hydrothermal Model

- Modification to the Oparin–Haldane biochemical model
- The last universal common ancestor of life (LUCA) was a hyperthermophile, a simple organism that lived in unusually hot conditions.

The Origin of Life

END

Hydrothermal Model

- Seawater mixes with molten magma and emerges as superheated steam, with the sulfur in it now concentrated as sulfide
- Emerging hot-water plume black
- Black smokers

The Origin of Life

Testing the Biochemical Model

The Origin of Life

Stanley Miller

- In 1953, then a student at the University of Chicago, made a model of the Precambrian atmosphere and ocean in a laboratory glass vessel

The Origin of Life

Stanley Miller

- Exposed a mixture of water, nitrogen, carbon monoxide and nitrogen to electric sparks, to mimic lightning,
- Found a brownish sludge in the bottle after a few days

The Origin of Life

Brownish Sludge

- Contained sugars, amino acids and nucleotides
- Researchers consider the mixture of gases that Miller used was strongly chemically reducing

The Origin of Life

Further Experiments

- in1950s and 1960s led to the production of polypeptides, polysaccharides and other larger organic molecules

The Origin of Life

Protocells of Sidney Fox

- FSU, Succeeded in creating cell-like structures, in which a soup of organic molecules became enclosed in a membrane
- Didn't survive for long

END

The Origin of Life

RNA World

The Origin of Life

Precellular Life

- The transition from non-living to living
- No a single Step
- What then could have been the transitional form of "precellular" life?
- widely accepted view "RNA World"

The Origin of Life

RNA

- RNA, or ribonucleic acid, is one of the nucleic acids and it has key roles in protein synthesis
- The genetic code
- There are several different forms of RNA

The Origin of Life

RNA the Precursor

- In 1968, Francis Crick suggested that RNA was the first genetic molecule
- RNA has unique property of acting both as a gene and an enzyme
- RNA could act as a precursor of life

The Origin of Life

RNA World

- Walter Gilbert first used the term "RNA world" in 1986
- The first evidence
- Sidney Altman and Thomas Cech

The Origin of Life

Ribozyme

- An RNA that could edit out unnecessary parts of the message it carried before delivering it to the ribosome.
- Cech called his discovery a ribozyme

The Origin of Life

Ribozyme

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The Origin of Life

Chasing Evidence

- Since 1990,
- Jack Szostak and colleagues argued that the first RNA molecules on the prebiotic ("before life") Earth were assembled from nucleotides dissolved in rock pools

The Origin of Life

Chasing Evidence

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The Origin of Life

RNA Replicase

- One RNA acted as a Template and another Enzyme
- Combine as an RNA replicase
- Szostak and colleagues proposed a second precellular structure, "A self-replicating vesicle"

The Origin of Life

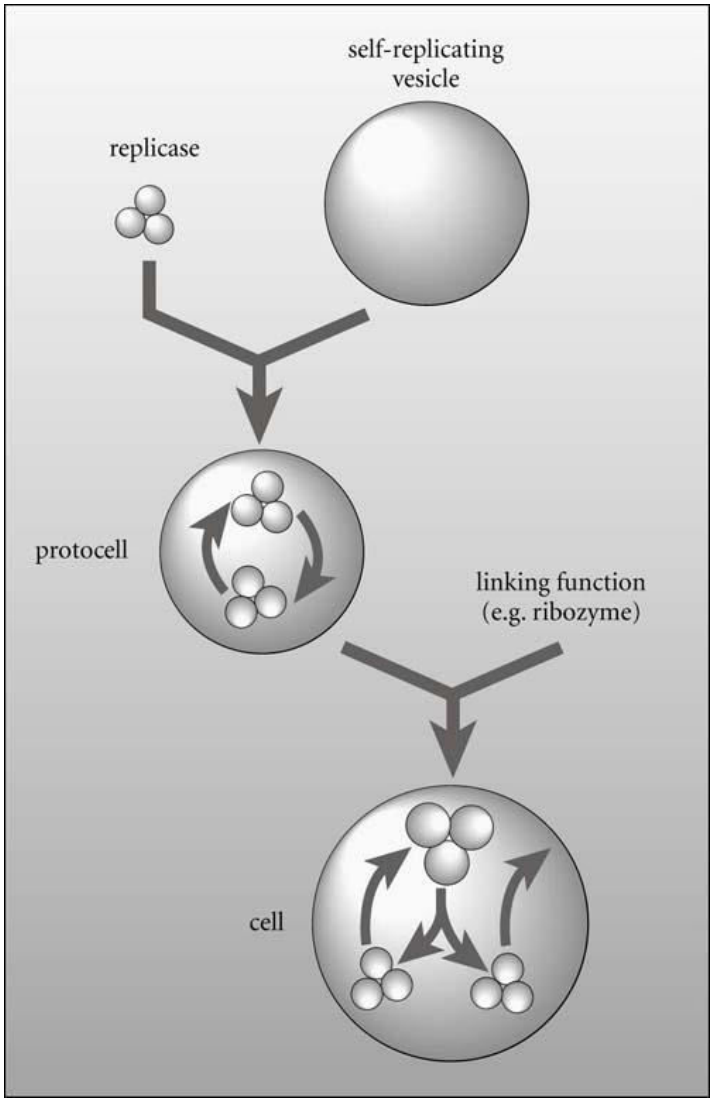
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Protocell

- A membrane-bound structure composed mainly of lipids that self-replicates
- The RNA replicase entered a self-replicating vesicle
- RNA replicase to function efficiently

The Origin of Life

The model behind "RNA world", where an RNA replicase and a self-replicating membrane-bound vesicle combine to form a protocell



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The Origin of Life

Evidence for the Origin of Life

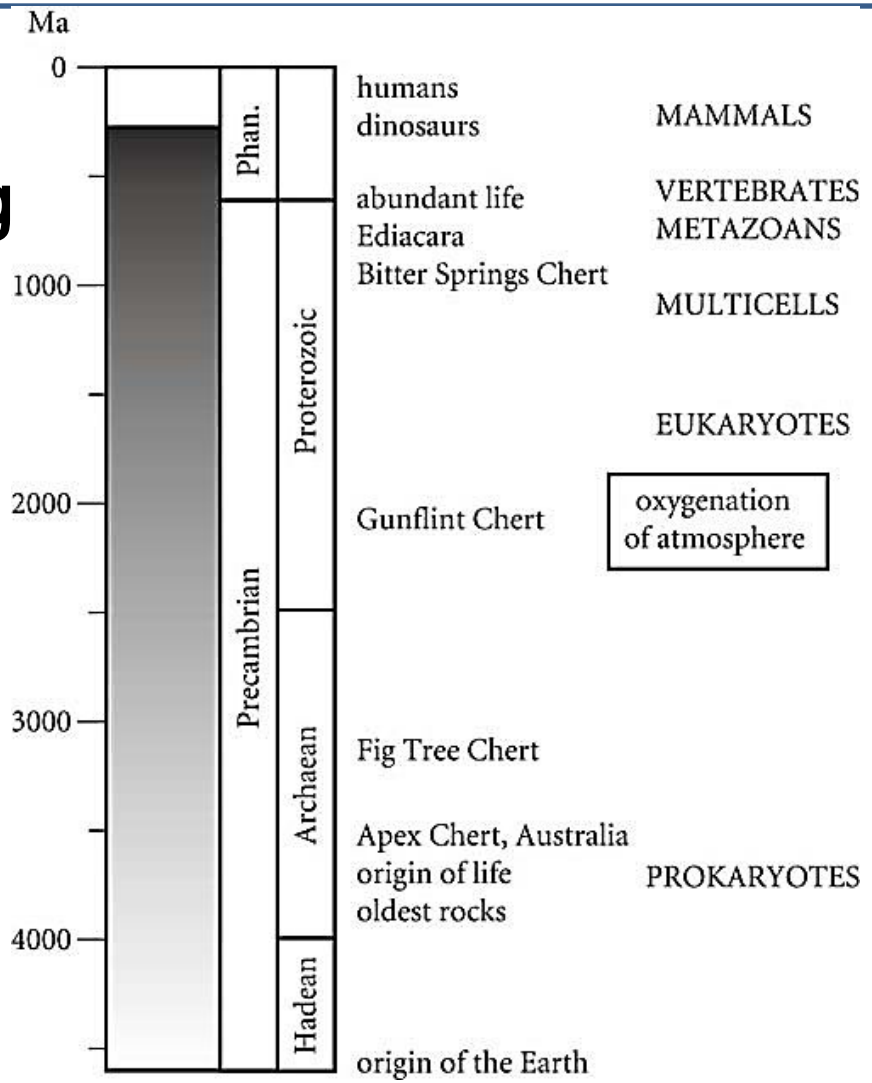
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The Early Precambrian world

- The Precambrian is divided into
- Hadean eon
- Archaean eon
- Proterozoic eon
- Hadean Eon spans from the origin of the Earth, 4.57 to about 4 Ga

The Origin of Life

Time scale showing major events in the history of the Earth and of life. Most of the time scale is occupied by the Precambrian



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The Origin of Life

Precambrian world

- Beginning of the Hadean
- Temperatures on surface were too high
- The crust was too unstable for any form of carbon-based life to exist.

The Origin of Life

Cooling of Earth

- As the Earth's surface cooled, the lithosphere, its rocky crust, began to differentiate as a cooler upper layer above the underlying asthenosphere.

The Origin of Life

Oldest Rocks

- The oldest rocks are from Canada and are dated at 3.8–4 Ga, and some mineral grains from Australia have even been dated to 4.4 Ga

The Origin of Life

The Archaean Eon

- Lasted from about 4 to 2.5 Ga.
- The oldest sedimentary rocks from the Isua Group in Greenland, dated at 3.8–3.7 Ga.

The Origin of Life

The Archaean Eon

- Rosing and Frei (2004)
- Reported values of δ13C in organic matter from the Isua Group rocks that match those of modern living organic matter.

The Origin of Life

The Great Oxygenation Event

- The Proterozoic Eon, from 2.5 Ga to 542 Ma
- 2.4 Ga, atm oxygen levels rose to one-hundredth or one-tenth of modern levels

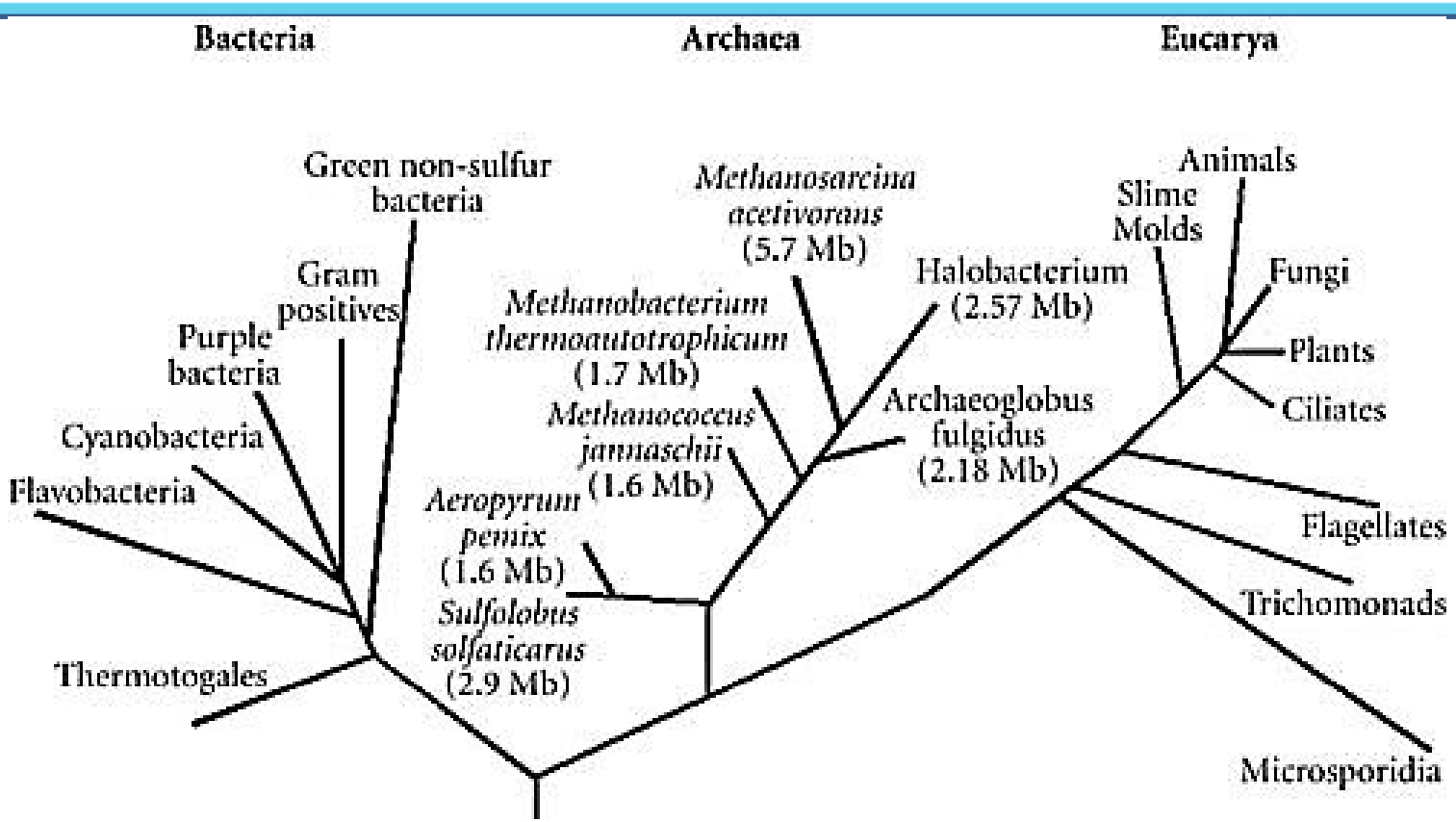
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A Second Rise of Oxygen

- around 0.8–0.6 Ga is indicated by increased levels of marine sulfate
- The two rises in oxygen levels, at the beginning and end of the Proterozoic

END

The Origin of Life



The Universal Tree of Life

based on molecular phylogenetic work.

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The Origin of Life

Life Diversifies: Eukaryotes

The Origin of Life

Eukaryote Characters

- Evidence about the earliest evolution of the three domains is scant
- Nucleus: DNA in chromosomes
- Cell organelles

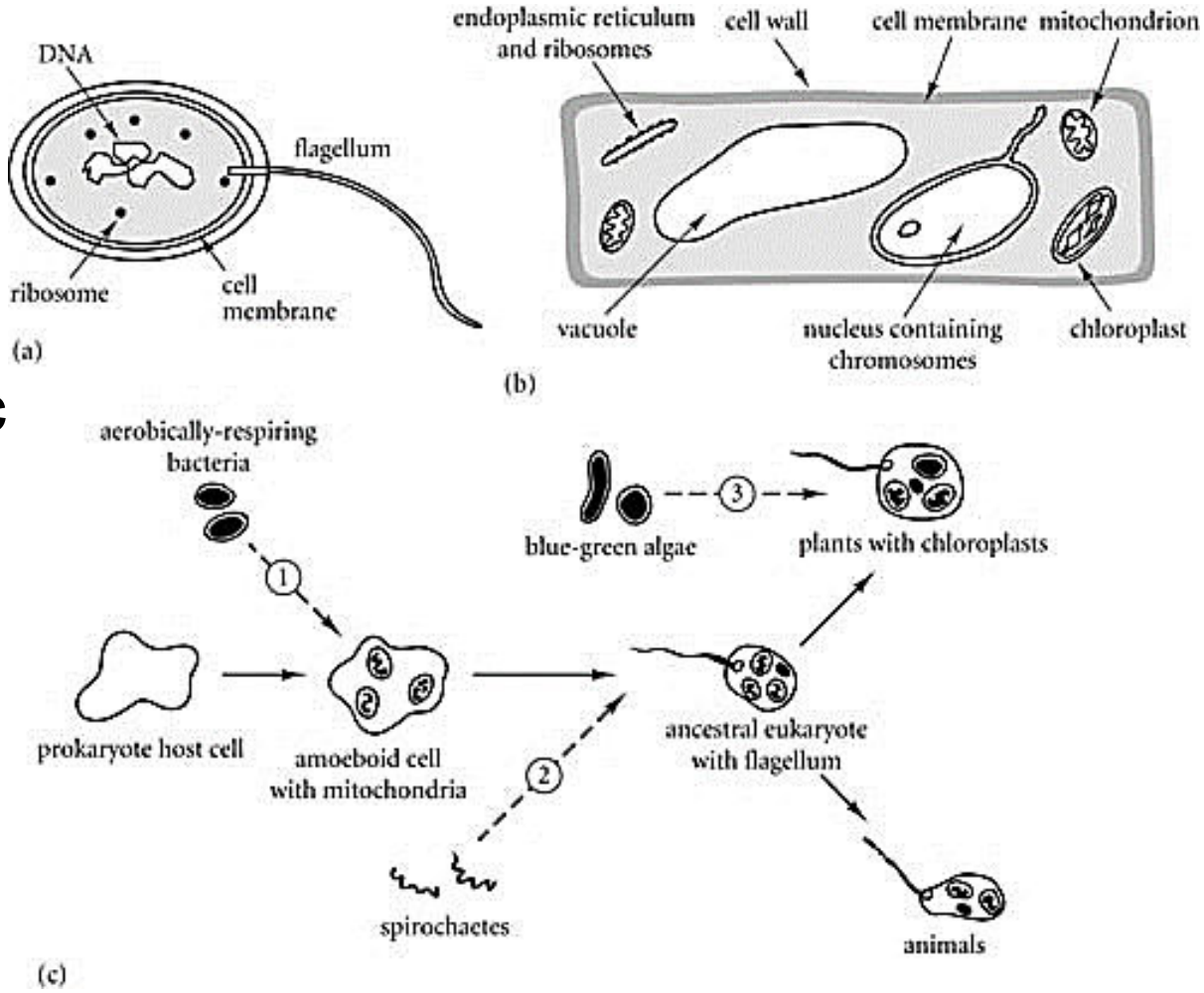
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Endosymbiotic Theory

- Margulis: 1970s.
- Prokaryote consumed, or was invaded by, prokaryotes
- Two species evolved together in a mutually beneficial way.

The Origin of Life

The Endosymbiotic Theory for the Origin of Eukaryotes



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Basal Eukaryotes

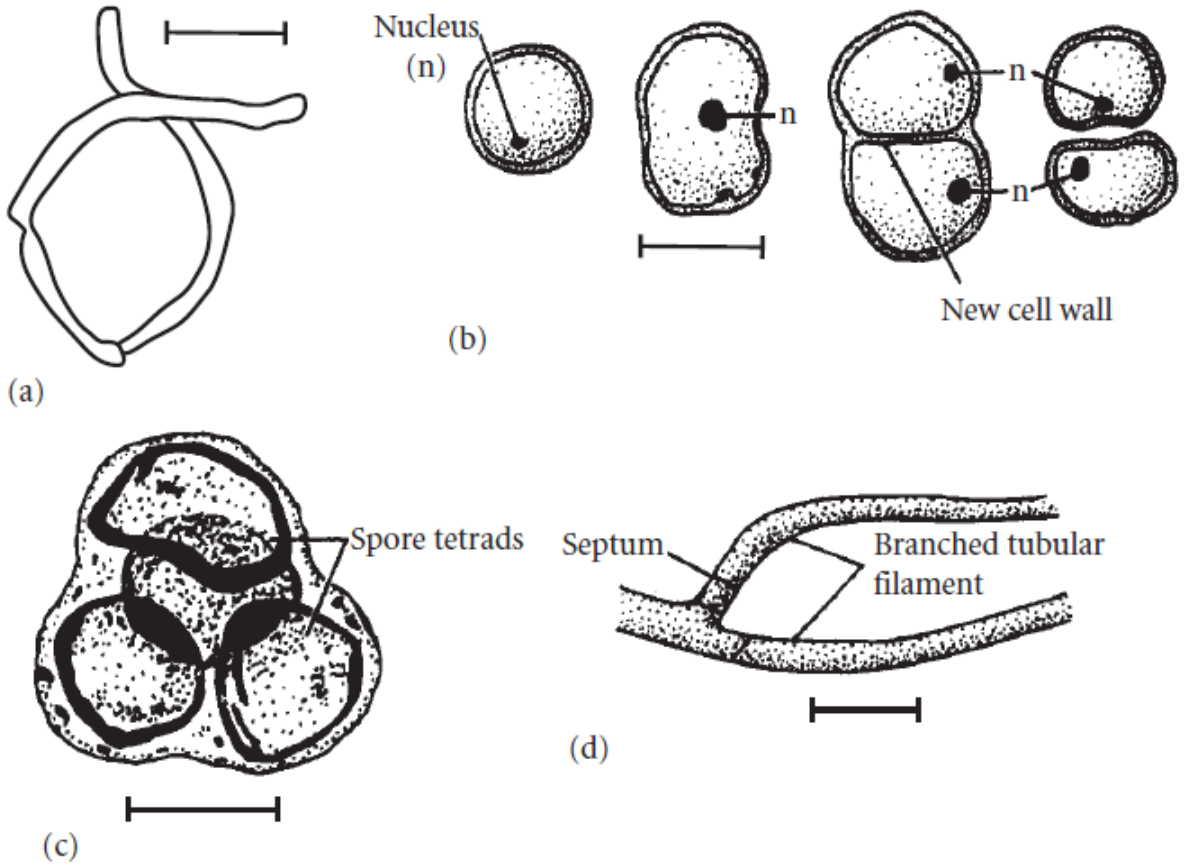
- The oldest eukaryote is controversial
  Lipid biomarkers indicate that eukaryotes were around at least by 2.7 Ga

The Origin of Life

Basal Eukaryotes

- The oldest eukaryote fossil may be Grypania,
- A coiled, spaghetti-like organism that has been reported from rocks as old as 1.85 Ga

The Origin of Life



Early fossil "Eukaryotes".

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The Origin of Life

Multicellularity and sex

- True multicellular organisms arose only among the eukaryotes.
- Allowed plants and animals to become large (kelp, reach lengths of tens of meters).

The Origin of Life

Multicellularity and sex

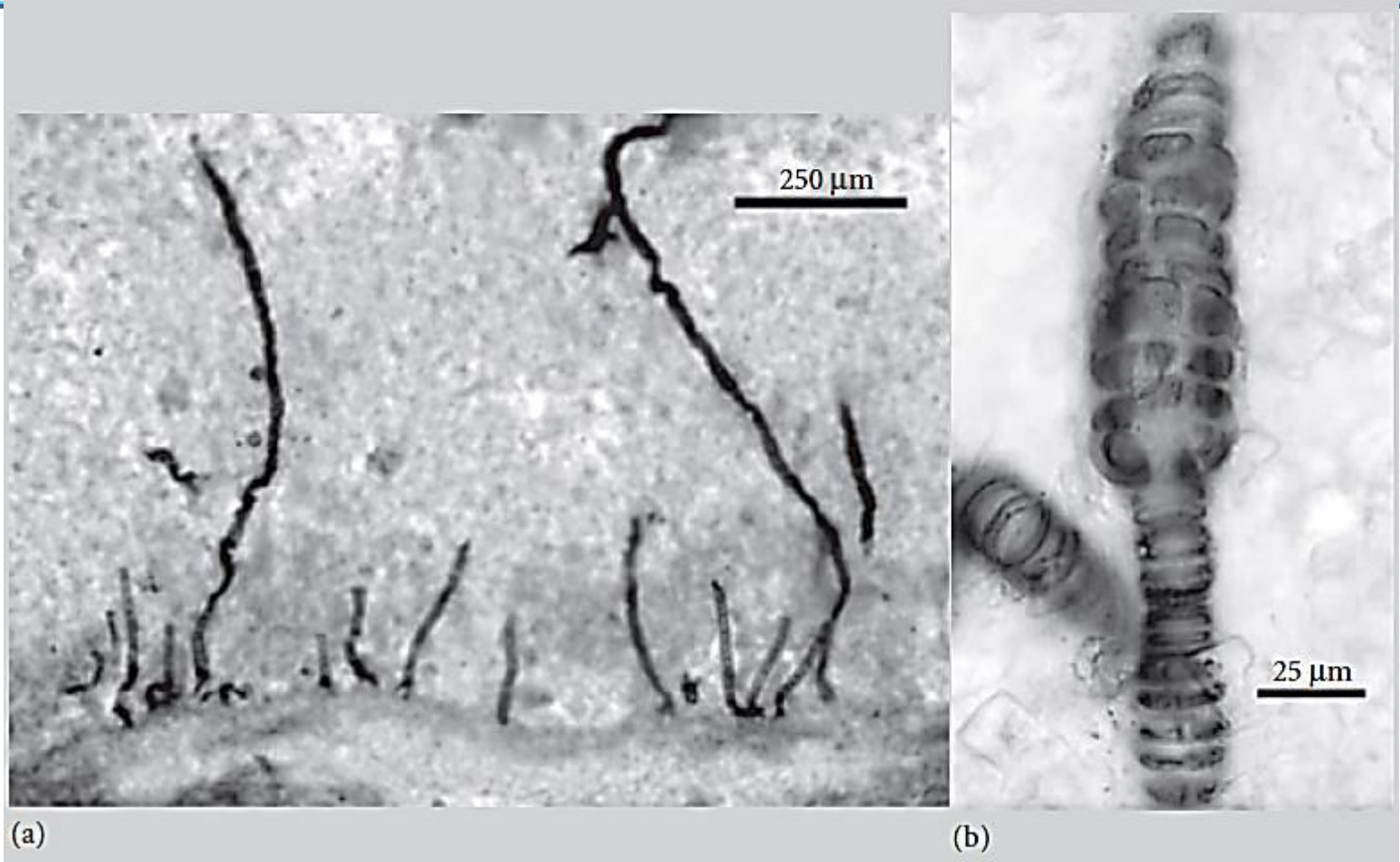
- Sexual reproduction involves the exchange of gametes (sperms and eggs) between organisms

The Origin of Life

Bangiomorpha

- One of the oldest multicellular organisms is Bangiomorpha
- 1.2 Ga
- Multicellular and a member of a modern group that engages in sex.

The Origin of Life



The oldest multicellular eukaryote, Bangiomorpha,

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Protists

Introduction

Protists

Micropaleontology

- A multidisciplinary science, focused on the study of microorganisms or the microscopic parts of larger organisms.

Protists

Prokaryotes

- Unicellular microbes lacking nuclei and organelles, include the carbonate producing cyanobacteria
- Promoted an oxygen-rich atmosphere

Protists

Protists

- Unicellular organisms with nuclei, a variety of organisms with external protective coverings (tests and cysts)
- Kingdoms Protozoa and Chromista

Protists

Fossilized Protists

- Split into
- Organic (acritarchs, dinoflagellates, chitinozoans),
- Calcareous (coccolithophores, foraminiferans)
- Siliceous (diatoms, radiolarians)

Protists

Foraminifera

- Single-celled animal-like protozoans
- Contain both benthic and planktonic form
- With chitinous, agglutinated tests throughout the Phanerozoic

Protists

Rock Formers

- Radiolarians: animal-like protozoans with siliceous tests,
- Diatoms: plant-like protozoans with silicic skeletons
- Both important rock formers.

END

Protists

Kingdom Protista

Protists

Kingdom Protista

- Single-celled organisms with nuclei and organelles
- Autotrophs
- Heterotrophs
- The Protista is a convenient grouping but it is not well defined

Protists

Kingdom Protista

- Subdividing the diversity of protists is equally problematic
- On the basis of Trophism
- Phylogenetic
- Mode of locomotion
- Flagellates and Amoebans

Protists

The First Protists

- Heterotrophs, but chloroplasts were acquired separately in at least six lineages, producing autotrophs,
- Lost secondarily even more often

Protists

The First Protists

- Heterotrophs, but chloroplasts were acquired separately in at least six lineages, producing autotrophs,
- Lost secondarily even more often

Protists

The First Protists

- Groups with microfossil records are widely scattered across the diversity of protists
- Modern molecular genetic and cytologic research

END

Protists

Protozoa

Protists

Protozoans

- Neither animal nor plant, but single-celled eukaryotes that show animal characteristics such as motility and heterotrophy
- Able to form cysts.

Protists

Foraminifera

- Foraminifera are shelled & heterotrophic
- Phanerozoic sedimentary rocks
- Considerable bio stratigraphic and paleoenvironment -al value.

Protists

Radiolaria

- The radiolarians are marine, unicellular, planktonic protists with delicate skeletons
- Usually composed of a framework of opaline silica

Protists

Radiolaria

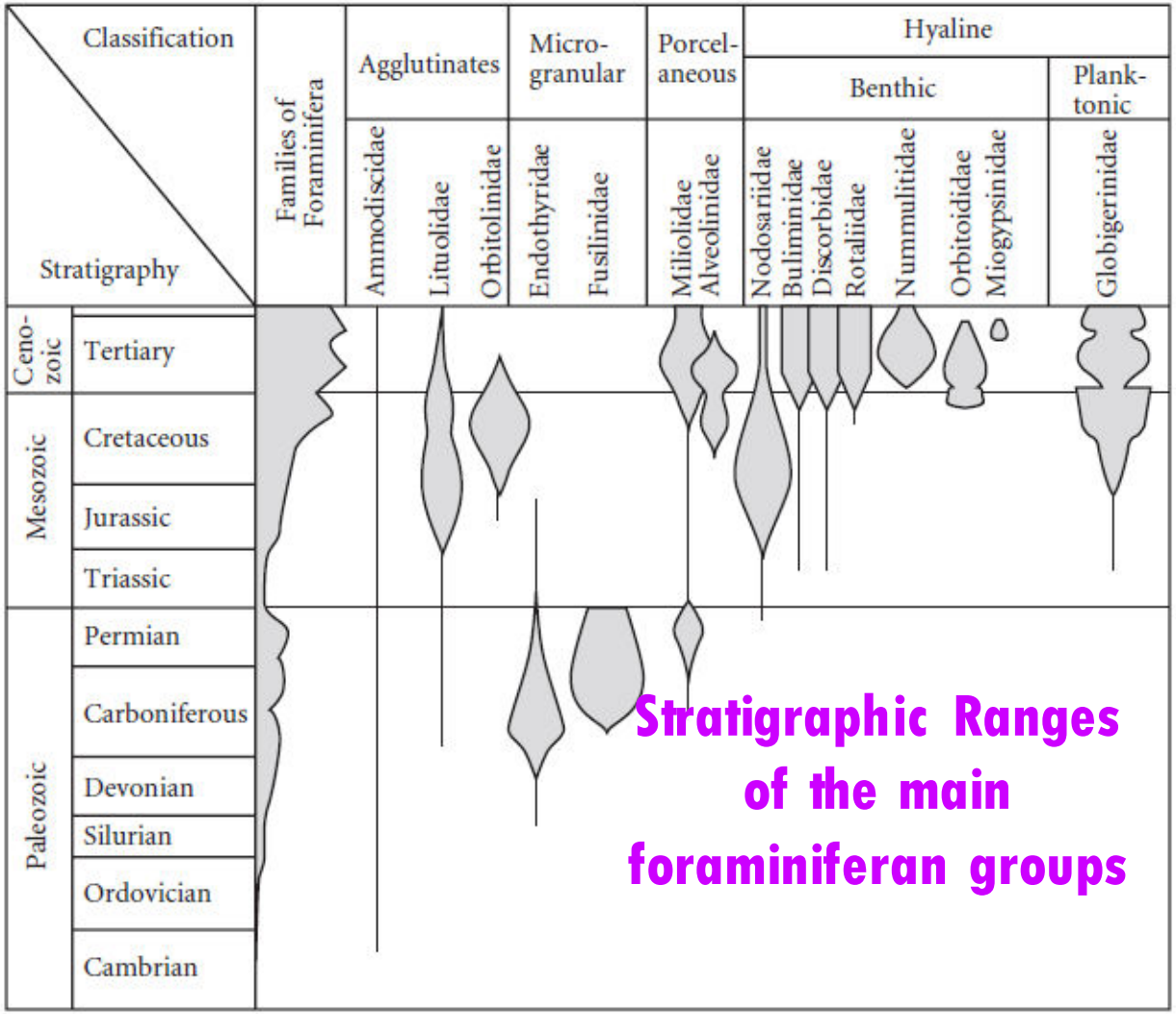
- Origin in the Mid Cambrian or earlier
- Became common in the Ordovician
- Often found in deep-sea cherts associated with major subduction zones

Protists

Radiolaria

- Radiolarian cherts commonly occur in oceanic facies preserved in mountain belts
- associated with ophiolites
- origins and destruction of ancient ocean systems

Protists



http://www.blackwellpublishing.com/paleobiology/figure.asp?chap=09&fig=Fig9-8&img=c09f008

Protists

Acritarchs

- A mixed bag of entirely fossil, hollow, organic-walled microfossils that are impossible to classify
- The acritarchs are probably polyphyletic

Protists

Acritarchs

- A mixed bag of entirely fossil, hollow, organic-walled microfossils that are impossible to classify
- The acritarchs are probably polyphyletic

Protists

Dinoflagellates

- The dinoflagellates, or "whirling whips", comprise a group of microscopic algae with organic-walled cysts.

Ophiolites are pieces of oceanic plate that have been thrusted (obducted) onto the edge of continental plates

Protists

Dinoflagellates

- Dinoflagellate biomarkers have been identified in Upper Proterozoic and Cambrian rocks.
- A number of Paleozoic acritarch taxa may in fact be dinoflagellates

Protists

Ciliophora

- Consist of some 8000 species of single-celled organisms that swim by beating cilia
- Tithonian (Upper Jurassic) to the Albian (Middle Cretaceous)

END

Protists

Chromista

Protists

Chromistans

- A paraphyletic group of eukaryotes that usually contains chloroplasts with chlorophyll c
- Chlorophyll c is absent from all known plant groups.

Protists

Chromistans

- The group includes various algae, the coccolithophores and the diatoms
- Primary producers, functioning as part of the phytoplankton

Protists

Nannoplankton

- Defined as plankton less than 63 µm across, the smallest standard mesh size for sieves
- Includes organic-walled and siliceous forms

Protists

Coccolithophores

- Dominant members of the fossil calcareous nannoplankton, and calcareous plates
- Coccoliths
- Dominate nannofossil assemblages.

Protists

Coccolithophores

- Calcareous nannoplankton
- Appeared: Late Triassic
- Increased: Jurassic and Cretaceous
- Reaching an acme of diversity: Late Cretaceous

Protists

END

Coccolithophores

- Severely affected: KT mass extinction
- Radiated: Early Paleogene and throughout the Cenozoic
- Extremely abundant: Modern oceans

Protists

Chromista:
Diatoms

Protists

Diatoms

- Unicellular autotrophs that are included among the Chrysophyte algae characterized by large green-brown chloroplasts.

Protists

Diatoms

- Both individuals and loosely integrated colonies
- Saline to freshwater
- Range of temperatures
- Common: Antarctic plankton

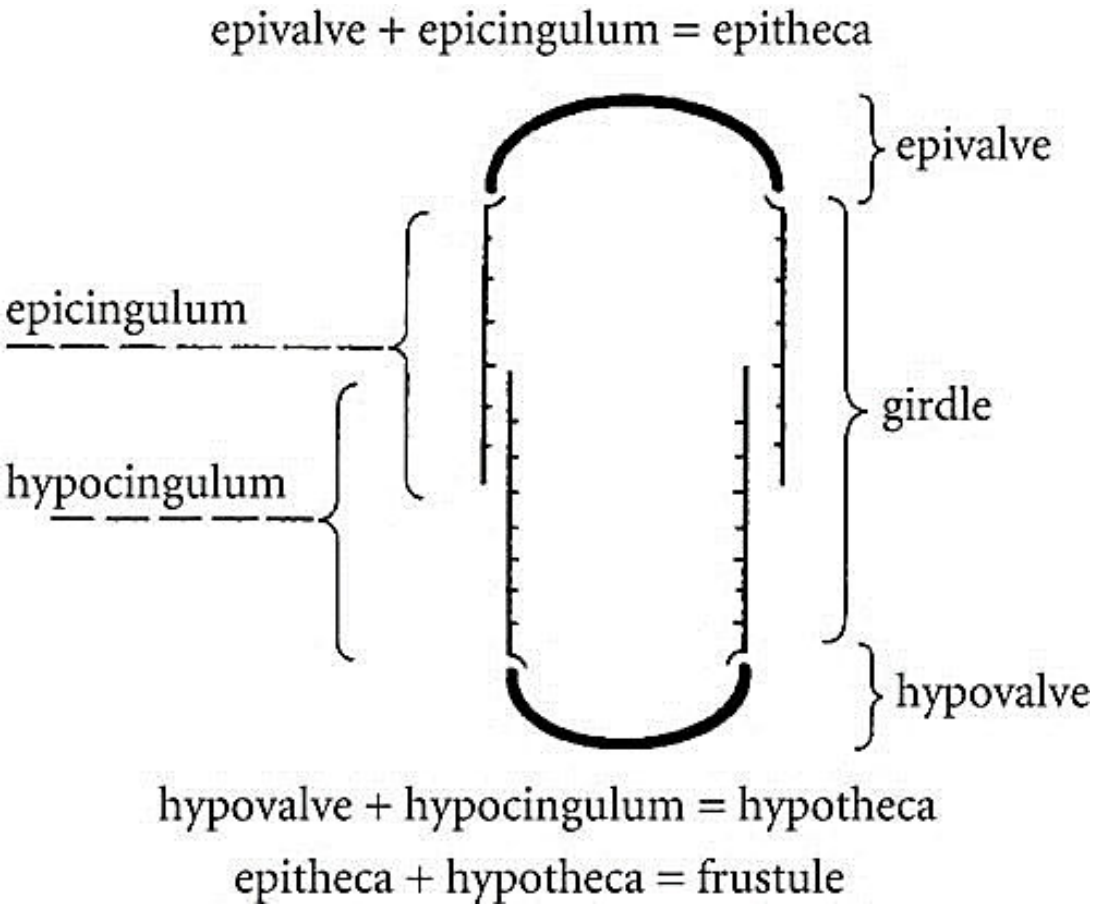
Protists

Diatoms

- The Centrales – prefer marine environments;
- The Pennales – more common in freshwater lakes
- Diatom frustules, Endospores: preserved in fossil record

Protists

Descriptive morphology of the diatoms



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Protists

Diatoms

- First diverse floras: mid-Cretaceous,10 families from Aptian rocks
- 100 genera of centric diatoms: the Upper Cretaceous.

Protists

END

Diatoms

- The first pennate diatoms appeared during Paleogene colonizing freshwater for the first time
- The group reached an acme during the Miocene.

Origin of the metazoans

Introduction

Origin of the metazoans

Fossil evidence

- Few basic body plans have appeared in the fossil record
- most animals have triploblastic architecture
- Three fundamental body layers.

Origin of the metazoans

Molecular Data

- Show three main groupings of animals:
- Deuterostomes
- Spiralians
- Ecdysozoans
- Spiralians and Ecdysozoans: the protostomes.

Origin of the metazoans

Three Groups

- The deuterostomes (echinoderm–hemichordate–chordate group)
- The Spiralians (mollusk–annelid–brachiopod–bryozoans group)

Origin of the metazoans

Three Groups

- The ecdysozoans (arthropod–nematode–priapulid plus other taxa group)

Origin of the metazoans

Five lines of evidence

- Body fossils
- Trace fossils
- fossil embryos
- The mol. Clock
- Biomarkers
- Suggest metazoans originated before Ediacaran 600 Ma

Origin of the metazoans

Coincidence or Design

- Snowball Earth by coincidence or design was a pivotal event in metazoan history; bilaterians evolved after the Marinoan glaciation.

Origin of the metazoans

The First Metazoans

- Probably similar to the demosponges, occurring first before the Ediacaran

Origin of the metazoans

The Ediacaran Biota

- A soft-bodied assemblage of uncertain affinities
- Reaching its acme: the Late Proterozoic
- Earliest ecosystem having large, multicellular organisms

Origin of the metazoans

Cambrian Explosion

- The Cambrian explosion generated a range of new body plans during a relatively short time interval.

Origin of the metazoans

Ordovician Radiation

- Marked by accelerations in diversification at the family, genus and species levels along with complexity in marine communities.

END

Origin of the metazoans

Origins

Origin of the metazoans

Precambrian life

- Evidence of metazoan body and trace fossils
- The investigation of minute fossil embryos
- Carefully calibrated molecular clocks
- Biomarkers

Origin of the metazoans

Metazoan Life

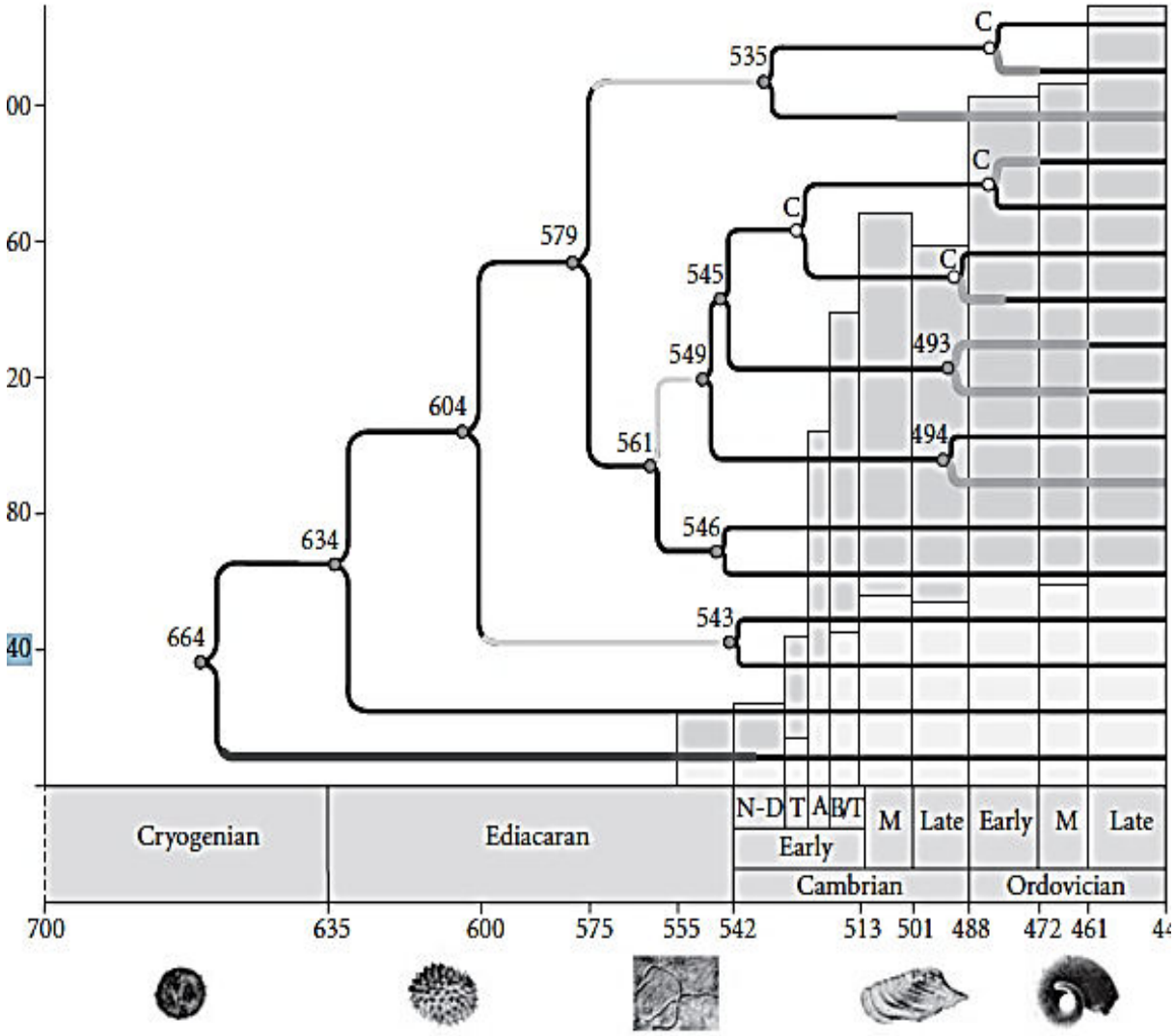
- Molecular data suggest metazoans have been around for at least 600 myr
- During which time as many as 35 separate phyla have evolved

Origin of the metazoans

Metazoan Life

- Recalibrated molecular clocks
- Animal origins be tracked back only to the Ediacaran, when there was also a sudden rise in oxygen levels in the deep ocean

Origin of the metazoans



Time scale and Tempo of early animal evolution

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Origin of the metazoans

Body Fossil Evidence

- Earliest metazoans occur in Ediacara biota 600–550 Ma
- Significant Proterozoic record for the cnidarians and sponges

Origin of the metazoans

Trace Fossil Evidence

- The oldest locomotory trace fossils are from about 550 Ma northwest Russia
- Fecal strings from rocks some 600 Ma suggesting
- Digestive system

Origin of the metazoans

Embryo Fossil Evidence

- Best studied example: Doushantuo Formation, South China
- 580 Ma
- Earliest body fossil evidence for metazoan life

Origin of the metazoans

Molecular Evidence

- Molecular clock
- the metazoan clade excluding the sponges and cnidarians, has been placed at anywhere between 900 and 570 Ma

Origin of the metazoans

Biomarker Evidence

- Biomarkers, essentially the biochemical fingerprints of life
- BMs associated with metazoan demosponges from rocks older than the Ediacaran

END

Origin of the metazoans

Invertebrate Body & Skeletal Plans

Origin of the metazoans

Invertebrate Body & Skeletal Plans

- Body plans are usually defined by the number and type of enveloping walls of tissue together with the presence or absence of a Celom

Origin of the metazoans

Invertebrate Body & Skeletal Plans

- The first metazoans were multicellular with one main cell type & peripheral collar cells or choanocytes, with a whip or flagellum

Origin of the metazoans

The Parazoan Body Plan

- Seen in sponges
- characterized by cells organized in two layers separated by jelly-like material punctuated by so-called wandering cells or amoebocytes

Origin of the metazoans

The Diploblastic Body Plan

- Cnidarians and the ctenophorans
- an outer ectoderm
- an inner endoderm
- Two layers are separated by acellular mesogloea

Origin of the metazoans

The Triploblastic Body Plan

- Seen in most other animal
- Has three layers of tissues from the outside in: the ectoderm, mesoderm and endoderm.

Origin of the metazoans

Bilateral Symmetry

- Defines the bilaterians.
- The development of the celom or body cavity characterizes Most of the animal groups found as fossils

Origin of the metazoans

Skeleton

- The skeleton is an integral part of the body plan of an animal, providing support, protection and attachment for muscles

Origin of the metazoans

Hydraulic Skeleton

- Many animals such as the soft-bodied mollusks (slugs) possess a hydraulic skeleton in which the movement of fluid provides support

Origin of the metazoans

Rigid Skeletons

- Mineralized material
- may be external (exoskeleton) in the case of most invertebrates
- Internal (endoskeleton) structures, in the case of a few mollusks

Origin of the metazoans

Skeleton Growth

- Growth is accommodated in a number of ways.
- Most invertebrate skeletons grow by the addition of new material, a process termed accretion.

Origin of the metazoans

END

Skeleton Growth

- Arthropods, however, grow by periodic bursts between intervals of ecdysis or molting

Origin of the metazoans

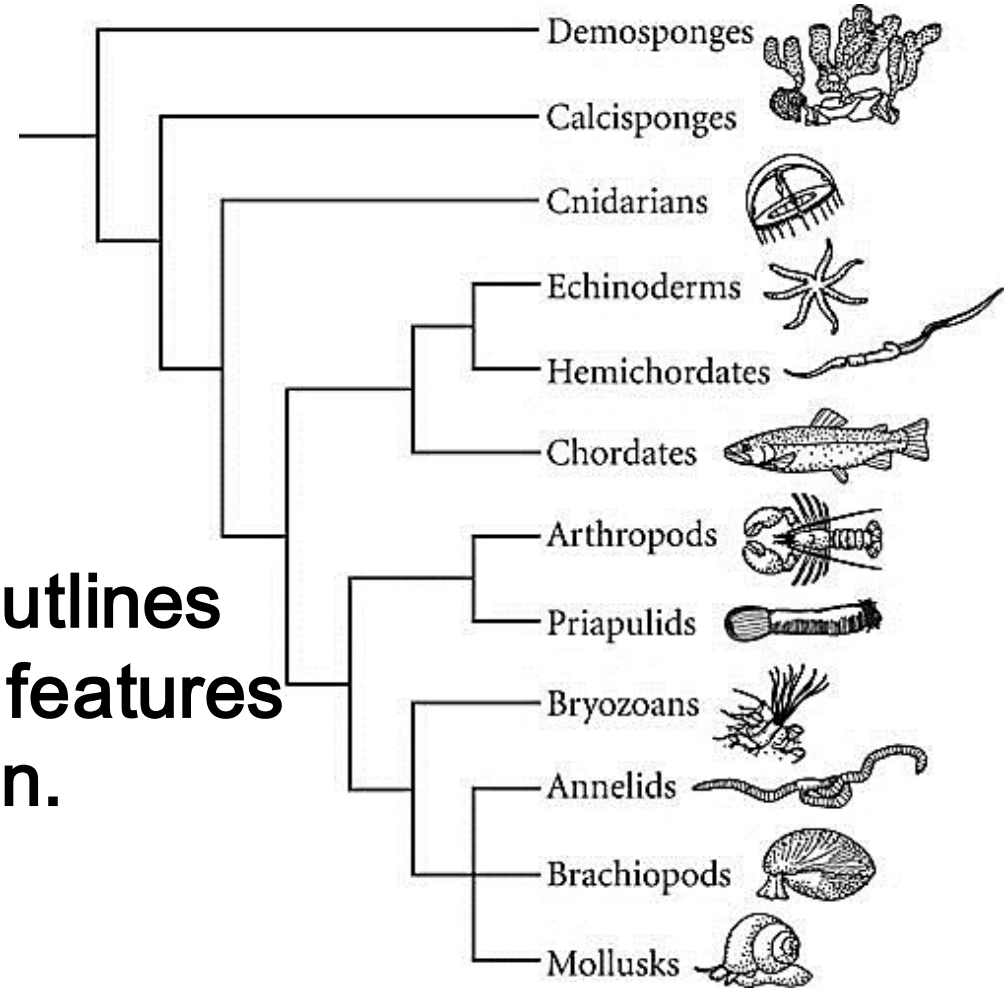
Classification &
Relationships

Origin of the metazoans

Classification

- Based on purely morphological data and embryology has problems
- Difficulties in establishing homologous characters and homoplasy

Origin of the metazoans



The locator tree outlines some of the main features of animal evolution.

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Origin of the metazoans

Basal Eumetazoans

- The demosponges and calcisponges are the simplest animals
- The cnidarians are the most basal eumetazoans

Origin of the metazoans

Three Groups

- Three robust bilaterian groupings are recognized mainly on molecular data
- Ecdysozoans,
- Spiralians
- Deuterostomes

Origin of the metazoans

Protostomes

- Ecdysozoans, spiralians comprise the protostomes.
- "First mouth" mouth develops directly from the blastopore
- By cell growth and migration

Origin of the metazoans

Deuterostomes

- "Second mouth" however, have a mouth arising from a secondary opening
- True blastopore develops as anus
- Echinoderm–Hemichordate-Chordate

Origin of the metazoans

Larval Types

- Nauplius larva: crustaceans
- Planula: cnidarians
- Trochophore larva: mollusks and the polychaetes
- Shelled veliger: mollusks

Origin of the metazoans

END

Larval Types

- Invertebrate larvae are occasionally identified in the fossil record.
- More advanced preparatory and high-tech investigative techniques

Origin of the metazoans

Four Key Faunas

Origin of the metazoans

Evolutionary Faunas

- The three great evolutionary faunas of
- Phanerozoic
- Cambrian
- Paleozoic
- Modern
- Developed during last 550 myr.

Origin of the metazoans

Evolutionary Faunas

- The Ediacara biota
- Faunas during Cambrian explosion & Ordovician radiation
- Set the scene for life on our planet.

Origin of the metazoans

Ediacara Biota

- The Ediacaran biota consisted of enigmatic tubular and frond-shaped,
- mostly sessile organisms that lived during the Ediacaran Period (ca. 635–542 Mya)

Origin of the metazoans

Ediacara Biota

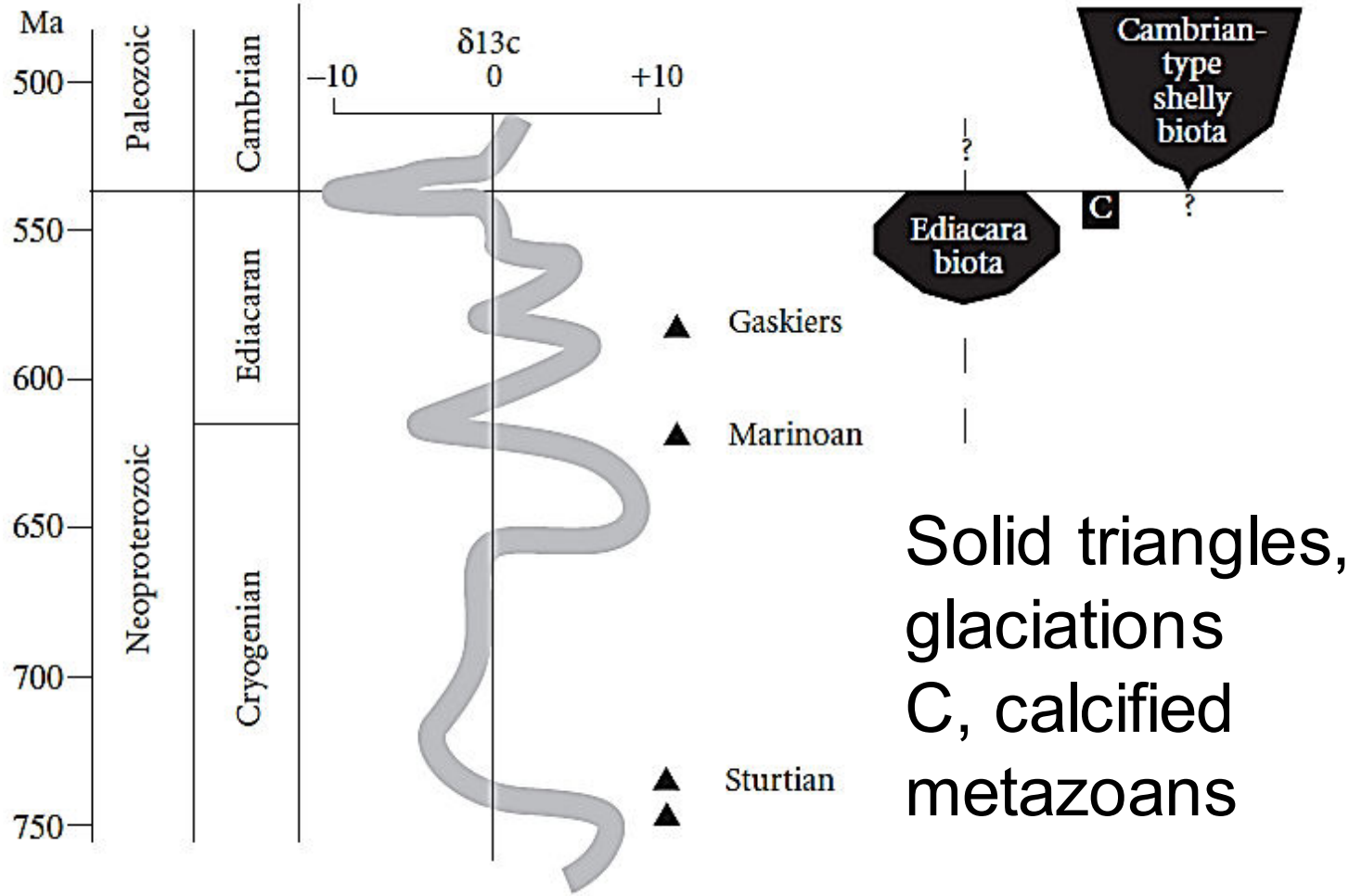
- First impressions of soft-bodied organisms
- Upper Proterozoic rocks of Namibia
- In the Pound Quartzite
- In the Ediacara Hills, north of Adelaide

Origin of the metazoans

Ediacara Biota

- Shallow-water siliciclastic sediments
- Soft-bodied, with high SA/V ratios
- Radial or bilateral symmetries
- Extinct about 550 Ma

Origin of the metazoans



Solid triangles, glaciations
C, calcified metazoans

Stratigraphic Distribution of the Ediacara Biota

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Origin of the metazoans

Cambrian Explosion

- Cambrian radiation
- Relatively short span event 20my
- 541 million years ago
- Most major animal phyla appeared

Origin of the metazoans

Cambrian Explosion

- Generated many entirely new and spectacular body plans
- Sequential appearance of successively more complex metazoans

Origin of the metazoans

Cambrian Explosion

- Phylogenetic analysis shows metazoans evolved monophyletically from flagellated colonial protists similar to modern choanoflagellates.

END

Origin of the metazoans

Ordovician Radiatio

Origin of the metazoans

Ordovician Radiation

- Great Ordovician biodiversification event (GOBE),
- Evolutionary radiation of animal life during the Ordovician Period, 40 my after the CE.

Origin of the metazoans

Ordovician Radiation

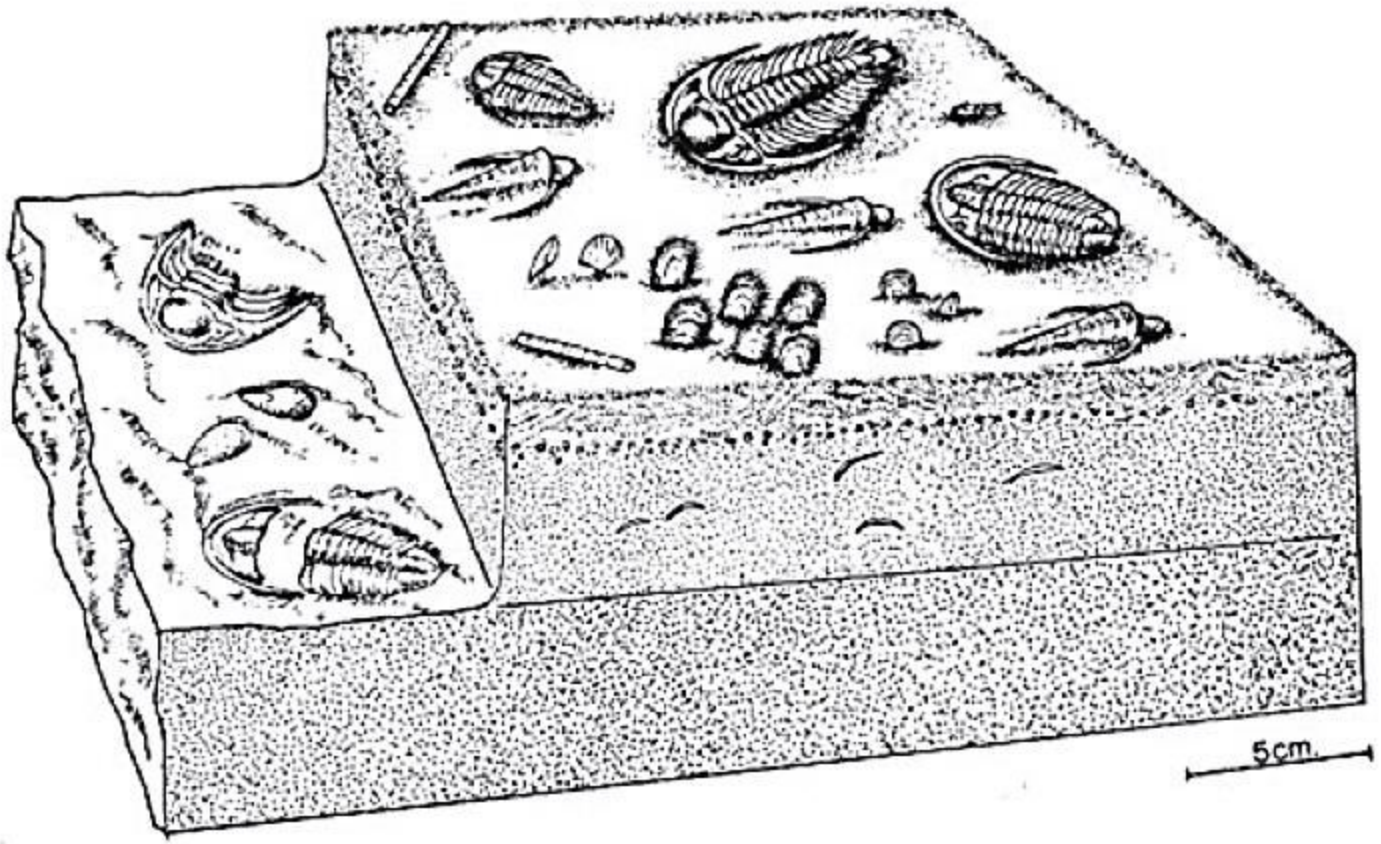
- Followed a series of Cambrian–Ordovician Extinction events
- Fauna went on to dominate the Palaeozoic
- Marine diversity increased

Origin of the metazoans

Ordovician Radiation

- The majority of "Paleozoic" taxa were derived from Cambrian stocks
- Interplay of many geological and ecological factors likely produced the diversification

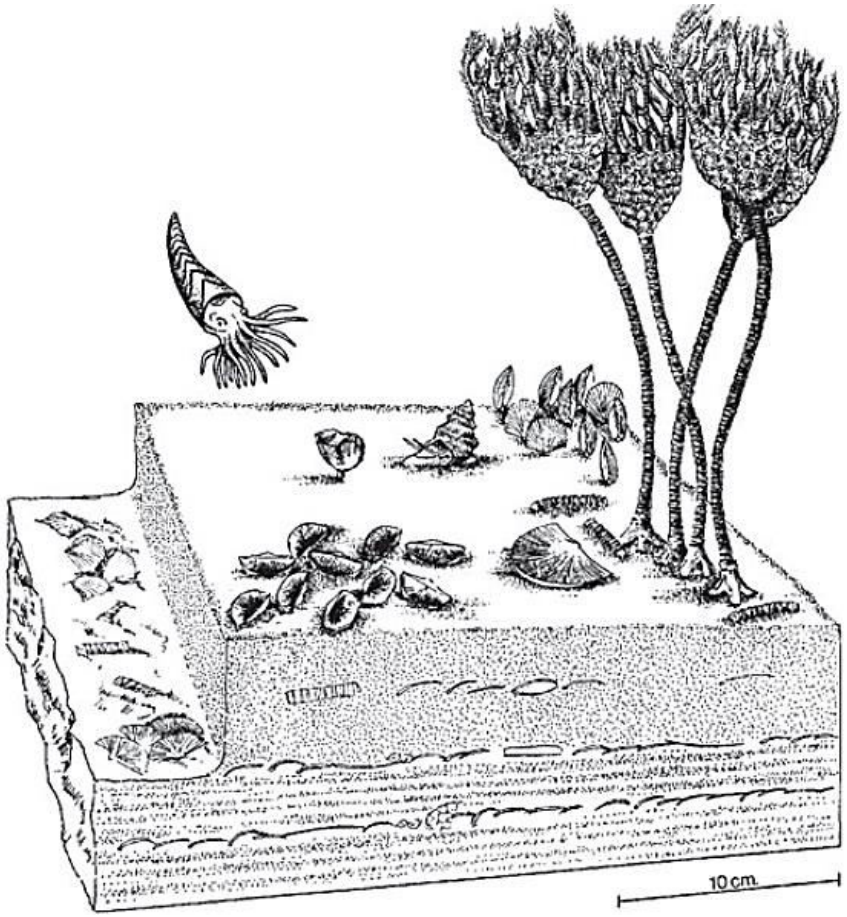
Origin of the metazoans



The Cambrian Seafloor

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Origin of the metazoans



The Ordovician Seafloor

http://www.blackwellpublishing.com/paleobiology/figure.asp?chap=10&fig=Fig10-17&img=c10f017

Origin of the metazoans

Ordovician Radiation

- Ordovician diversification generated witnessed a staggering increase in biodiversity at the family, genus and species levels.

END

Origin of the metazoans

Soft-bodied
Invertebrates

Origin of the metazoans

Inadequate Fossil Record

- Out of 25 or so commonly recognized animal phyla fewer than nine (35%) have an adequate fossil record

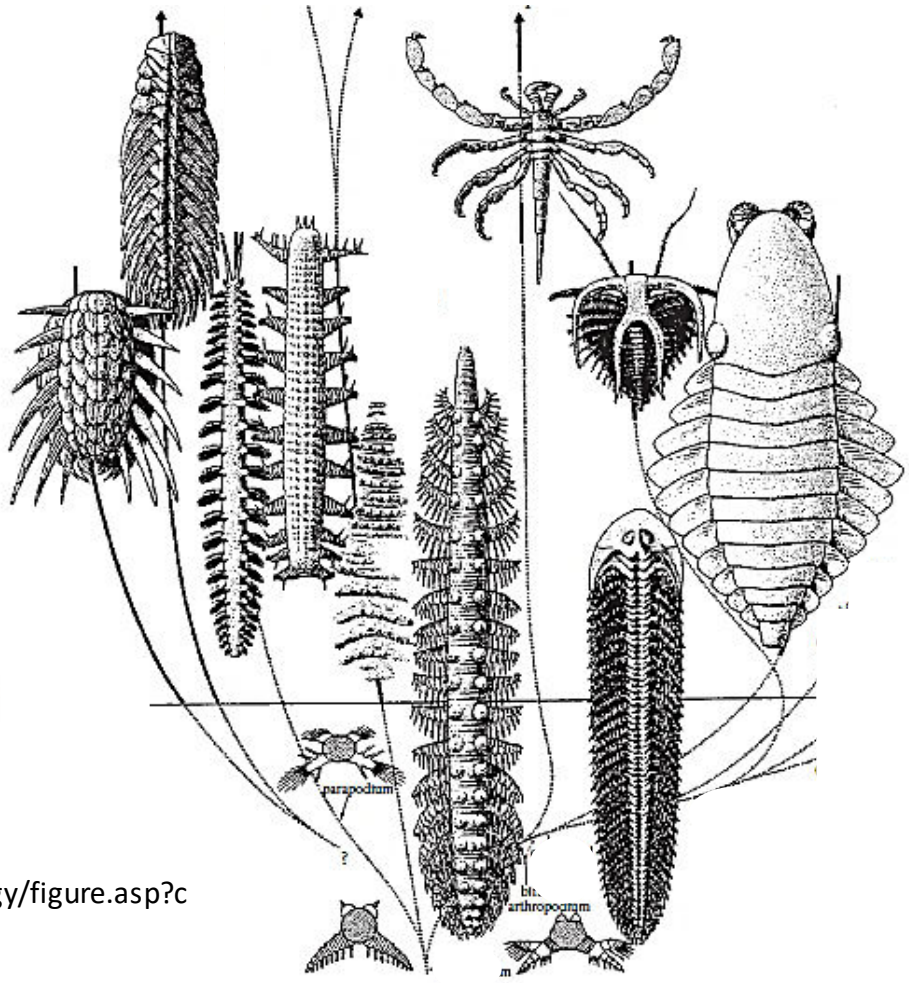
Origin of the metazoans

Inadequate Fossil Record

- A number of larger soft-bodied phyla lack a preservable skeleton
- Soft-bodied forms are preserved in fossil Lagerstätten

Origin of the metazoans

Significance of the diverse worm-like animals at the Precambrian–Cambrian boundary and the postulated origins of some major clades



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Origin of the metazoans

Burgess Shale

- Burgess Shale and other exceptionally preserved faunas suggests that many of these soft-bodied groups dominated certain marine paleocommunities

Origin of the metazoans

Platydendron

- The platyhelminths or the flatworms are bilateral animals
- *Platydendron* from the Middle Cambrian Burgess Shale has been ascribed to the platyhelminthes

Origin of the metazoans

Polychaetes

- A diverse polychaete fauna has been described from the Burgess Shale
- Even contains *Canada spinosa*
- Similar to some living polychaetes

Origin of the metazoans

Polychaetes

- Paddle worms: most complete fossil record
- Record is enhanced by the relatively common preservation of elements of the phosphatic jaw: scolecodonts

END

The Basal Metazoans

Introduction

The Basal Metazoans

Parazoa

- Parazoans are metazoans composed of multicellular complexes with few cell types and lack variation in tissue or organs
- I.e. The sponges (Phylum Porifera)

The Basal Metazoans

Parazoa

- Two groups, the Porifera and the Cnidaria, form the basal parts of the metazoan tree, diverging during Neoproterozoic
- Important parts of the planet's reef ecosystems

The Basal Metazoans

Sponges

- Sponges are filter-feeding members of the sessile benthos
- Contains a variety of grades of functional organization

The Basal Metazoans

Sponges

- Sponge reefs dominated: Phanerozoic
- Calcareous grades
- Siliceous sponges were important reef builders: Mesozoic

The Basal Metazoans

Stromatoporoids

- A grade of organization within the Porifera with a secondary calcareous skeleton
- Important in reefs during the mid-Paleozoic and mid-Mesozoic

The Basal Metazoans

Archaeocyaths

- Cambrian organisms of sponge grade
- Mainly solitary
- Developed branching, modular growth
- Successfully built reefs in unstable environments

The Basal Metazoans

Cnidarians

- The simplest of the higher metazoans with a radial diploblastic body plan and stinging cells or cnidoblasts
- Sea anemones, jellyfish and hydra along with Corals

The Basal Metazoans

Reef-Type Structures

- Reef-type structures were already present in the Late Precambrian hosting large, robust, colonial organisms

The Basal Metazoans

Reef Development

- Reef development through time has waxed and waned
- Dominated at different times by different groups of reef-building organisms

The Basal Metazoans

Coloniality

- Evolved many times
- One hypothesis suggests that a Precambrian colonial organism may have been a source for the bilaterians

END

The Basal Metazoans

Porifera

The Basal Metazoans

Porifera

- Dr Robert Grant "Porifera is a unique group in its own right"
- Unique porous structure and cellular body plan
- Lack true tissues

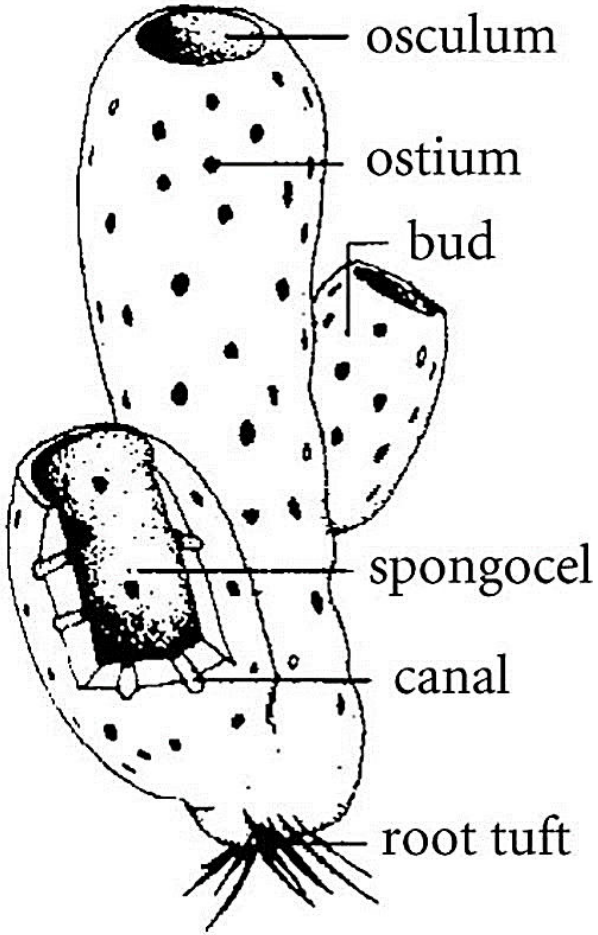
The Basal Metazoans

Porifera

- Most lack symmetry, true differentiated tissues, and organs, although their cells, like those of the protists, can switch function.

The Basal Metazoans

Basic Sponge Morphology



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The Basal Metazoans

Skeleton

- Sponges are skeletal organisms. Skeletons are composed of a colloidal jelly or spongin

The Basal Metazoans

Chamber Organization

- Three levels

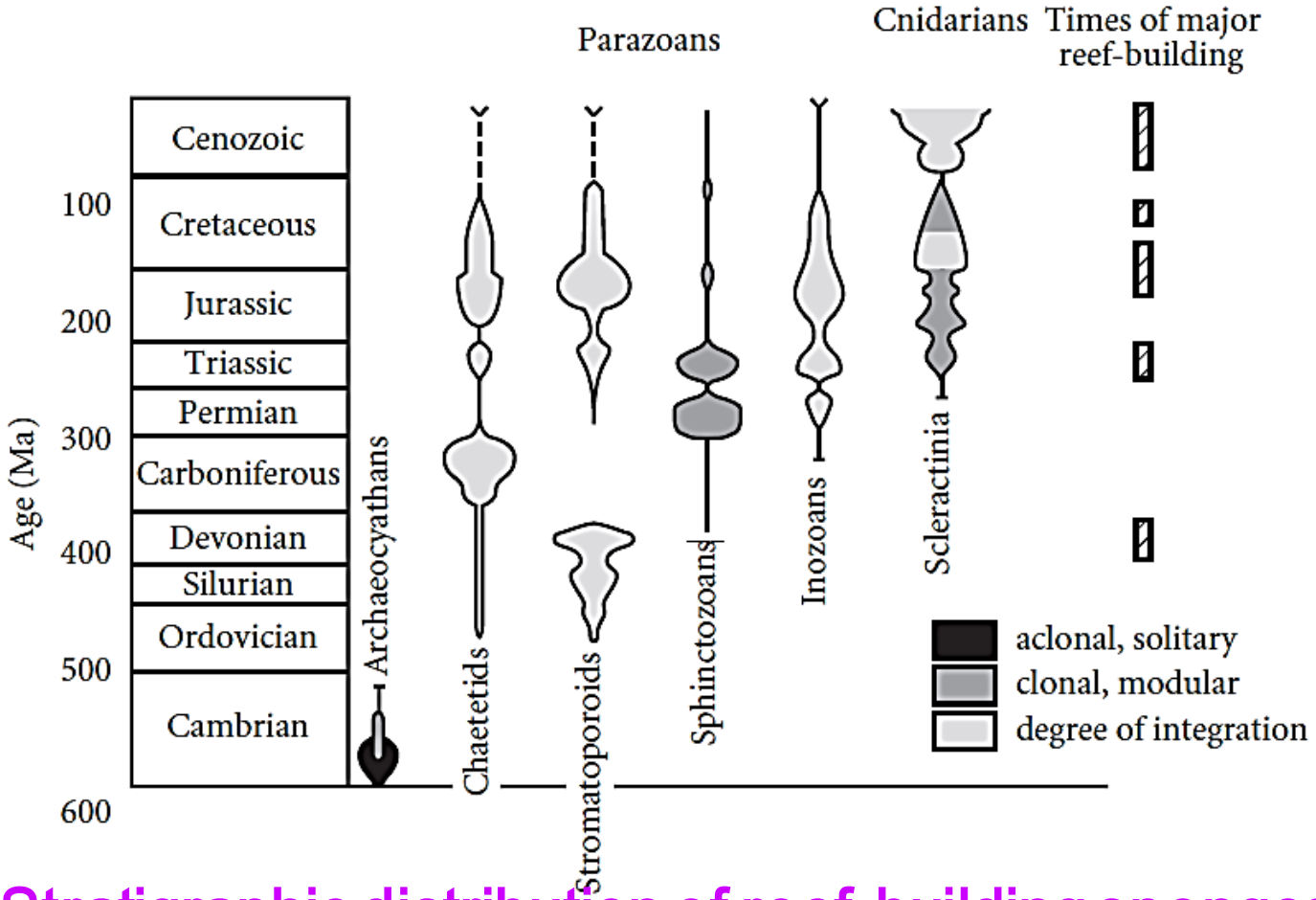
1. Ascon: Sacs with a single chamber lined by flagellate cells
2. Sycon: a number of simple chambers with a single central paragaster.

The Basal Metazoans

Chamber Organization

3. Leucon: most common where a series of sycon chambers access a large central paragaster

The Basal Metazoans



Stratigraphic distribution of reef-building sponges and related parazoans

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The Basal Metazoans

Stromatoporoidea

- Common in the fossil record from the Ordovician through the Devonian.
- Especially abundant in the Silurian and Devonian.

The Basal Metazoans

Stromatoporoidea

- These invertebrates were important reef-formers throughout the Paleozoic and the Late Mesozoic.

The Basal Metazoans

Archaeocyatha

- "Ancient cups" are one of only a few major animal groups that are entirely extinct.
- Reef-building marine organisms of warm tropical and subtropical waters

The Basal Metazoans

Archaeocyatha

- Lived during the early (lower) Cambrian Period.
- It is believed that the center of the Archaeocyatha origin are now located in East Siberia

END

The Basal Metazoans

Cnidaria

The Basal Metazoans

The Cnidarians

- "Nettle-bearers"
- Include the
- Sea anemones, jelly fish, corals
- The least complex of the true metazoans (eumetazoans)
- Few tissue types in a radial plan.

The Basal Metazoans

The Cnidarians

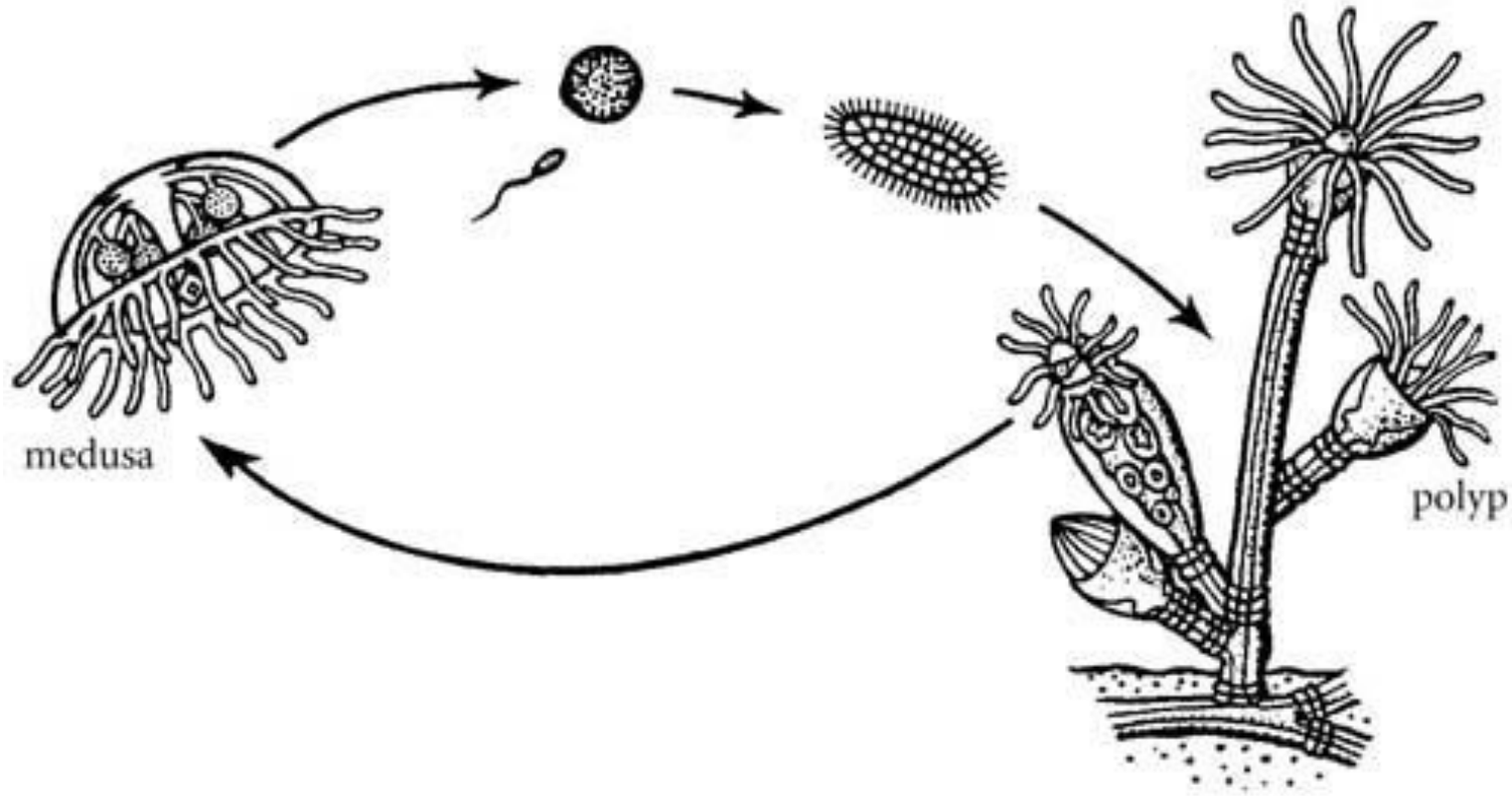
- Although there are no specialized organs and only a few tissue types,
- They are more complex than the parazoans

The Basal Metazoans

The Cnidarians

- Two basic life strategies
- Polyps: sessile
- Medusae: swim,
- Medusoids and polyps appear different, they are essentially same inverted structures

The Basal Metazoans



Cnidarian life cycles

generalized view of the life of the hydrozoan Obelia, alternating between the conspicuous polyp and medusa stages

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The Basal Metazoans

Class Hydrozoa

- Polymorphic forms
- Undivided enteron, solid tentacles, colonies
- Chondrophora oldest cnidarians
- Ediacaran to Recent

The Basal Metazoans

Class Scyphozoa

- Jellyfish
- Lagerstätten.
- The extinct Conulata
- Appeared: Cambrian
- Extinct: Mid Triassic
- Ediacaran to Recent

The Basal Metazoans

Class Anthozoa

- Marine, sessile, colonial forms
- Mobile planula larvae
- Octocorallia class includes corals, sea anemones
- Produce spicules microfossils
- Ediacaran to Recent

The Basal Metazoans

Class Cubozoa

- The sea wasps and box jellyfish
- Both medusae and polyps
- Restricted to tropical and subtropical latitudes
- Carboniferous to Recent

The Basal Metazoans

Corals

- Most diverse and most threatened ecosystems
- The coral reef
- Shallow-water coral reefs in zone extending 30° north and south of equator
- 30m or 18 °C

END

Spiralians

Spiralians:
Lophophorates

Spiralians

The Spiralians

- Morphologically diverse clade of protostome animals
- Molluscs
- Annelids
- Platyhelminths and other taxa

Spiralians

Lophophores

- Lophophores, a filamentous feeding organ
- Three spiralian invertebrate groups
- Brachiopods
- Bryozoans
- Phoronids

Spiralians

Brachiopods

- Twin-valved shellfish
- Lophophore and a pedicle,
- Linguliformeans: organophosphatic shells
- Craniiformeans & rhynchonelliformeans: calcareous shells

Spiralians

Paleozoic Communities

- Orthides and Strophomenides
- Rhynchonellides and Terebratulides

were in Lower-diversity post-Paleozoic brachiopod assemblages

Spiralians

Paleozoic Communities

- Brachiopods dominated the filter-feeding benthos of the Paleozoic
- Never fully recovered during the end-Permian mass extinction

Spiralians

Living Brachiopods

- Living brachiopods are relatively rare, occupying mostly cryptic and deep-water habitats.

Spiralians

Bryozoans

- Colonial invertebrates with Lophophores
- Displaying marked non-genetic variation across a wide range of environments

Spiralians

Stenolaemata

- Dominated Paleozoic bryozoan faunas
- Only the cyclostomes surviving the combined effects of the end-Permian and end-Triassic mass extinctions

Spiralians

END

Cyclostomes

- Cyclostomes continued to decline after the end-Cretaceous extinction event
- Cheilostomes radiated to dominate Cenozoic assemblages

Spiralians

Lophophorates:
Brachiopoda

Spiralians

Brachiopoda

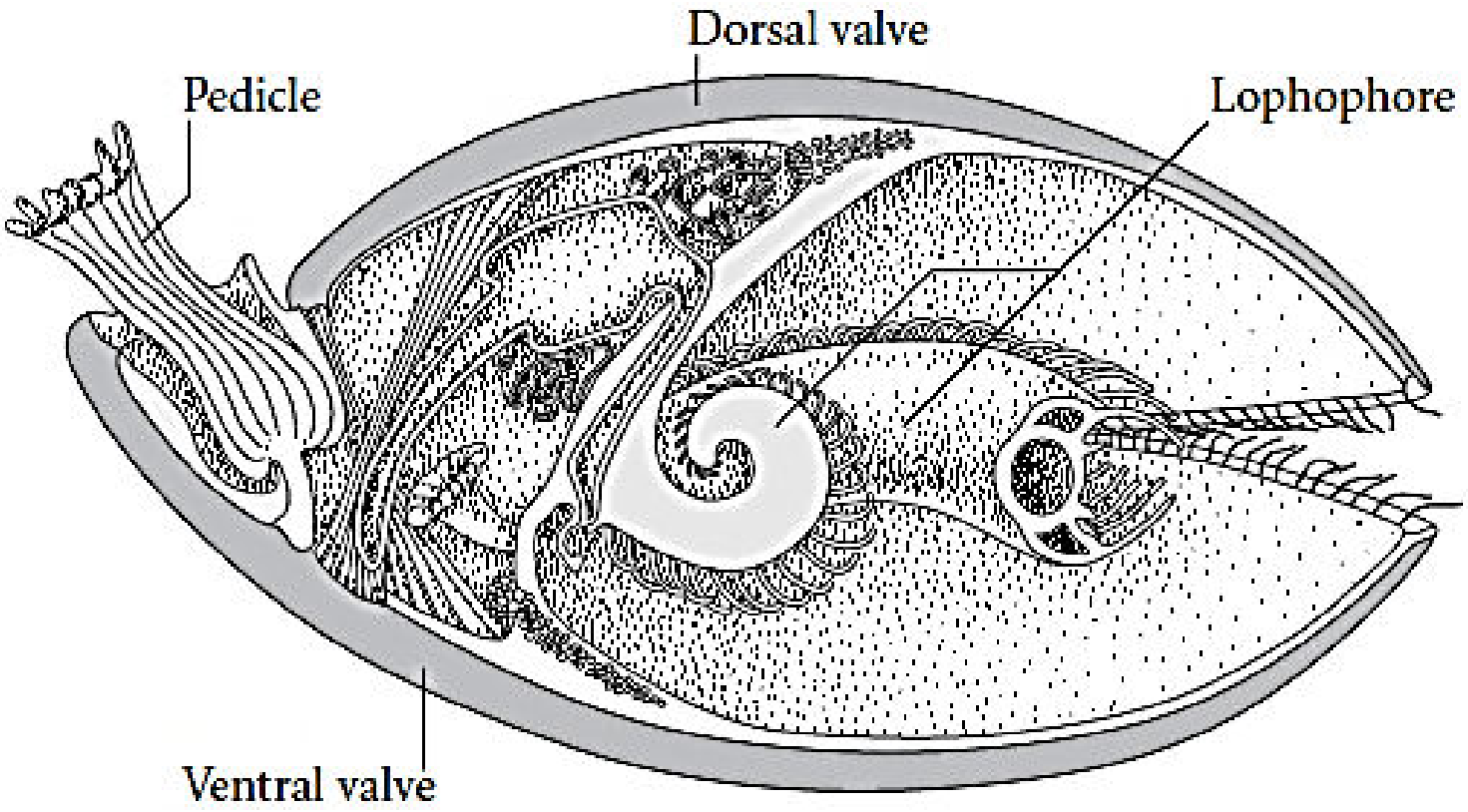
- Most successful invertebrate phyla
- Appeared: Early Cambrian
- Diversified: Paleozoic
- Dominate the low-level, suspension feeding benthos

Spiralians

Morphology

- The brachiopod soft parts are enclosed by two morphologically different shells or valves with variety of muscles
- The ventral or pedicle valve is larger

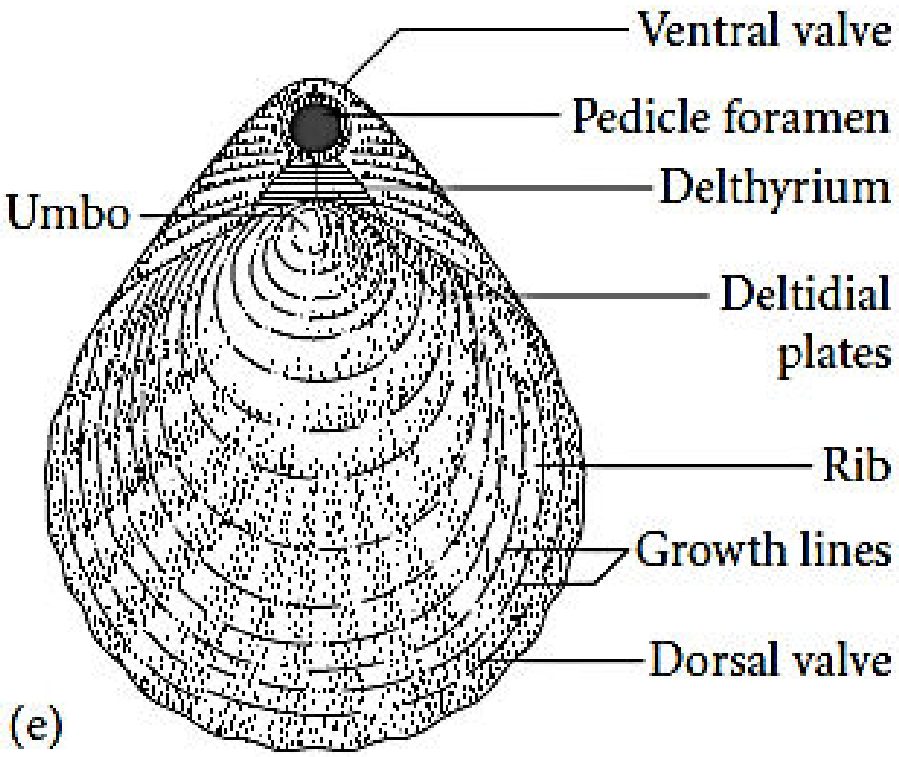
Spiralians



Internal Features of a Terebratulide

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Spiralians



External Terminology of a Typical Articulate

http://www.blackwellpublishing.com/paleobiology/figure.asp?chap=12&fig=Fig12-1&img=c12f001

Spiralians

Distribution in time

- Cambrian
- Paleozoic
- Modern brachiopod faunas
- Fundamentally different
- A dominance of different orders

Spiralians

Cambrian Faunas

- Dominated by a range of non-articulated groups
- with groups of disparate articulated taxa
- Chileides
- Naukatides
- Obolellides etc

Spiralians

Ordovician Radiation

- Deltidiodont, Orthides and Strophomenides
- First evolved: Early Ordovician island complexes
- Dominated the shelf benthos & basis of the Paleozoic brachiopod fauna.

END

Spiralians

Brachiopoda: Five Extinction Events

Spiralians

Five Main Extinctions

- The brachiopods experienced five main extinction events
- Followed by recoveries and radiations of varying magnitudes

Spiralians

The End-Ordovician Event

- Occurred in two phases
- glaciation & loss of almost 80% of brachiopod families
- Decline of deltidiodont groups
- Dominance of the spiriferides pentamerides

Spiralians

Late Devonian events

- Climate change
- Removed the Atrypides & Pentamerides
- Severely affected Orthides & Strophomenides
- Spiriferides & Rhynchonellides survived in deeper-water environment

Spiralians

Carboniferous Era

- Intervals of spectacular experimentation
- Mimicked corals or developed extravagant clusters of spines
- Reduced shells, presenting soft tissues to outside environment.

Spiralians

End-Permian Mass Extinction

- Demise of over 90% of brachiopod species,
- Disaster taxa including lingulids
- later diversified with in a few clades dominated by the rhynchonellides and terebratulides

Spiralians

End-Triassic Event

- Removed the majority of the remaining spiriferides and the last strophomenides
- Dominance of Rhynchonellide and terebratulide groups

Spiralians

Modern Brachiopods

- Exhibit a remarkable range of adaptations based on a simple body plan and a well defined role in the fixed, low-level benthos.

END

Spiralians

Lophophorates

Bryozoa

Spiralians

Bryozoans

- All species are colonial.
- Skeletons fragment easily after death
- least well-known invertebrates
- 6000 living & 16,000 fossil species, marine

Spiralians

Bryozoans

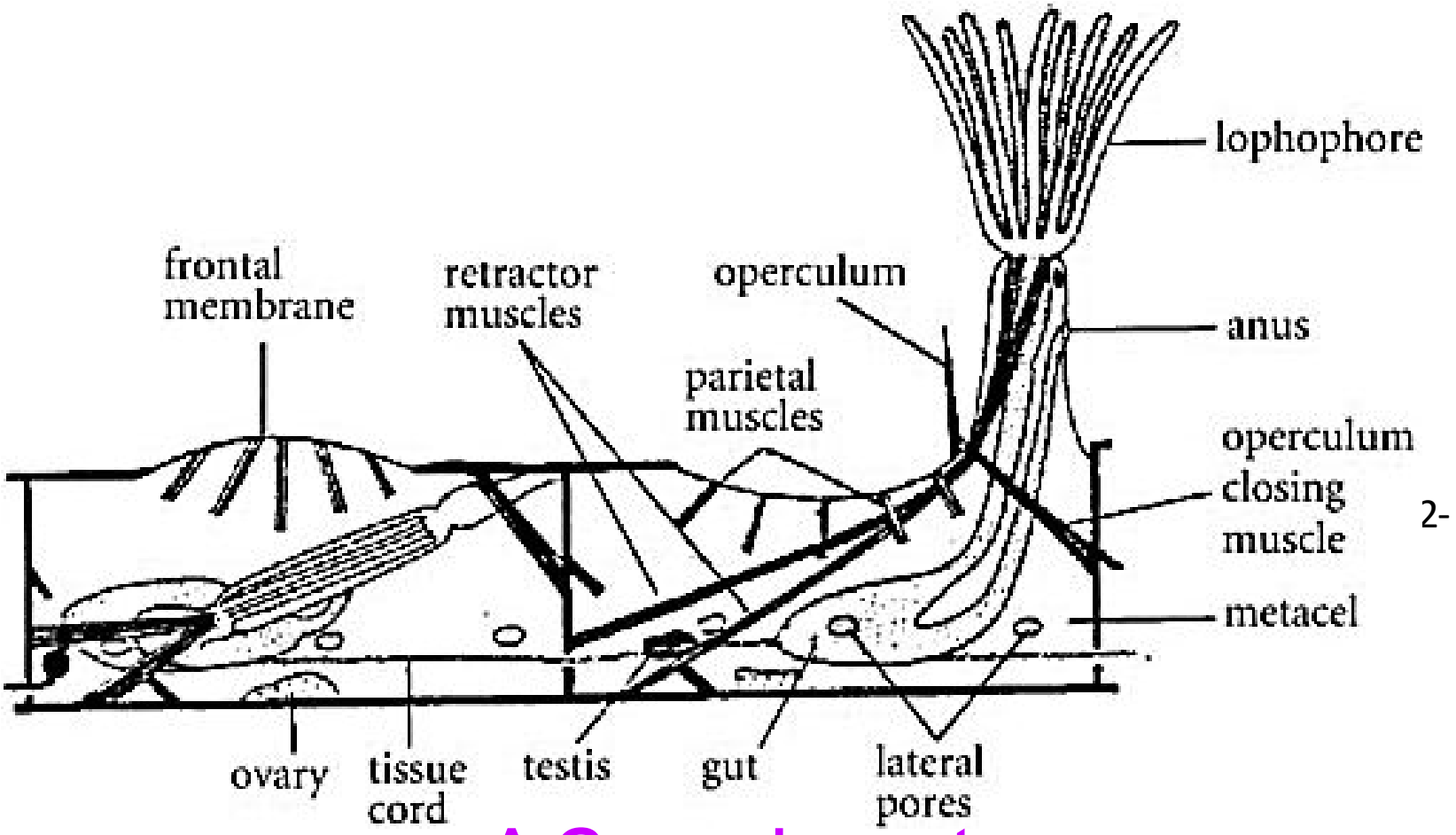
- Superficially resembling the corals and hydroids,
- “Moss animals”
- Filter feeders
- Sieve food particles using a retractable lophophore,

Spiralians

Bryozoans

- The genus *Bowerbankia* is a relatively simple Bryozoan
- Useful for illustrating the general anatomy of bryozoan zooids.

Spiralians



A Gymnolaemate

http://www.blackwellpublishing.com/paleobiology/figure.asp?chap=12&fig=Fig12-15&img=c12f015

Spiralians

Morphology

- Each living zooid is enclosed by a body wall/cystid
- The lophophore, with its beating cilia, A ring of 10 tentacles direct food to mouth leading into a U shaped gut
- Egestion: anus

Spiralians

Oldest Bryozoans

- Occur in the Tremadocian Stage of the Lower Ordovician
- Primitive, soft-bodied bryozoans existed during the Cambrian but have not been fossilized

Spiralians

Oldest Bryozoans

- Stenolaemata dominated Paleozoic bryozoan faunas
- Genera such as Monticulipora, Prasopora and Parvohallopora are typical of Ordovician assemblages

Spiralians

Old Bryozoans

- Cheilostomes
- Polymorphic zooids,
- Appeared: during the Late Jurassic;
- Common in the Late Cretaceous and Paleogene of the Baltic and Denmark.

Spiralians

END

Brachiopods and Bryozoans

- Members of the filter-feeding Paleozoic evolutionary fauna.
- Brachiopods a minor part of the Recent marine fauna but bryozoans continue to flourish

Spiralians

Spiralians: Mollusks

Spiralians

Mollusks

- The Phylum Mollusca can be traced back to at least the Late Precambrian, when *Kimberella* probably fed on algae in Ediacaran communities

Spiralians

Early Mollusks

- Characterized by some short-lived, unusual forms
- Molluskan features: mantle, mineralized shell and radula
- Members of the small shelly fauna

Spiralians

Bivalves

- Characterized by a huge variety of shell shapes, dentitions and muscle scars
- Adapted for a wide range of life strategies in marine and some freshwater environments

Spiralians

Gastropods

- Most gastropods undergo torsion in early life; they have a single shell, often coiled.
- The group adapted to a wide range of environments from marine to terrestrial.

Spiralians

Cephalopods

- The most advanced mollusks, with a head, senses and a nervous system
- Nautiloids, Ammonoids and the Coleoids
- Carnivorous

Spiralians

Mollusks

- During the Mesozoic developed a number of protective strategies
- Robust armor
- Deep infaunal life
- Multiformity of shape and color

Spiralians

END

Annelid worms

- Annelid worms were a sister group to the mollusks
- Their jaws, the scolecodonts, are relatively common in Paleozoic faunas

Spiralians

Mollusks: Introduction

Spiralians

Phylum Mollusca

- The Phylum Mollusca includes
- The slugs, snails, squids, cuttlefish and octopuses
- All manner of marine shellfish such as clams, mussels and oysters

Spiralians

Phylum Mollusca

- The most common marine animals today,
- Occupy wide range of habitats
- Abyssal depths of the oceans and intertidal mudflats
- Forests, lakes and rivers.

Spiralians

Phylum Mollusca

- Mollusks are usually
- Unsegmented
- Soft-bodied animals
- With a body plan based on four features

Spiralians

Four Features

1. The head contains the sensory organs, and a rasping feeding organ, the radula
   Composed of chitin and designed to scrape and in some cases drill.

Spiralians

Four Features

2. The foot is primitively a sole-like structure on which the animal crawls, but is considerably modified in many mollusks.

Spiralians

Four Features

3. The visceral mass of the digestive, excretory, reproductive and circulatory organs is enclosed in the celomic cavity.

Spiralians

Four Features

4. The mantle is a sheet of tissue lying dorsally over the visceral mass that is responsible for secreting the shell.

Spiralians

Molluskan shells

- Secreted as calcium carbonate, mainly aragonite, with an organic matrix and an outer organic layer.

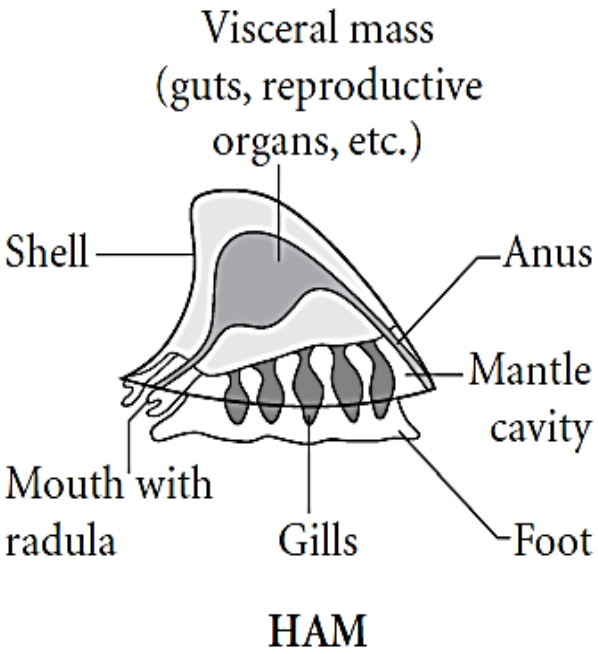
Spiralians

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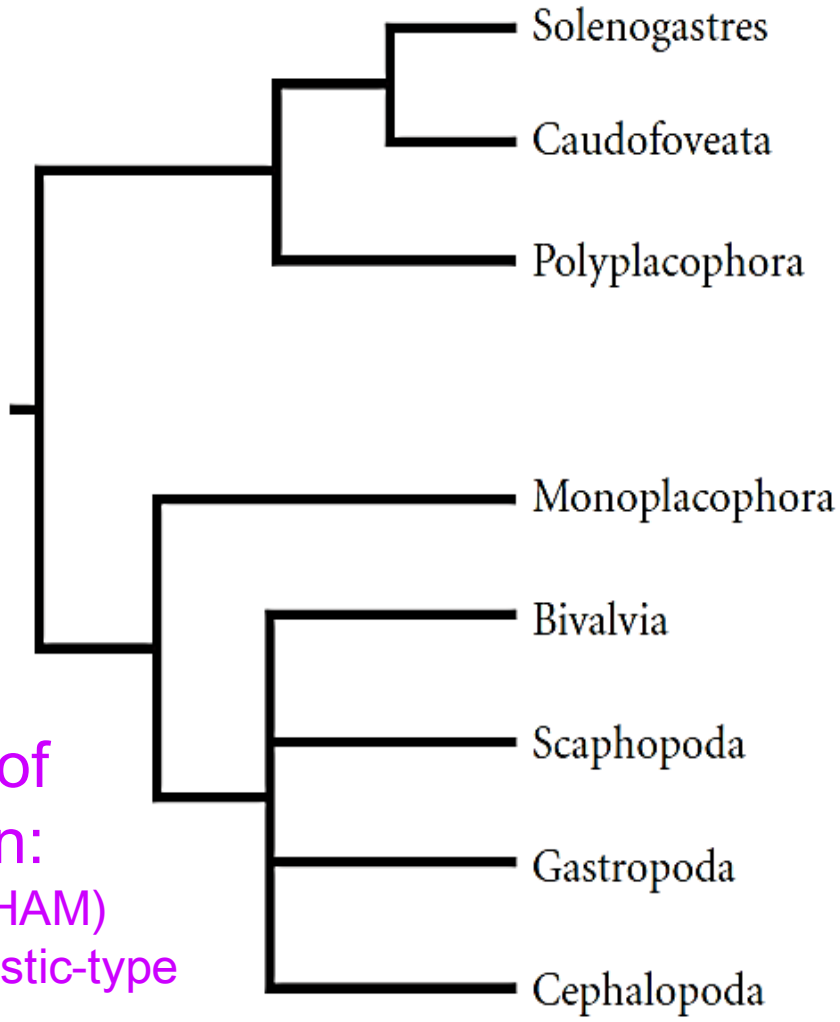
Molluskan shells

- Visualizing molluskan evolution
- Hypothetical ancestor Archemollusk,
- Merged with a recent cladogram for the phylum

Spiralians



HAM



Pseudocladogram of molluskan evolution:

hypothetical archemollusk (HAM) evolution integrated with a cladistic-type framework

http://www.blackwellpublishing.com/paleobiology/jpg/300_96dpi/c13f001.jpg

Spiralians

Early Mollusks

Spiralians

Early Mollusks

- Early Cambrian: A time of experimentation
- A variety of short-lived, often bizarre molluskan groups
- Dominating many faunas

Spiralians

Early Mollusks

- First mollusks descended from forms like living flatworms
- Spiculate animals
- Radula and gills on the posterior
- Similar to modern soft-bodied Aplacophorans

Spiralians

Aplacophorans

- A group of shell-less mollusks.
- Aplacophorans and shelled mollusks shared a common ancestor probably during the Late Precambrian
- Kimberella

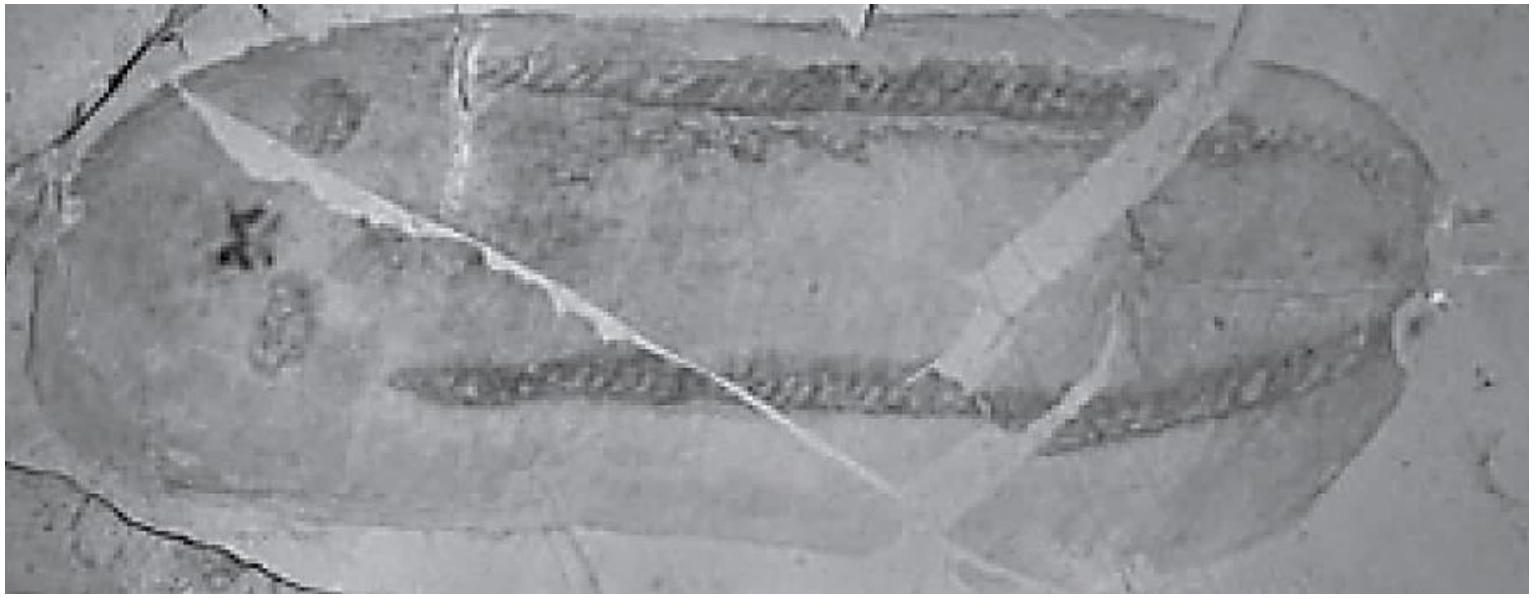
Spiralians

The body fossils and trace fossils place Kimberella near the base of the Molluskan clade and suggest a deep origin for the phylum

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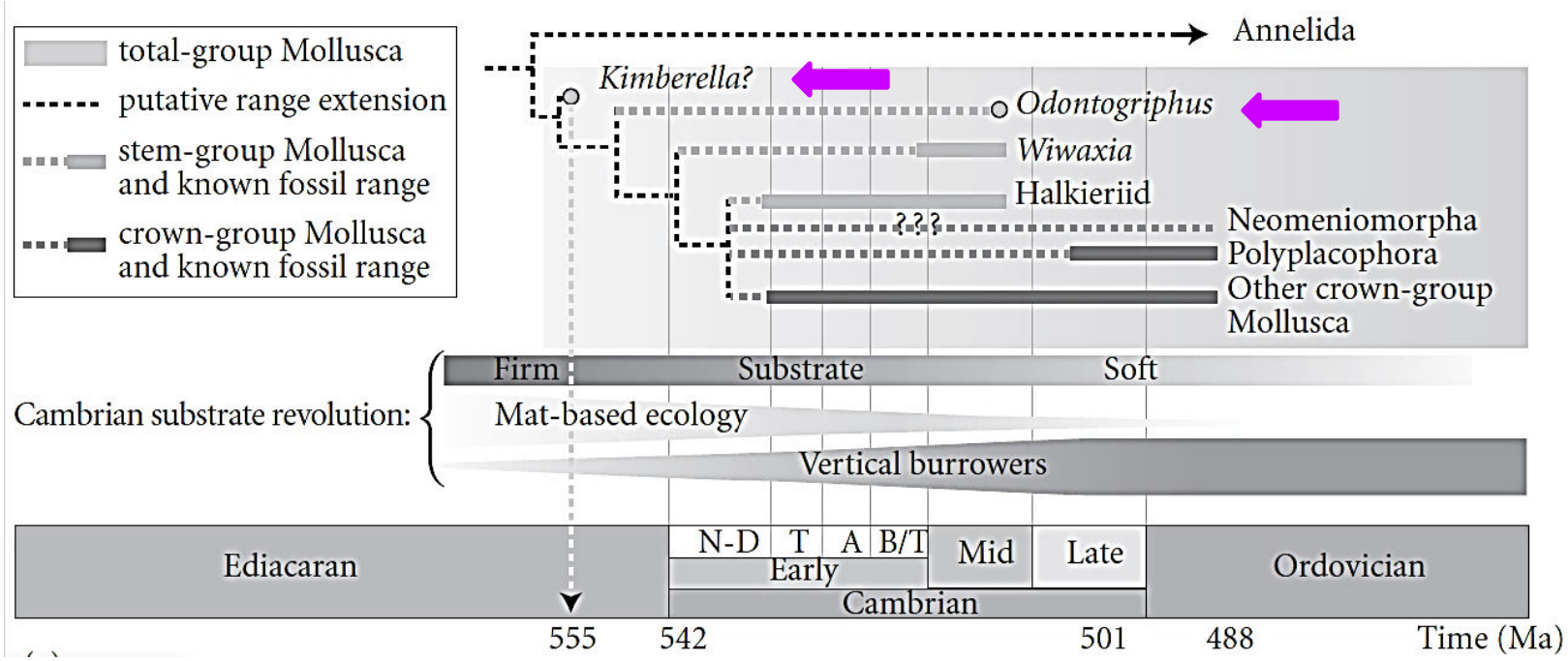
Spiralians



Odontogriphus possesses a radula, a broad foot and a stiffened dorsum, so placing it firmly within the mollusks, close to Kimberella

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Spiralians



Phylogeny and stratigraphic ranges of early mollusks mapped onto some ecological changes.

N-D, Nemakit-Daldynian; T, Tommotian; A, Atdabanian; B/T, Botomian. (a, courtesy of Ben Waggoner; b, c, courtesy of ten-Bernard Caron.)

http://www.blackwellpublishing.com/paleobiology/figure.asp?chap=13&fig=Fig13-2&img=c13f002

Spiralians

Halkieria

- Cambrian rocks of the Danish island of Bornholm.
- Articulated specimens from the Early Cambrian Sirius Passet fauna from North Greenland

Spiralians

Halkieria is a mollusk, possessing most of the features that define the phylum, but a number of characters (such as the shells at the anterior and posterior of the animal) have formed the basis for a new class of mollusk, the Diplacophora

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Spiralians

Origins

- Halkieriid mollusks
- Lower Cambrian
- The hyoliths – long, conical, calcareous shells with an operculum-covered aperture
- Cambrian to Permian

END

Spiralians

Mollusks:
Class Bivalvia

Spiralians

Bivalves

- Among the commonest shelly components of beach sands
- Many taxa are farmed and harvested for human consumption
- Pearls: a valuable by-product

Spiralians

Bivalves

- Spectacular variety of shell shapes and life strategies
- Spanning the entire Phanerozoic
- All based on a simple bilaterally symmetric exoskeleton

Spiralians

Morphology

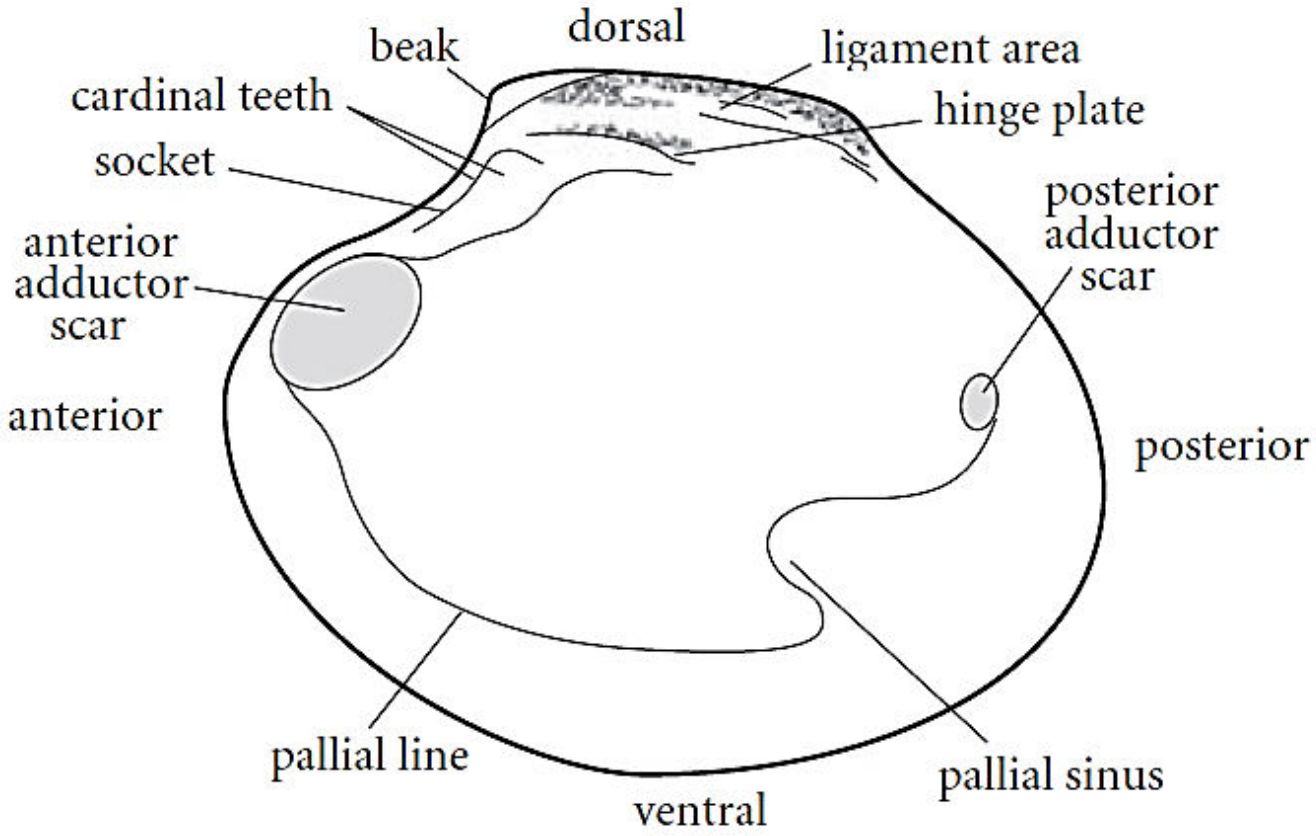
- Bivalves are twin-valved shellfish superficially resembling the brachiopods
- Common in modern seas
- Bivalve shells: calcium carbonate, usually aragonite

Spiralians

Morphology

- Two valves are virtually mirror images of each other.
- Termed lamellibranchs or pelecypods, but they were first named Bivalvia by Linnaeus in 1758

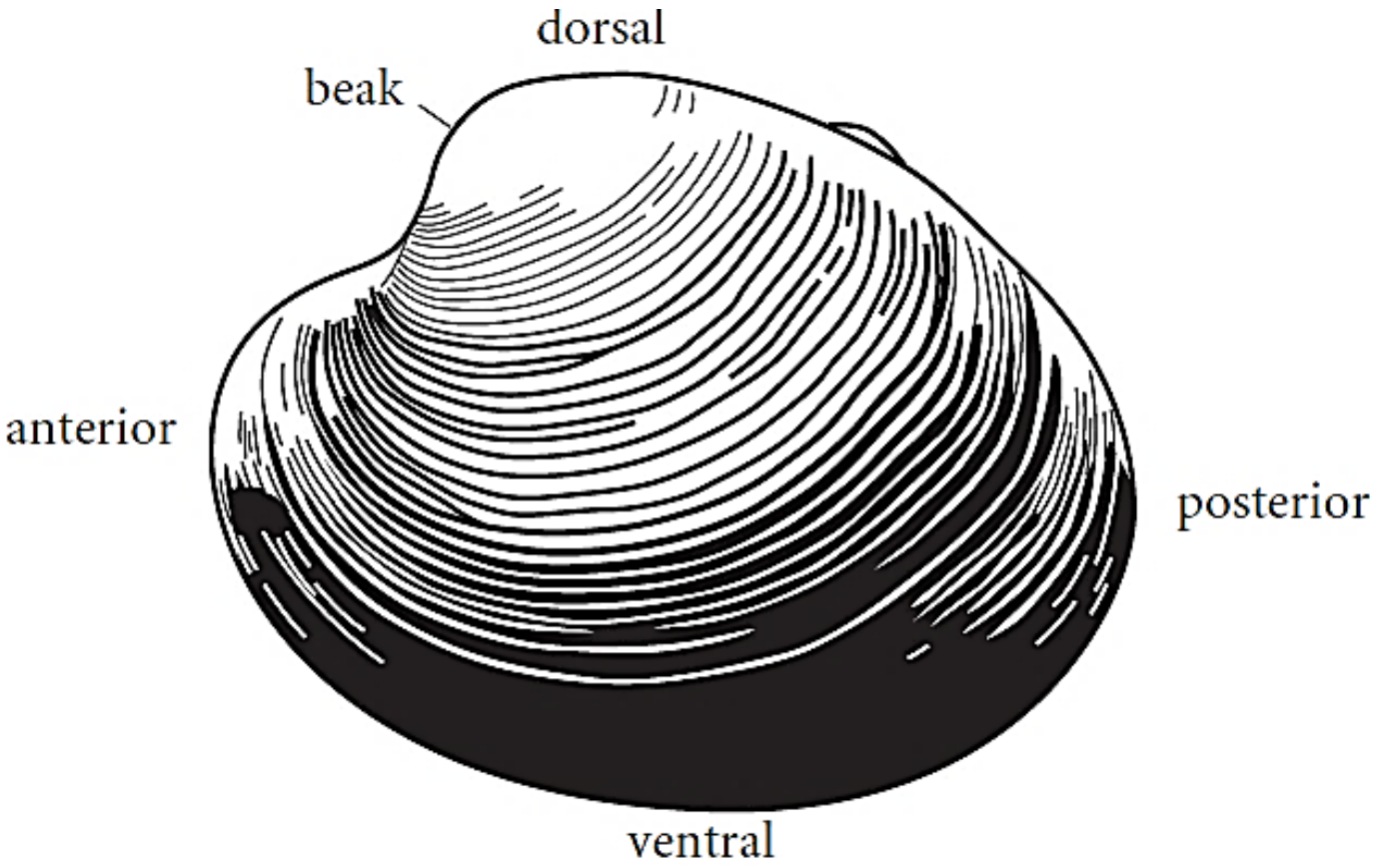
Spiralians



Bivalve morphology based on a living bivalve: internal features of the right valve

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Spiralians



Bivalve morphology based on a living bivalve: External features of the left valve

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Spiralians

Evolution

- Basal Cambrian
- Two Early Cambrian genera

1. Pojetaia from Australia and China
2. Fordilla from Denmark, North America and Siberia

Spiralians

Evolution

- Evolved from Rostroconchs
- The class evolved rapidly in the Early Ordovician to include basal forms of all bivalve infrasubclasses
- Tremadocian and Floian radiation

Spiralians

END

Evolution

- Major diversification: Paleozoic
- Evolved extensive siphons helping deep-burrowing life modes
- Early Mesozoic radiation: featured siphonate forms

Spiralians

Mollusks: Class Gastropoda

Spiralians

The Gastropods

- The "belly-footed" mollusks
- Most varied and abundant of the molluskan classes today
- Snails and slugs
- Forms both with and without a calcareous shell

Spiralians

The Gastropods

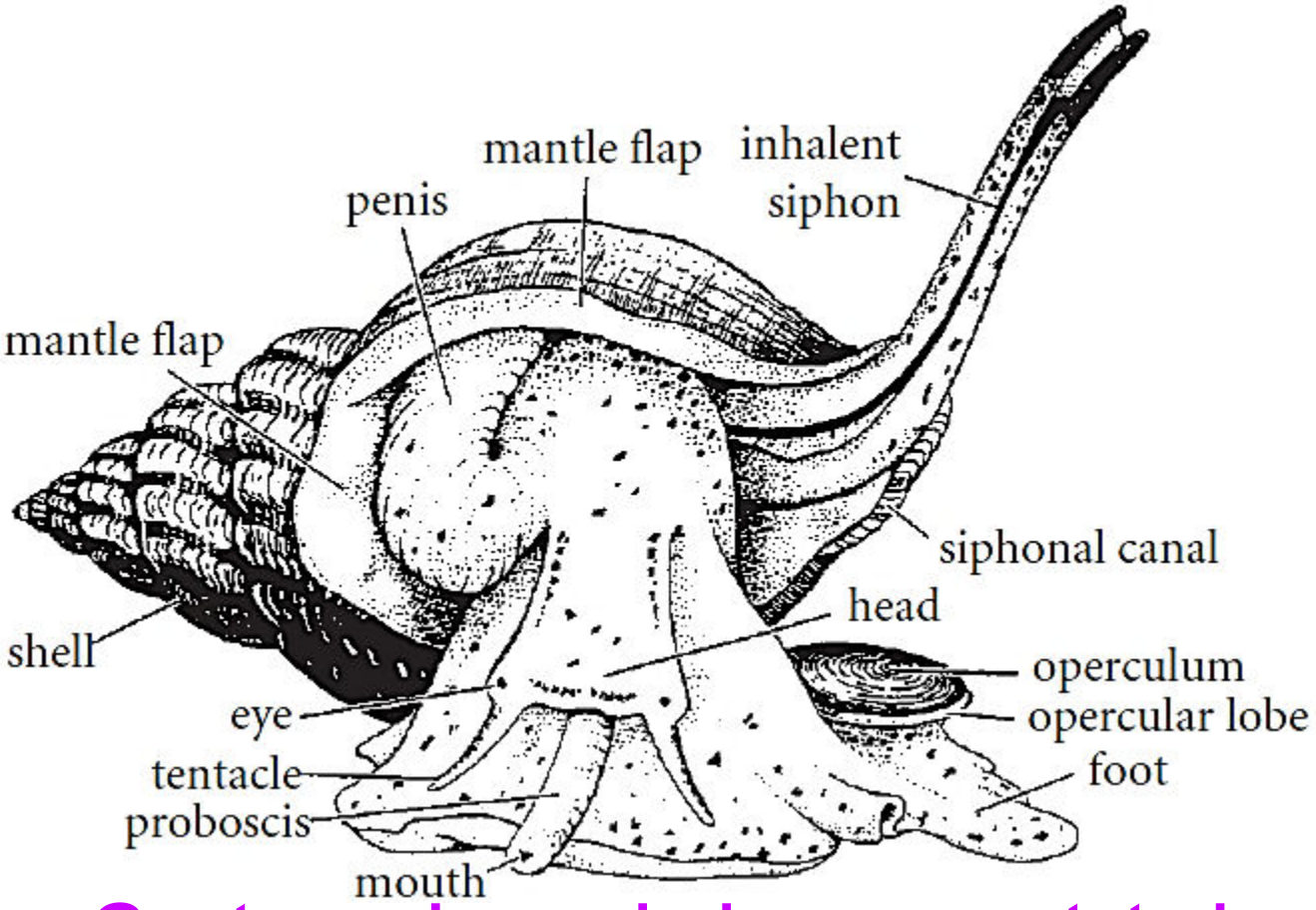
- Entire Phanerozoic,
- Evolved
- Swimming strategies: creeping, floating and
- Trophic styles: razing, predatory and parasitic.

Spiralians

Torsion

- Mantle cavity containing gills and anus, excretory and reproductive openings comes to lie above head.
- Seems to be disadvantageous loss of one of the gills

Spiralians



Gastropod morphology: annotated reconstruction of a living gastropod

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Spiralians

Evolution

- No general agreement
- Derived from a monoplacophoran-type ancestor by torsion and development of an exogastric condition
- Shell is coiled away from the head

Spiralians

Evolution

- An origin from among coiled forms such as *Pelagiella* may link the monoplacophoran grade through the Tommotian *Aldanella* to the gastropods.

Spiralians

END

Evolution

- Most Paleozoic gastropods: herbivores or detritus feeders
- Classic study of Late Tertiary snails from Lake Turkana, Kenya, Williamson (1981): punctuated changes in 14 separate lineages

Spiralians

Mollusks:
Class Cephalopoda

Spiralians

Cephalopods

- Most highly organized of the mollusks
- The greatest complexity of any of the spiralians
- Well-defined head with the foot modified into tentacles,
- “Head-footed

Spiralians

Cephalopods

- High metabolic and mobility rates
- Developed nervous system, sharp eyesight & an advanced brain
- Adaptations for a carnivorous predatory life mode.

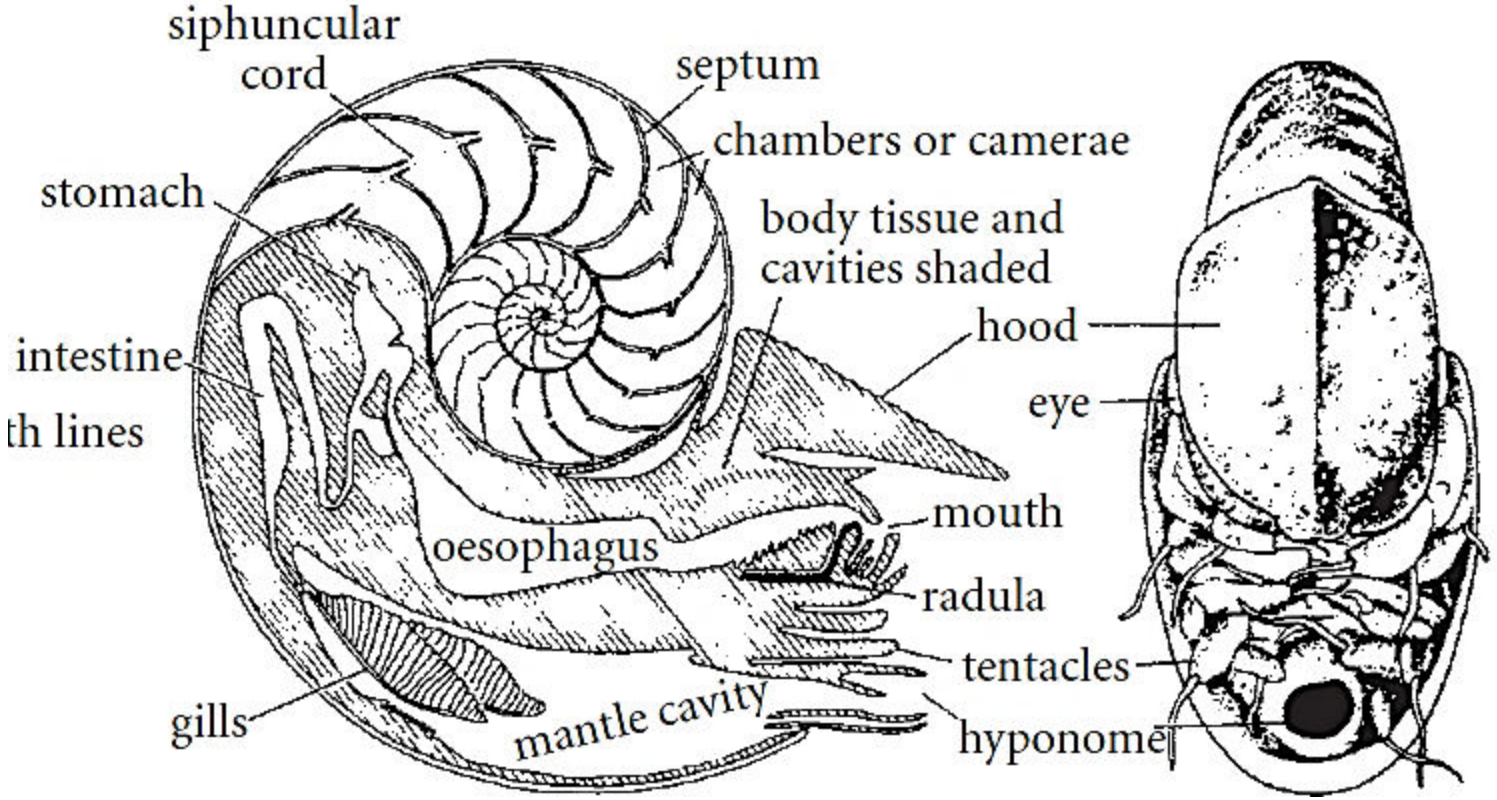
Spiralians

Cephalopods

- A tripartite division
- Three subclasses include

1. *Nautiloidea*, with straight or coiled external shells with simple sutures (Late Cambrian to Recent)

Spiralians



Features of the shell and internal morphology of a living Nautilus

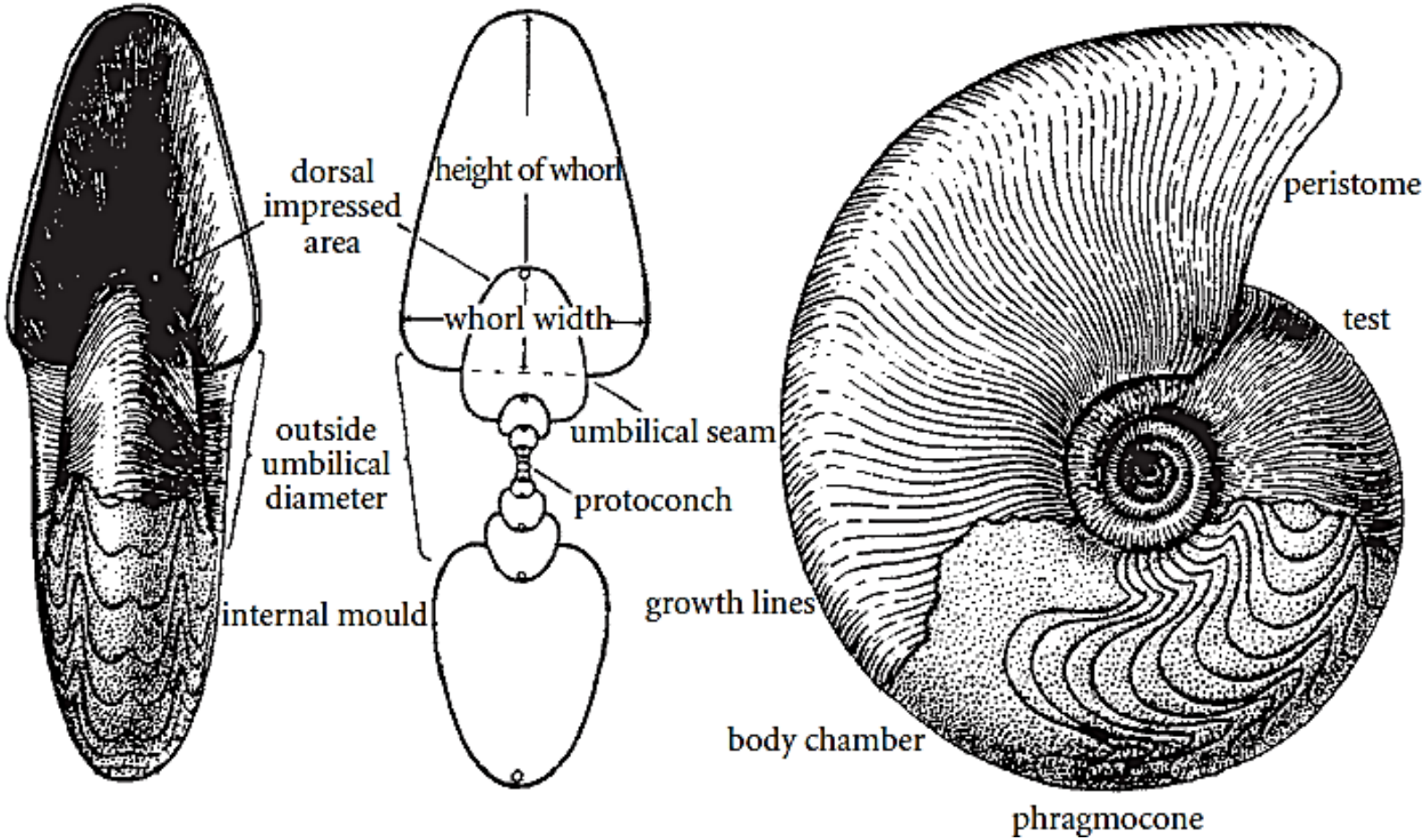
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Spiralians

Cephalopods

2. *Ammonoidea*, with coiled, commonly ribbed external shells with complex sutures (Early Devonian to latest Cretaceous, possibly earliest Paleogene)

Spiralians



External Morphology of the ammonoids

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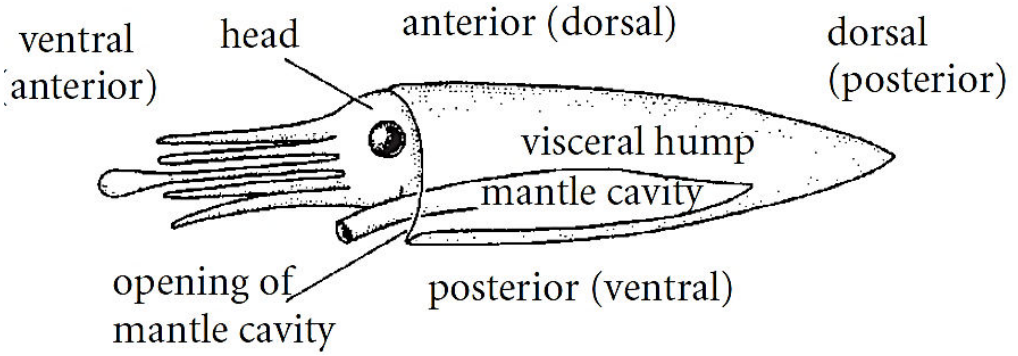
Spiralians

Cephalopods

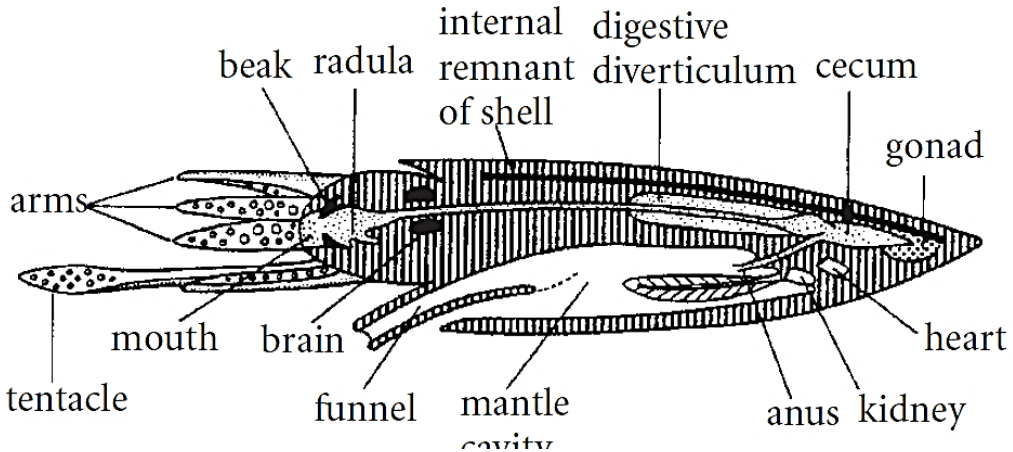
3. *Coleoidea*, with straight or coiled internal skeletons (Carboniferous To Recent).
   - Belemnites (extinct), the squids and cuttlefish
   - The octopods: lost their skeleton.

Spiralians

Reconstruction of a living belemnite,



Soft-part morphology of the belemnites



Coleoid morphology

http://www.blackwellpublishing.com/paleobiology/figure.asp?chap=13&fig=Fig13-14&img=c13f022

Spiralians

Mollusks:
Class Scaphopoda

Spiralians

Scaphopods

- Are generally rare as fossils
- The Scaphopoda, or elephant-tusk shells, have a single, slightly curved high conical shell, open at both ends

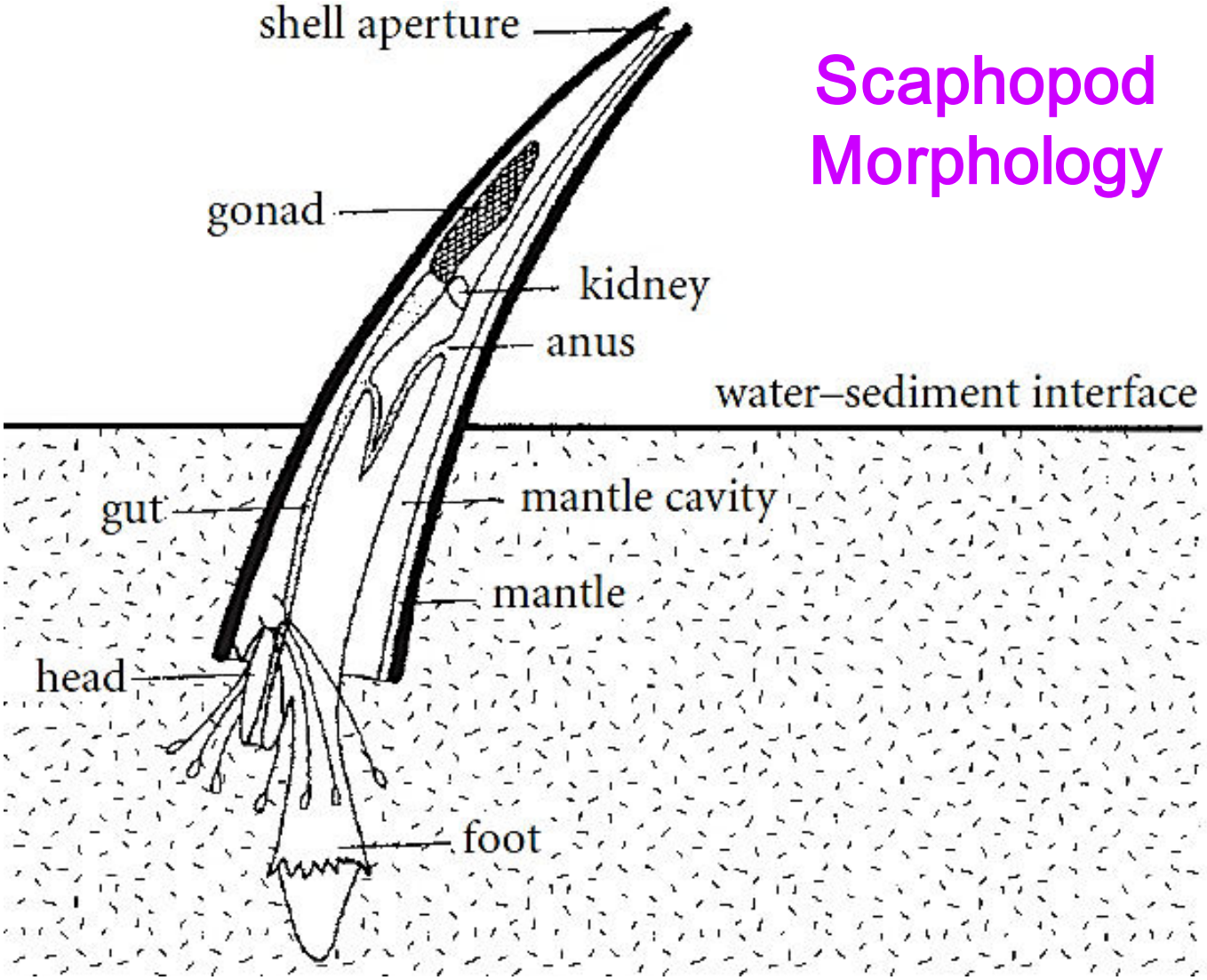
Spiralians

Scaphopods

- They lack gills and eyes,
- Have a mouth equipped with a radula
- A foot, surrounded by tentacles similar to that of the bivalves, adapted for burrowing.

Spiralians

Scaphopod Morphology



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Spiralians

Scaphopods

- Carnivorous, feeding on small organisms, foraminiferans
- Spend life in quasi-infaunal positions within soft sediment in deeper-water environments

Spiralians

END

Scaphopods

- The first scaphopods appeared during the Devonian and apparently had similar lifestyles to living forms such as *Dentalium*

Spiralians

Mollusks:
Class Rostroconcha

Spiralians

Rostroconchs

- Relatively recently a small class of mollusks
- Superficially resembling bivalves but lacking a functional hinge,
- Documented from the Paleozoic

Spiralians

Rostroconchs

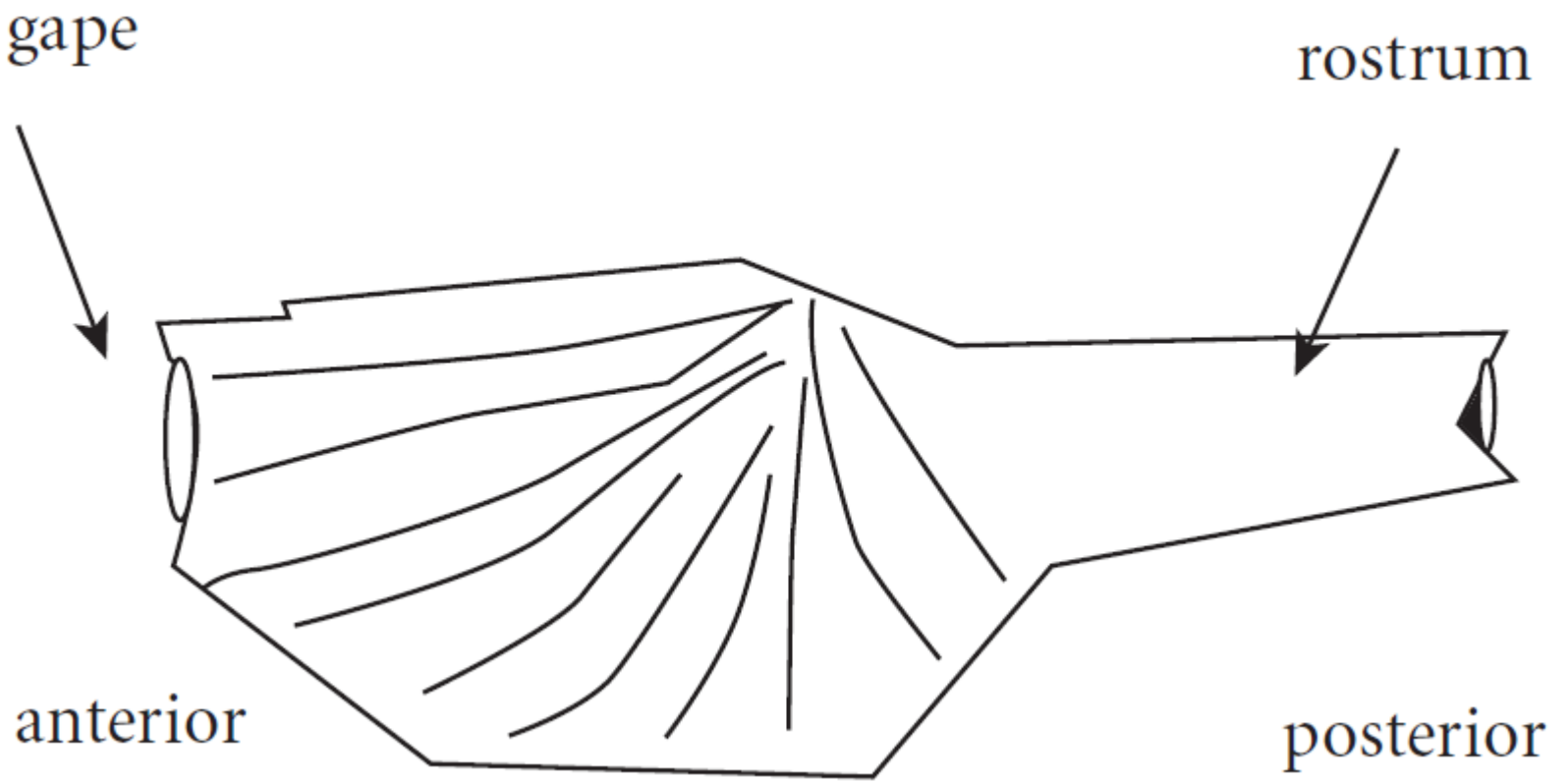
- Over 35 genera
- Originally described as bivalved arthropods
- Had a foot that emerged through the anterior gape between the shells

Spiralians

Rostroconchs

- The two shells are fused along the middorsal line
- Posteriorly the shells are extended as a platform or rostrum

Spiralians



Rostroconch Morphology

http://www.blackwellpublishing.com/paleobiology/figure.asp?chap=13&fig=Fig13-25&img=c13f025

Spiralians

Evolution

- Appeared first: Early Cambrian
- *Heraultipegma* and *Watsonella* dominated faunas of the Tommotian
- Diversified: During Ordovician
- Reach an acme: Katian

Spiralians

Evolution

- Similar to bivalves ecological niches
- Decline in abundance and diversity
- Final extinction: end of the Permian
- Except Conocardiodes (*Arceodomus*)

Spiralians

Evolution

- The Rostroconchs occupy a pivotal position in molluskan evolution
- Developed from within the monoplacophoran plexus with a loss of segmentation

Spiralians

Evolution

- Generated both the bivalves and the scaphopods
- The gastropods and cephalopods were probably derived independently from a separate monoplacophoran ancestor

Spiralians

Evolution

- Missing link between the univalved and bivalved molluskan lineages
- Their unlikely morphology may have contributed towards their late discovery

END

Spiralians

Evolutionary Trends Within The Mollusca

Spiralians

Phanerozoic

- A spectacular variety of mollusk morphotypes and life modes evolved during the Phanerozoic, from the simple body plan of the archemollusk

Spiralians

Cambrian

- Despite the diversity of early mollusks in the Cambrian, the phylum was not notably conspicuous in the tiered suspension-feeding

Spiralians

Paleozoic Bivalves

- Common in nearshore environments, associated with *lingulide brachiopods*
- Inhabited deeper-water clastic environments

Spiralians

Paleozoic Bivalves

- Late Paleozoic: a variety of carbonate environments
- End of the Paleozoic: shallow-water belts displaced the Paleozoic associations seaward.

Spiralians

Mesozoic and Cenozoic Bivalves

- Late Mesozoic and Cenozoic: The significant radiation of infaunal taxa
- A response to increased predation

Spiralians

Paleozoic Gastropods

- The majority were Eogastropoda
- Dominated shallow-water marine environments and some carbonate reef settings

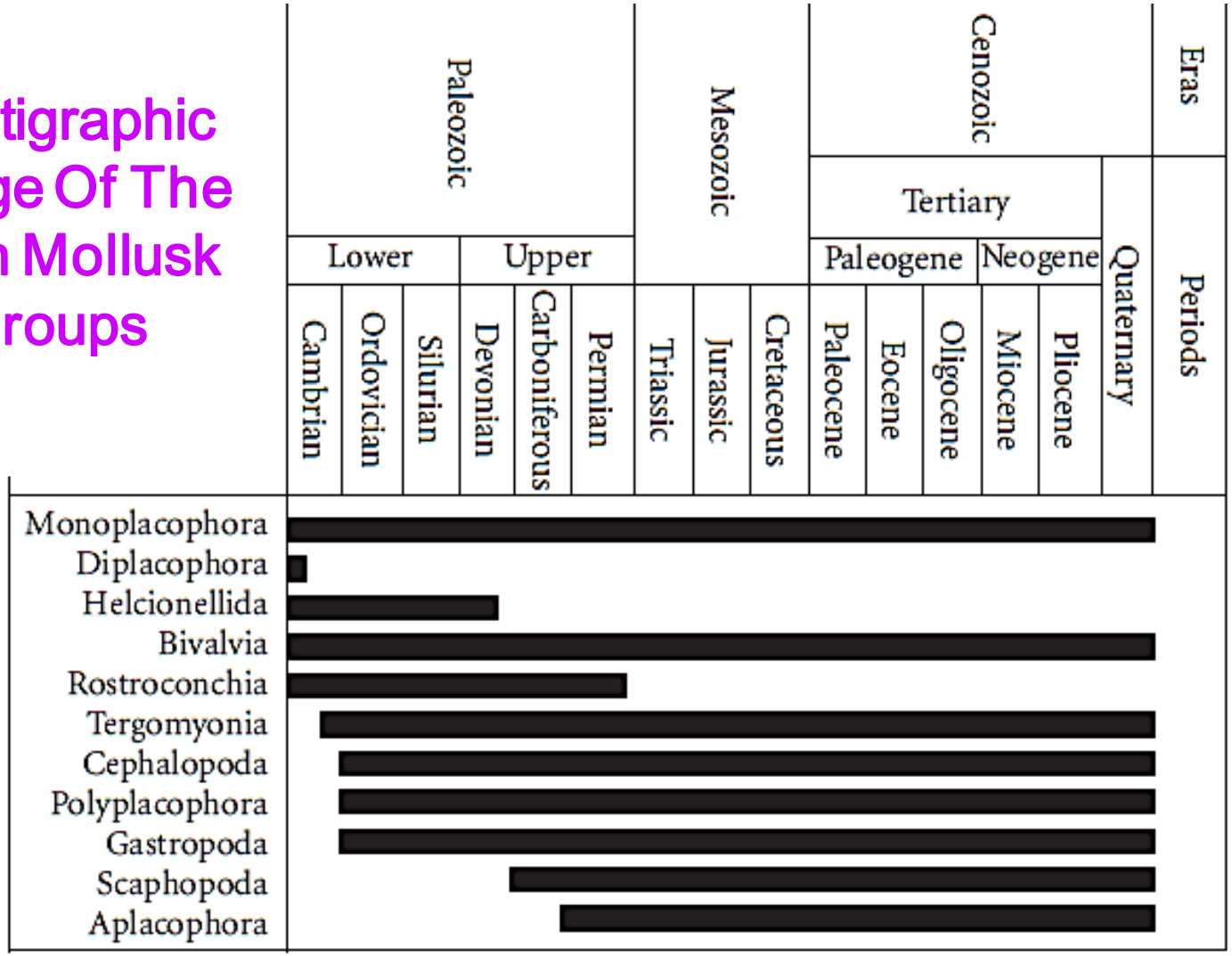
Spiralians

Mesozoic Gastropods

- Mesogastropoda, grazed on algal-coated hard substrates
- The Cenozoic: the acme of the group with the radiation of the siphonal carnivorous neogastropods

Spiralians

Stratigraphic Range Of The Main Mollusk Groups



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Spiralians

END

Cephalopods

- Radiation: during the Mesozoic
- The development of a chambered shell with a siphuncle, which gave them considerable control over attitude and buoyancy

Ecdysozoa: Arthropods

Introduction

Ecdysozoa: Arthropods

Arthropods

- Such as lobsters, spiders, beetles and trilobites
- Have legs, a segmented body plan with jointed appendages and the ability to molt
- From ek- ‘out, off’ + duein ‘put’

Ecdysozoa: Arthropods

Arthropods

- The first major faunas: Early Cambrian
- Appear bizarre
- Probably were no more morphologically different to each other than living faunas

Ecdysozoa: Arthropods

Arthropods

- Arthropod-like animals in the Ediacara biota: Ancient origins
- Trilobites appeared: Early Cambrian and during the Paleozoic

Ecdysozoa: Arthropods

Trilobites

- Evolved advanced visual systems
- Enrolment structures
- In Order to pursue a variety of benthic and pelagic life styles

Ecdysozoa: Arthropods

The Largest Arthropods

- Chelicerates and included the giant *eurypterids*
- Patrolled marine marginal environments during the Silurian and Devonian.

Ecdysozoa: Arthropods

Myriapods

- Earliest terrestrial body fossils in the Mid Ordovician
- Trackways of euthycarcinoids (i.e. stem-group mandibulates) show that these moved onto land even earlier (Late Cambrian).

Ecdysozoa: Arthropods

Insects

- Insects first appeared during the Early Devonian and diversified rapidly
- There are probably 10 million species of living insects

Ecdysozoa: Arthropods

Flight in Insects

- Insects had probably already evolved flight before the Mid Carboniferous, when giant dragonflies patrolled the forests.

Ecdysozoa: Arthropods

The Crustaceans

- Include many familiar groups such as crabs, lobsters and shrimps
- Together with the barnacles and ostracodes.

Ecdysozoa: Arthropods

END

History of Phylum

- Exceptionally preserved fossils from the faunas of
  1. Cambrian Burgess Shale
  2. Chengjiang
  3. Sirius Passet

Ecdysozoa: Arthropods

Arthropods

Ecdysozoa: Arthropods

Arthropods

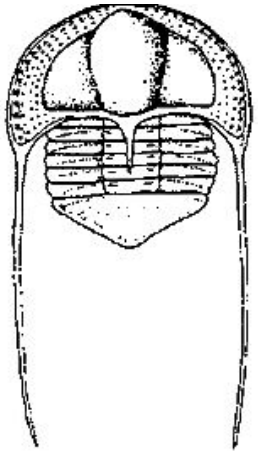
- Arthropods are spectacularly diverse group of legged invertebrates
- Accounting for about three-quarters of all species living on the planet today

Ecdysozoa: Arthropods

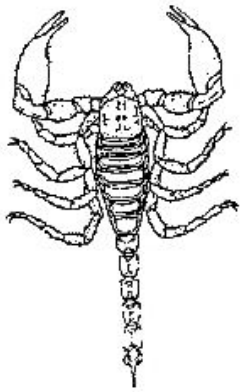
Basic Body Plan

- Segmented, with jointed appendages
- Adapted for feeding, locomotion and respiration
- Exoskeleton appeared during the Early Cambrian

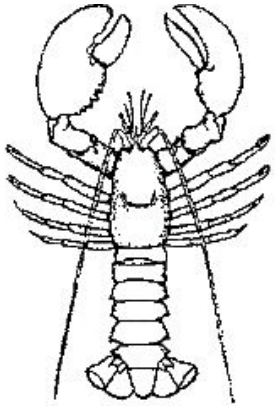
Ecdysozoa: Arthropods



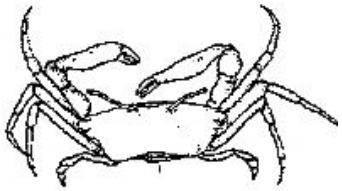
Trilobitomorph: trilobite



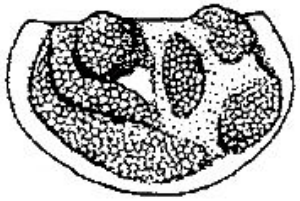
Chelicerate: scorpion



Crustacean: decapod



Crustacean: decapod



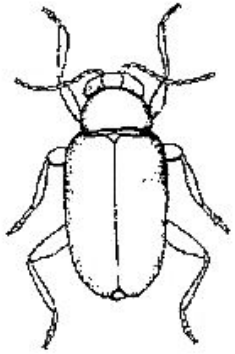
Crustacean: ostracode



Crustacean: barnacle



Insect: palaeopteran



Insect: coleopteran

All Members of This Phylum have Both Segmented Bodies And Appendages

http://www.blackwellpublishing.com/paleobiology/figure.asp?chap=14&fig=Fig14-1&img=c14f001

Ecdysozoa: Arthropods

Basic Body Plan

- Differentiated into a head, thorax and abdomen
- Often the head and thorax fused: Cephalothorax
- Hard mouthparts, mandibles: Ability to process wide variety of foods

Ecdysozoa: Arthropods

Arthropod Exoskeleton

- Organic substance, chitin
- Hardened or sclerotized by $CaCO_3$ or $Ca_3(PO_4)_2$
- Potential for preservation: Excellent

Ecdysozoa: Arthropods

Arthropod Exoskeleton

- Acts as a base for the attachment of locomotory muscles
- Permitting rapid movement
- Not usually mineralized

Ecdysozoa: Arthropods

Arthropod Exoskeleton

- Metamorphosis: Many arthropods
- Molting or Ecdysis: All the main groups
- Exuviae, or cast-off coverings: previous skeleton or cuticle of the animal

Ecdysozoa: Arthropods

Five Subphyla

- During a geological history of at least 540 million years, the five subphyla of arthropod have adapted to life in marine, freshwater and terrestrial environments

Ecdysozoa: Arthropods

Five Subphyla

1. Trilobitomorpha: Cambrian to Permian
2. Chelicerata: Cambrian to Recent
3. Myriapoda: Ordovician, Silurian to Recent

Ecdysozoa: Arthropods

END

Five Subphyla

4. Hexapoda: Devonian to Recent

5. Crustacea: Cambrian to Recent

Ecdysozoa: Arthropods

Subphylum
Trilobitomorpha

Ecdysozoa: Arthropods

Trilobitomorphs

- Highly derived arthropods
- lack specialized mouthparts
- Tagmata comprising the cephalon, thorax and pygidium
- Trilobitomorph appendages

Ecdysozoa: Arthropods

Trilobitomorphs

- Mainly the trilobites and over 15,000 species are known
- Common: throughout the Paleozoic
- Extinction: end of the Permian

Ecdysozoa: Arthropods

Earliest Arthropods

- Earliest arthropods were trilobites
- Marine Cambrian strata are usually correlated on the basis of trilobite assemblages

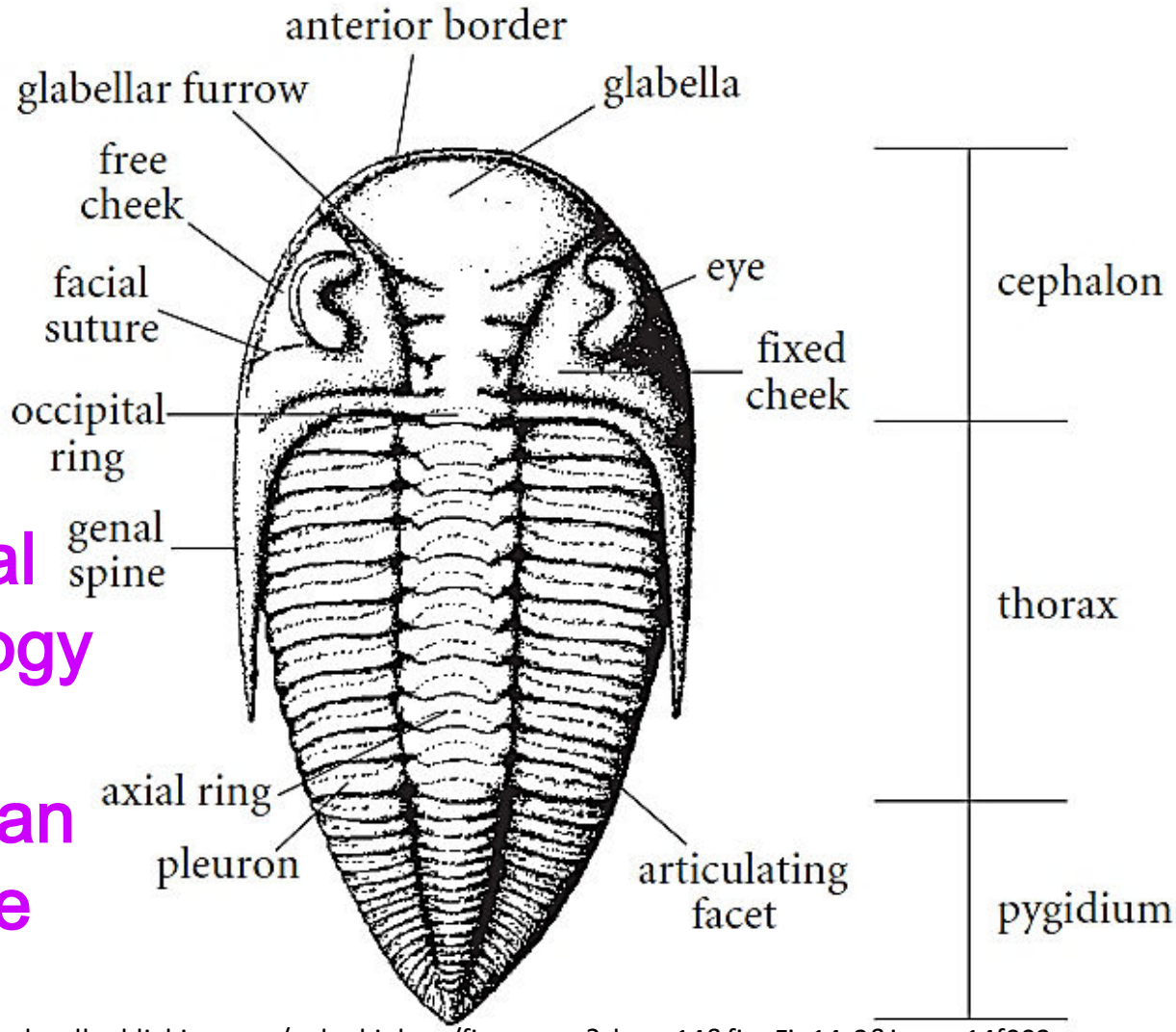
Ecdysozoa: Arthropods

Morphology

- Trilobite ("three-lobed")
- Exoskeleton is divided into three lobes
- The axial lobe protects digestive system, whereas the two pleural lobes cover appendages.

Ecdysozoa: Arthropods

External Morphology of the Ordovician Trilobite



http://www.blackwellpublishing.com/paleobiology/figure.asp?chap=14&fig=Fig14-3&img=c14f003

Ecdysozoa: Arthropods

Evolution

- Trilobite faunas: basis for many paleogeographic reconstructions of the Cambrian and Ordovician world
- The Cambrian, complex biogeographic patterns

Ecdysozoa: Arthropods

Evolution

- Some Early Paleozoic trilobite communities may also be interpreted as showing an onshore–offshore spectrum

Ecdysozoa: Arthropods

Evolution

- Trilobites (such as *Choubertella* and *Schmidtiellus*) first appeared in the Early Cambrian
- Survived until the end of the Permian

Ecdysozoa: Arthropods

Extinction

- Last genera, such as *Pseudophillipsia*, disappeared: end of the Permian
- History of 350 million years, the basic body plan was essentially unchanged

Ecdysozoa: Arthropods

END

Macroevolution Studies

- Trilobites have provided key evidence in studies of macroevolution
- Controversy over punctuated equilibrium and phyletic gradualism

Ecdysozoa: Arthropods

Subphylum Chelicerata

Ecdysozoa: Arthropods

Chelicerates

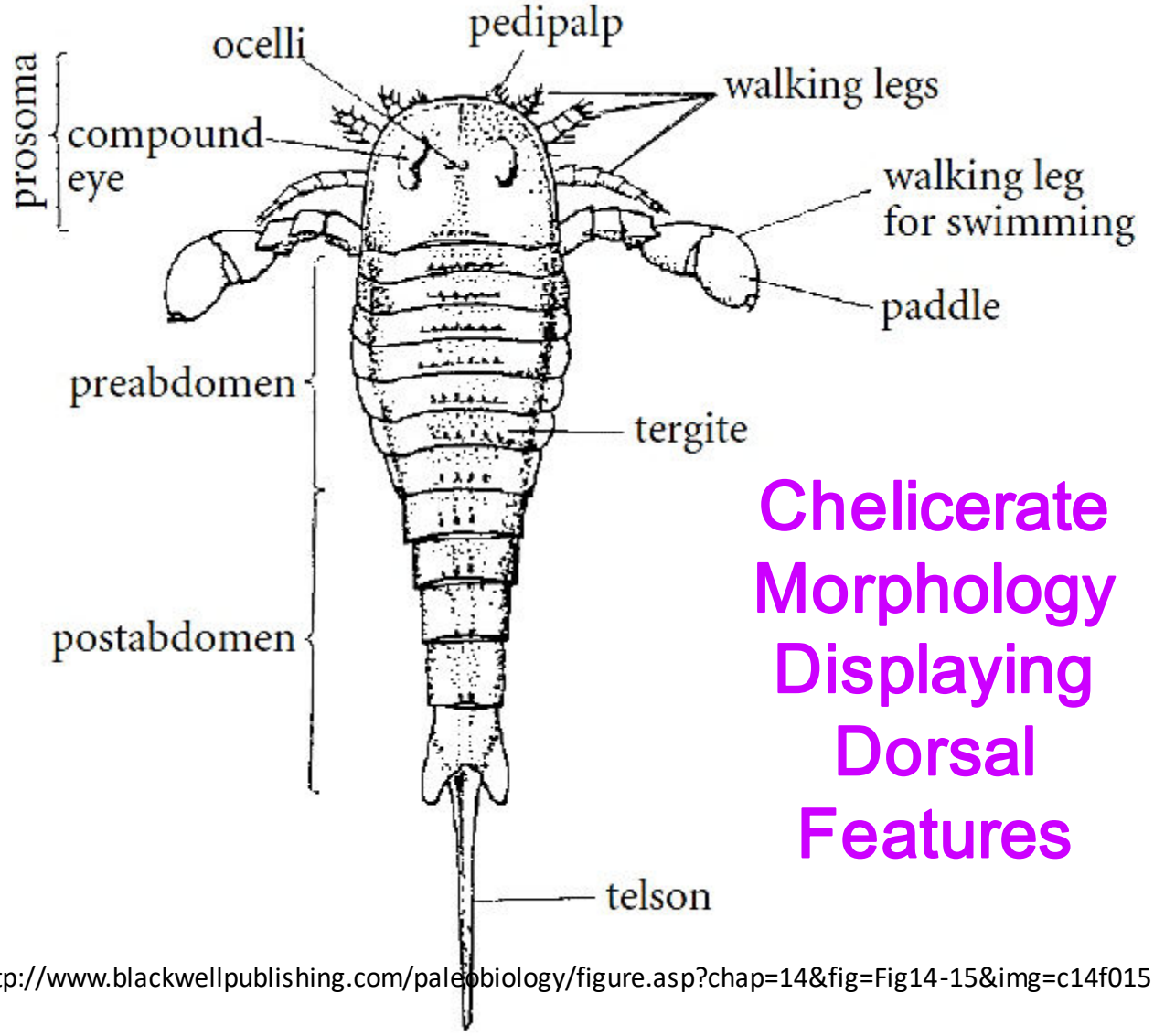
- Diverse and heterogeneous
- Mites, scorpions and spiders.
- The familiar horseshoe crab *Limulus*, together with the extinct sea scorpions, the *eurypterids*

Ecdysozoa: Arthropods

Chelicerates

- Prosoma (head and thorax) with 6 segments with appendages.
- An opisthosoma (abdomen) with 12 segments,
- Pair of chelicerae (pincers) on the first segment of the prosoma

Ecdysozoa: Arthropods



Chelicerate Morphology Displaying Dorsal Features

http://www.blackwellpublishing.com/paleobiology/figure.asp?chap=14&fig=Fig14-15&img=c14f015

Ecdysozoa: Arthropods

Chelicerate Groups

- Traditionally two main groups

1. The merostomes, including the aquatic horseshoe crab *Limulus* and the giant sea scorpions or *eurypterids*

Ecdysozoa: Arthropods

Chelicerate Groups

2. The arachnids: mainly comprise the terrestrial spiders and scorpions.
   - This split of the subphylum into marine and non-marine has been challenged

Ecdysozoa: Arthropods

Evolution

- The bizarre *Sanctacaris* from the Middle Cambrian Burgess Shale may be a basal outgroup to the clade *Chelicerata*

Ecdysozoa: Arthropods

Evolution

- Xiphosure (Horse Shoe crab) -like taxa, such as *Eolimulus,* have been described from the Lower Cambrian
- A trend towards larger size and a shorter fused abdomen

Ecdysozoa: Arthropods

END

Evolution

- 50 genera of eurypterids have been described
- The group was most abundant during the Silurian and Devonian
- Arachnids
- Book lungs or tracheae or both

Ecdysozoa: Arthropods

Subphylum Myriapoda

Ecdysozoa: Arthropods

Myriapods

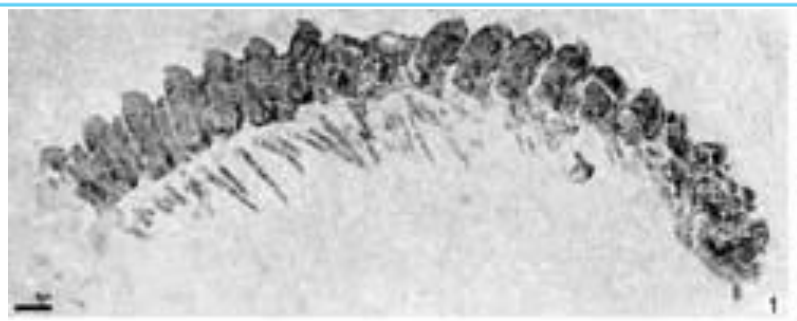
- A varied group comprising the millipedes, centipedes, symphylans and pauropods
- First appeared during the Mid Silurian

Ecdysozoa: Arthropods

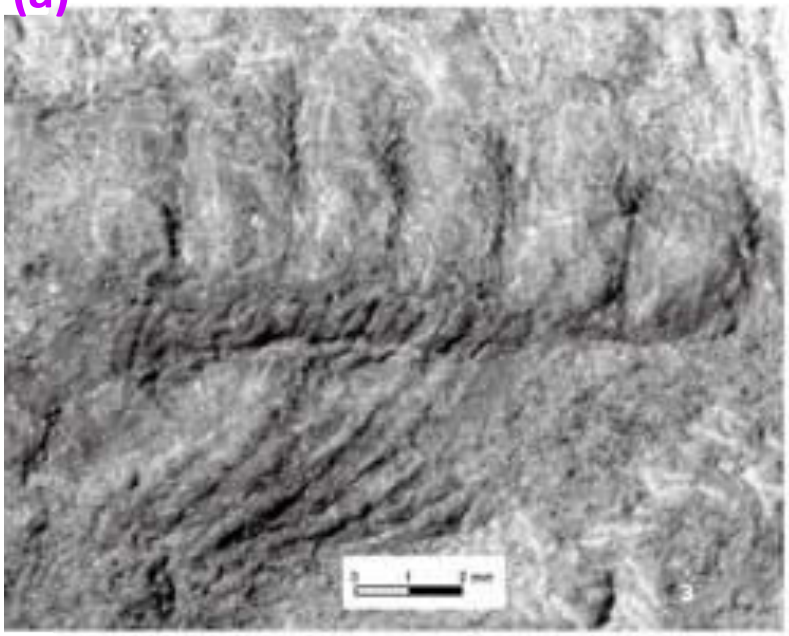
Invasion of land

- The first animals to colonize the land.
- Heather Wilson and Lyall Anderson (2004) have described the few Silurian and Devonian taxa from Scotland

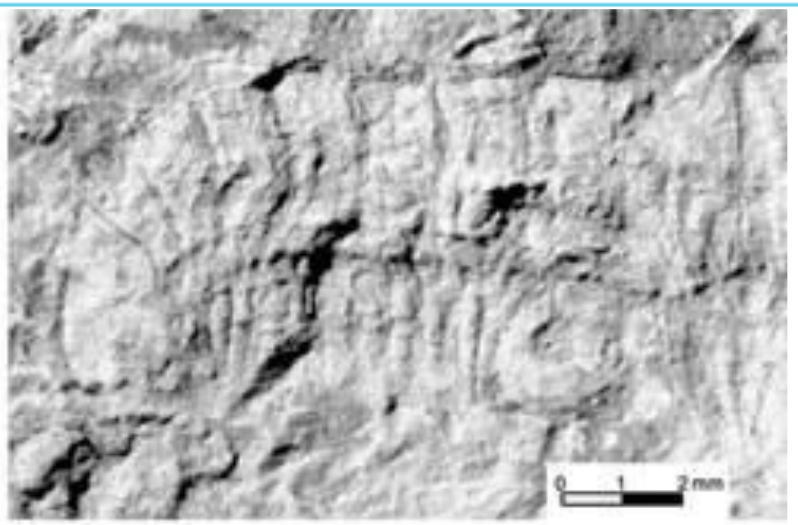
Ecdysozoa: Arthropods



(a)



(c)



(b)

The Millipedes:
(a) *Archidesmus* (Lower Devonian),
(b) *Cowiedesmus*(Middle Silurian) and (c) *Pneumodesmus* (Middle Silurian), from Scotland.

http://www.blackwellpublishing.com/paleobiology/figure.asp?chap=14&fig=Fig14-18&img=c14f018

Ecdysozoa: Arthropods

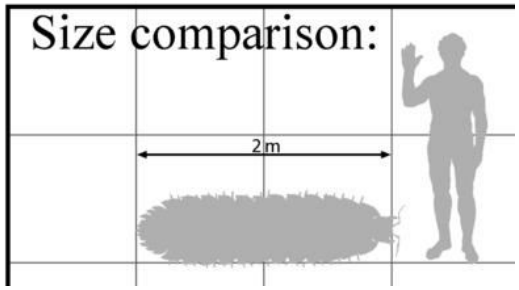
Invasion of land

- The giant *Arthropleura,* nearly 2 m long, hoovered their way through the lush, green vegetation of the Late Carboniferous forests.

Ecdysozoa: Arthropods



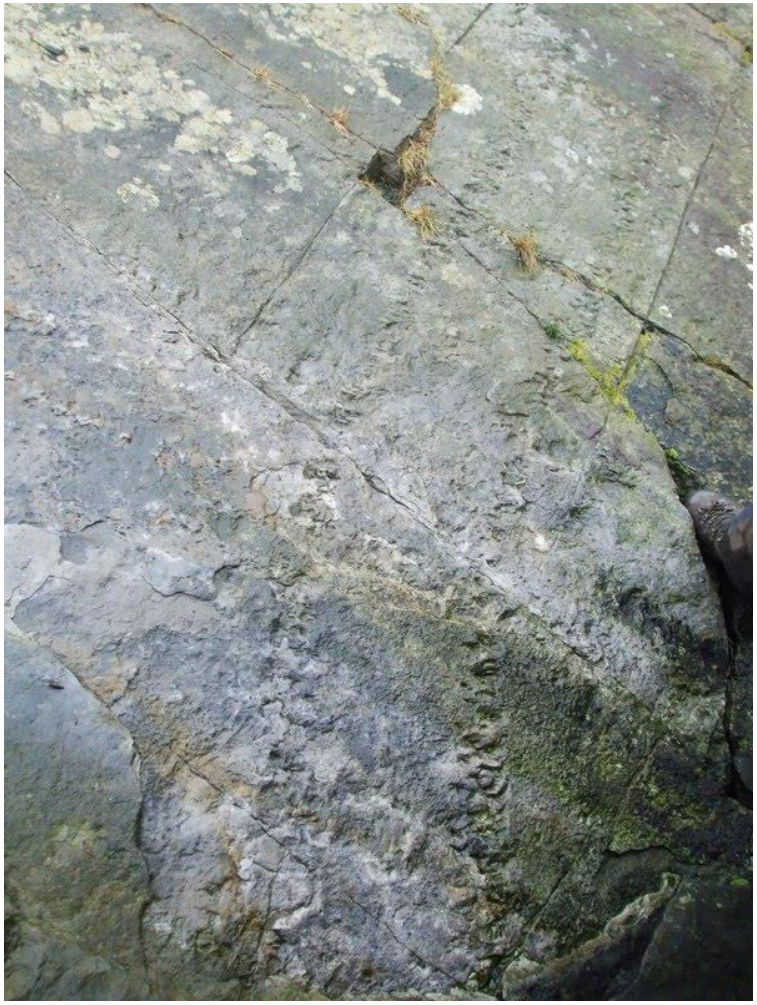
The giant
Arthropleura



https://commons.wikimedia.org/wiki/File:Arthropleura_NT_small.jpg#/media/File:Arthropleura_NT_small.jpg

Ecdysozoa: Arthropods

Fossil footprints of Arthropleura, Laggan Harbour, Isle of Arran, Scotland.



https://commons.wikimedia.org/wiki/File:Fossil_Footprints_Arthropleura_-_geograph.org.uk_-_1994274.jpg#/media/File:Fossil_Footprints_Arthropleura_-_geograph.org.uk_-_1994274.jpg

Ecdysozoa: Arthropods

Invasion of land

- Trackways from Middle Ordovician rocks in the English Lake District (Johnson et al. 1994) suggests that arthropods were on land about 50 myr earlier

Ecdysozoa: Arthropods

Invasion of land

- Trace fossils from Cambrian rocks in Ontario push arthropod life on land back (MacNaughton et al. 2002) suggesting the presence then of large, amphibious Euthycarcinoids

Ecdysozoa: Arthropods

END

Euthycarcinoids

- An enigmatic group of arthropods with antennae and mandibles
- In phylogeny somewhere near the origin of myriapods and insects

Ecdysozoa: Arthropods

Subphylum Hexapoda

Ecdysozoa: Arthropods

Hexapoda

- Can be divided into pterygotes (with wings) and apterygotes (without wings)
- Include the springtails, dragonflies, cockroaches and locusts

Ecdysozoa: Arthropods

Hexapoda

- The group may prove to have as many as 10 million living species when the rich faunas of the tropics have been completely described.

Ecdysozoa: Arthropods

Hexapoda

- Unbranched appendages, a simple gut, a single pair of antennae and a pair of mandibles, together with a toughened head capsule.
- Insects: six limbs

Ecdysozoa: Arthropods

The Oldest Insect

- Probably the springtail *Rhyniella praecursor* from the Lower Devonian Rhynie Chert of the Orcadian Basin of northeast Scotland

Ecdysozoa: Arthropods

Insects Diversified

- Had been earlier than had been thought (Labandeira 2006)
- The group probably originated in freshwater during the Late Silurian

Ecdysozoa: Arthropods

Insects Diversified

- Fresh water origin may account for the poor fossil record of the group before the Devonian (Glenner et al. 2006).

Ecdysozoa: Arthropods

Insects Diversified

- Early and Mid Devonian faunas are known from Rhynie, Gaspé, Québec and Gilboa, New York State
- Coincided with the diversification of land plants.

Ecdysozoa: Arthropods

Insects Flight

- By the Late Carboniferous a very diverse insect fauna had evolved, with forms such as the dragonflies and mayflies capable of powered flight

Ecdysozoa: Arthropods

Insects & Plants

- End of the Permian: familiar insect orders had appeared.
- Late Mesozoic and Cenozoic: significant coevolutionary relationships between plants and insects

Ecdysozoa: Arthropods

Insects & Mammals

- Miocene: Fossil hair trapped in amber together with the sand fly *Lutzomyia*
- These blood suckers were already feeding on mammals (Peñalver & Grimaldi 2006).

END

Ecdysozoa: Arthropods

Subphylum
Crustacea

Ecdysozoa: Arthropods

Crustaceans

- A hard, crusty carapace.
- First appeared in the Cambrian
- Aquatic, mainly marine, with gills, mandibles, two pairs of antennae and stalked compound eyes.

Ecdysozoa: Arthropods

Crustaceans

- The heavily armored crabs and lobsters typify this diverse subphylum
- The barnacles and ostracodes are crustaceans with a notable geological record.

Ecdysozoa: Arthropods

Crustaceans

- Eight main classes
- Ostracodes are considered part of the microfauna
- Only two groups, the Cirripedia and the Malacostraca have significant geological records.

Ecdysozoa: Arthropods

Cirripedes

- Or barnacles have shells, or capitula (singular, capitulum), consisting of several plates and these animals are adapted to an encrusting lifestyle

Ecdysozoa: Arthropods

Cirripedes

- Acorn barnacles and goose barnacles
- Contrasting life strategies.
- Acorn barnacles During the Late Cretaceous
- goose barnacles: Pseudoplankton

Ecdysozoa: Arthropods

Malacostracans

- Include two subclasses,
- Phyllocarids and Eumalacostracans
- The phyllocarids have large bivalved carapaces, seven abdominal somites and a telson with a pair of furcae

Ecdysozoa: Arthropods



The First Crustaceans: *Canadaspis* from the Burgess Shale

https://en.wikipedia.org/wiki/Canadaspis#/media/File:Canadaspis_perfecta_Burgess_Shale.jpg

Ecdysozoa: Arthropods

END

Ostracodes

- Ostracodes are crustacean arthropods, abundant and widespread in aquatic environments.
- Appeared first during the Early Cambrian

Deuterostomes

Echinoderms
and Hemichordates

Deuterostomes

Deuterostomes;

- The first opening to develop in the embryo is the anus and a second forms the mouth.
- A dipleurula larva & body cavity that developed from an extensions of the embryonic gut

Deuterostomes

Echinoderms

- Sea urchins starfish & sea cucumbers
- Water vascular system
- A mesodermal skeleton of calcitic plates
- Pentameral symmetry and tube feet.

Deuterostomes

Echinoderms

- During the Cambrian radiation many bizarre forms evolved.
- The spindle-shaped *Helicoplacus* may be part of the stem group for the entire phylum

Deuterostomes

Echinoderms

- Pelmatozoans: fixed echinoderms
- The blastoids, crinoids and cystoids;
- The crinoids; four classes, the Inadunata, Flexibilia, Camerata and Articulata

Deuterostomes

Echinoderms

- The echinoids were part of the mobile benthos.
- During the Mesozoic irregular groups, adapted for burrowing, evolved from the more regular forms that characterized the Paleozoic.

Deuterostomes

Asteroids

- Asteroids (starfish) were more important in post-Paleozoic rocks; their Triassic radiation may have inhibited the re-radiation of some key groups of brachiopod

Deuterostomes

Carpoids

- Traditionally classed with the echinoderms
- Some have argued they were ancestral to chordates
- They were probably stem-group echinoderm

Deuterostomes

Hemichordates

- Graptolites are hemichordates closely related to the living rhabdopleurids with similarly constructed rhabdosomes and ultrastructure

Deuterostomes

Hemichordates

- Graptolites probably pursued benthic (dendroids), planktic (dendroids and graptoloids) and automobile (graptoloids) lifestyles.

Deuterostomes

Hemichordates

- Graptoloids evolved rapidly and were widespread
- The ideal zone fossils in rocks of Ordovician-Silurian and Early Devonian age.

END

Deuterostomes

Echinoderms

Deuterostomes

Echinoderms

- One of the most abundant marine animal groups
- As fossils they can sometimes be rather robust
- an unusual five-fold symmetry & Water Vascular System

Deuterostomes

Water Vascular System

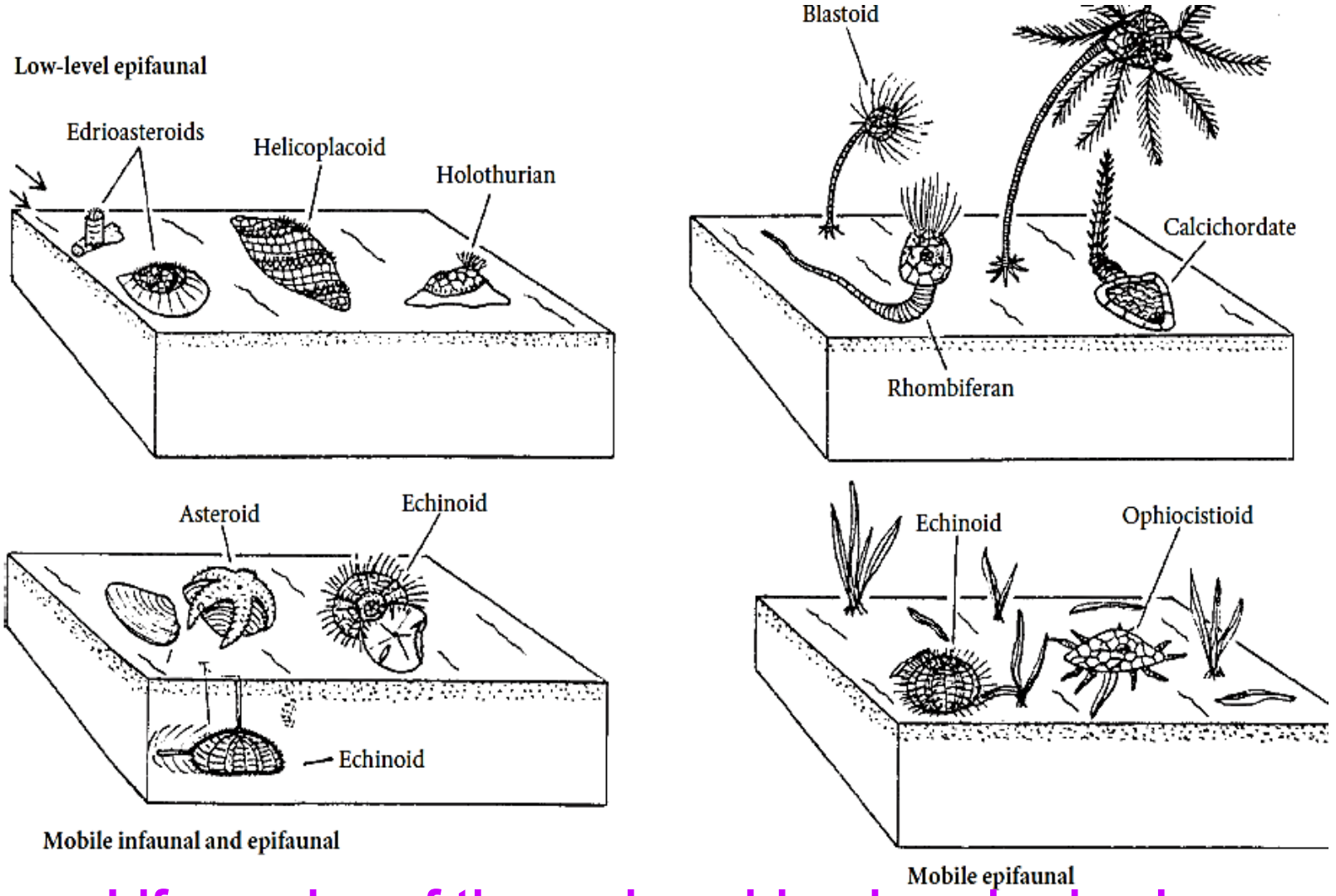
- Water is forced around the plumbing by muscular action
- Tube feet: are often modified for food processing, locomotion and respiration

Deuterostomes

Echinoderms

- The 6000 or so living echinoderm species include familiar forms such as sea lilies, sea urchins, sand dollars, starfish and sea cucumbers

Deuterostomes



Life modes of the main echinoderm body plans.

http://www.blackwellpublishing.com/paleobiology/figure.asp?chap=15&fig=Fig15-1&img=c15f001

Deuterostomes

Echinoderms

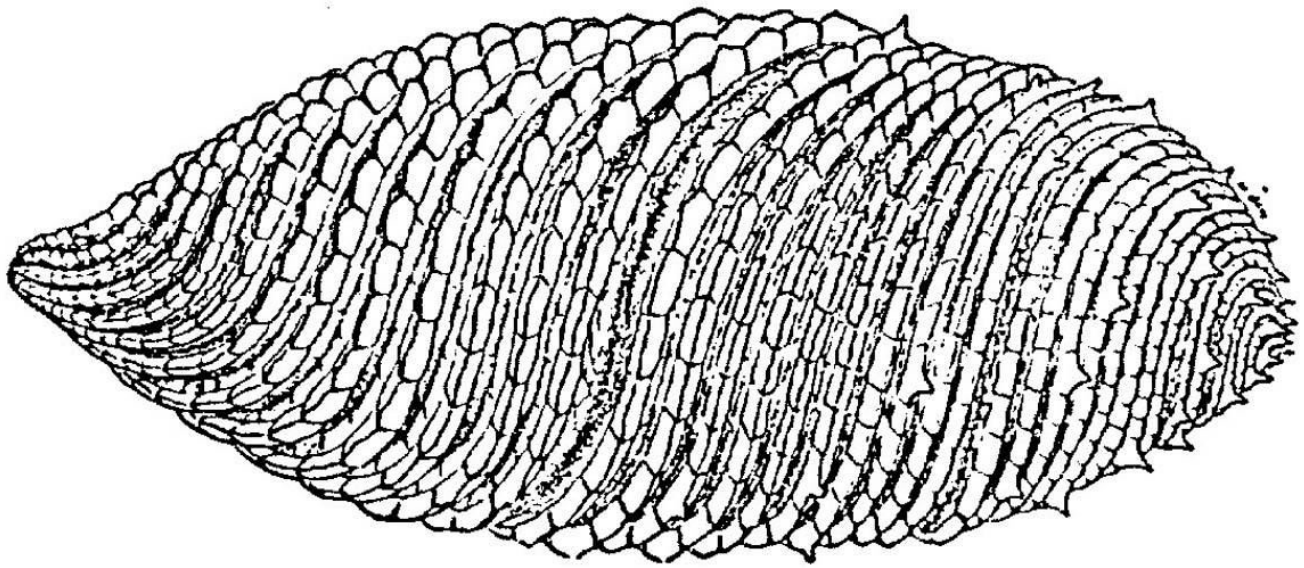
- One of the most remarkable Lagerstätte occurs in the Upper Ordovician succession of the Craighead inlier, north of the Girvan valley, southwest Scotland.

Deuterostomes

Echinoderms

- During the Early Cambrian radiation: bizarre forms appeared suddenly with very different morphologies
- One non-pentameral group The helicoplacoids

Deuterostomes



Helicoplacus from the Lower Cambrian (×10).

http://www.blackwellpublishing.com/paleobiology/figure.asp?chap=15&fig=Fig15-1&img=c15f002

Deuterostomes

Classification

- Class Eocrinoide
  Cambrian to Silurian
- Class Paracrinoidea
  Ordovician
- Class Blastoidea
  Ordovician to Permian
- Class Diploporita
  Cambrian to Devonian

Deuterostomes

Classification

- Class Rhombifera
  Cambrian to Devonian
- Class crinoidea
  Ordovician to recent
- Class Asteroidea
  Ordovician to Recent
- Class Echinoidea
  Ordovician to Recent

END

Deuterostomes

Hemichordates

Deuterostomes

Hemichordates

- Form a small phylum of only a few hundred species
- Common fossil representatives, the graptolites in ancient seas of the Early Paleozoic

Deuterostomes

Graptolites

- Extinct
- life styles are difficult to interpret
- small, soft-bodied animals with bilateral symmetry and a lack of segmentation

Deuterostomes

Hemichordata

- Two classes
- First, the tiny, mainly colonial, pterobranchs in sessile benthos
- Second, the larger infaunal acorn or tongue worms, the enteropneusts

Deuterostomes

Hemichordata

- Closely related to cephalochordates and urochordates or tunicates,
- Molecular data suggest that the latter two groups are more closely related to the chordates

Deuterostomes

Hemichordates

- Graptolites probably pursued benthic (dendroids), planktic (dendroids and graptoloids) and automobile (graptoloids) lifestyles.

Deuterostomes

Hemichordates

- Graptoloids evolved rapidly and were widespread
- The ideal zone fossils in rocks of Ordovician-Silurian and Early Devonian age.

Deuterostomes

Classification

- Class Graptolithina
- Order Dendroidea
  Cambrian to Carboniferous
- Order Tuboidea
  Ordovician to Silurian
- Order Camaroidea
  Ordovician

Deuterostomes

Classification

- Order Crustoidea
  Ordovician
- Order Dithecoidea
  Cambrian to Silurian
- Order Graptoloidea
  Ordovicianto Devonian

END

Fishes & Basal Tetrapods

Introduction

Fishes & Basal Tetrapods

Vertebrates

- Subgroup of the Phylum Chordata, a major deuterostome clade.
- The evolution of fishes from the Cambrian to the present day, and the Paleozoic tetrapods.

Fishes & Basal Tetrapods

Vertebrates

- Characterized by a skeleton made from bone (apatite)
- The oldest vertebrates are small fish-like creatures from the Early Cambrian of China.

Fishes & Basal Tetrapods

Ancient Fishes

- Armored fishes were abundant in Devonian seas and lakes.
- After the Devonian, the cartilaginous and bony fishes radiated in several phases.

Fishes & Basal Tetrapods

Conodonts

- Conodonts commonly occur as tooth-like elements that are useful in biostratigraphy, as are some other fish teeth and scales (ichthyoliths).

Fishes & Basal Tetrapods

Tetrapods

- Arose during the Devonian from lobe-finned fish ancestors, and fish-eating amphibians
- Diversified in the Carboniferous.

Apatite is a group of phosphate minerals, usually referring to hydroxylapatite, fluorapatite and chlorapatite, with high concentrations of OH^-, F^- and Cl^- ions, respectively, in the crystal.

Fishes & Basal Tetrapods

Reptiles

- The first reptiles were small insect eaters.
- Synapsids dominated ecosystems on land during the Permian and Triassic.

Fishes & Basal Tetrapods

Reptiles

- These groups were heavily hit by the end-Permian mass extinction event, and diapsid reptiles, most notably the dinosaurs, were key forms through the Mesozoic

Fishes & Basal Tetrapods

Importance of Vertebrates

- Humans are such a successful species
- The huge diversity and abundance of species of bony fishes, birds and mammals.

Fishes & Basal Tetrapods

END

Importance of Vertebrates

- Insects and microbes, are also abundant and diverse
- Vertebrates include the largest animals on land, in the sea and in the air.

Fishes & Basal Tetrapods

Origin of the Vertebrates

Fishes & Basal Tetrapods

The Skeleton

- Bone and cartilage.
- Bone: A network of collagen fibers on which needle-like crystals of hydroxyapatite (a form of apatite, $CaPO_4$) accumulate

Fishes & Basal Tetrapods

The Skeleton

- Cartilage: Flexible tissue, usually unmineralized, contains collagen and elastic tissues.
- In humans; early skeleton, cartilage, progressively mineralizes by apatite.

Fishes & Basal Tetrapods

The Skeleton

- The first vertebrates had a cartilaginous skeleton.
- Some of the oldest fish grow very large because the skeleton can grow with the animal
- Advantageous than exoskeletons

Fishes & Basal Tetrapods

Jawless Fishes

- Two key defining characters : the head and neural crest tissues
- The neural crest appears in the early embryo as a strip of cells where the backbone will develop

Fishes & Basal Tetrapods

Jawless Fishes

- The first vertebrates had no jaws
- Fossil sites at Chengjiang in China: Early Cambrian.
- Late Cambrian and Ordovician: commonest vertebrates were the conodont animals.

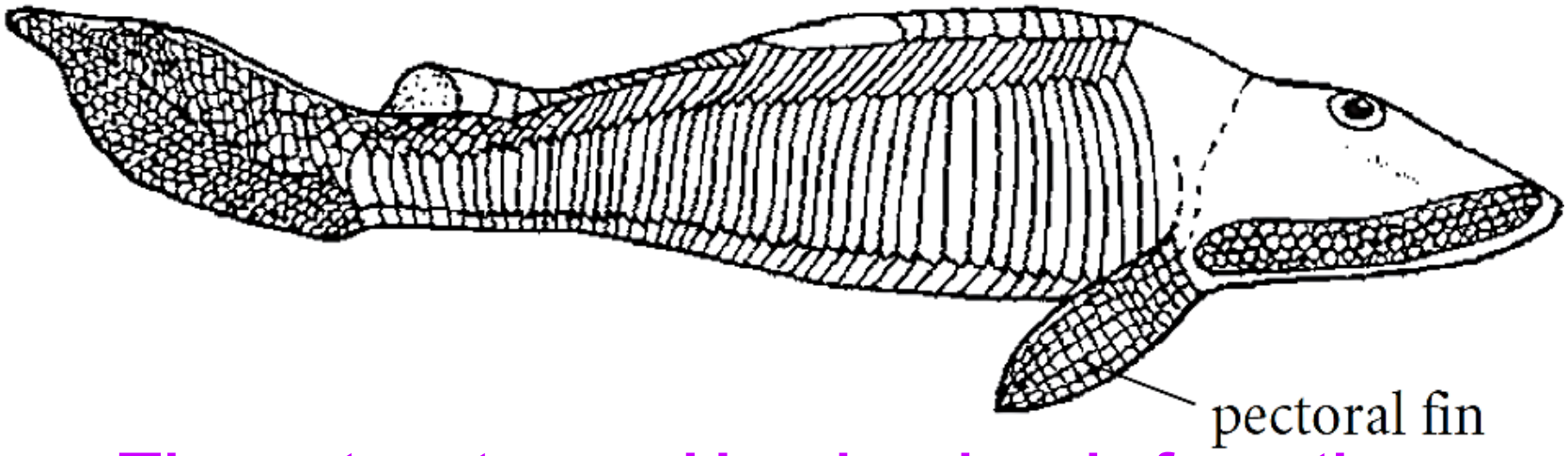
Fishes & Basal Tetrapods

Jawless Fishes

- Ostracoderms
- Were jawless, generally armored
- Had their heyday in the Devonian.
- *Hemicyclaspis*
- Today, the 50 or so species of lampreys and hagfishes, eel-shaped animals.

Fishes & Basal Tetrapods

Jawless Fishes



The osteostracan Hemicyclaspis from the Devonian

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Fishes & Basal Tetrapods

END

Conodonts:

- Animals of mystery
- The commonest early vertebrates (Sweet & Donoghue 2001).
- For over 150 years a mystery, known only from their jaw elements – no one knew which animal had produced them.

Fishes & Basal Tetrapods

Conodonts

Fishes & Basal Tetrapods

Conodonts

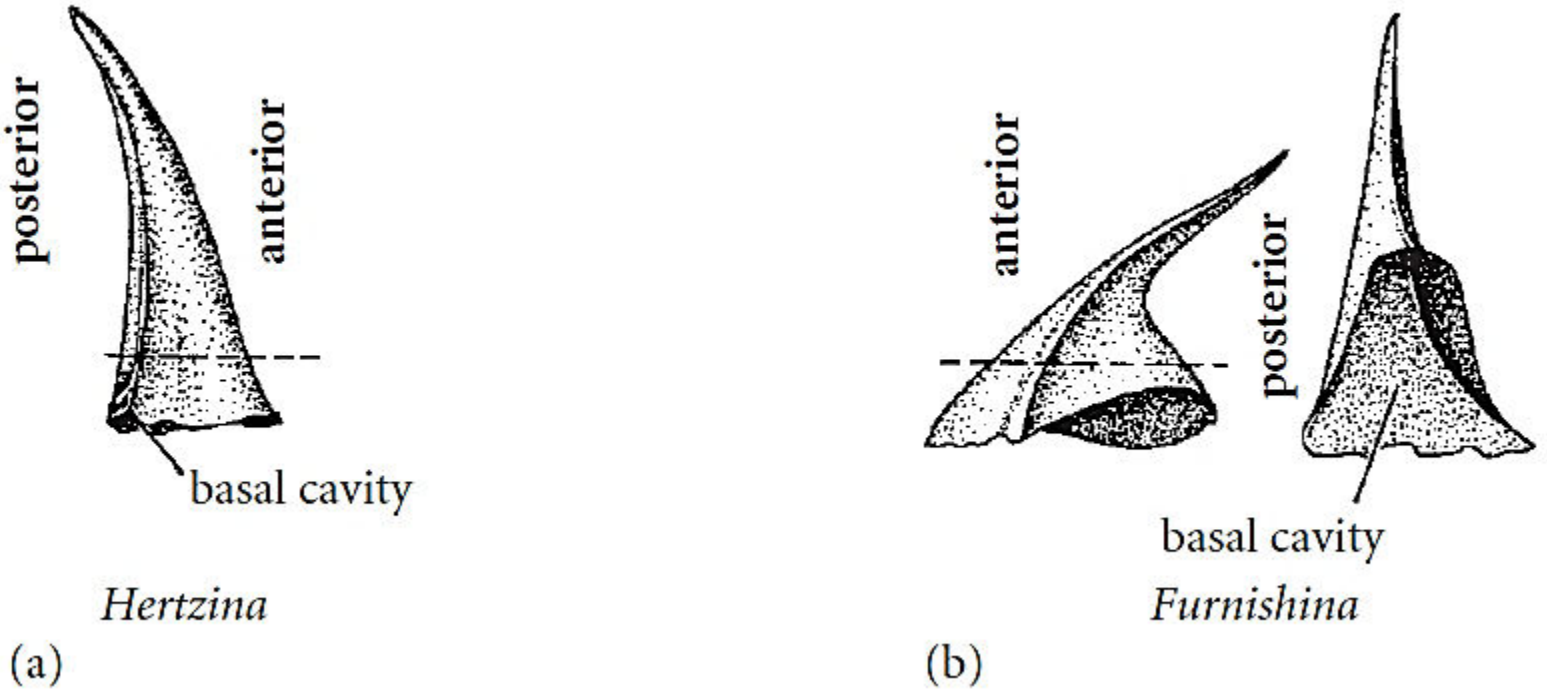
- First identified by the Latvian embryologist and paleontologist Christian Pander in 1856.
- They occur as phosphatic tooth-like microfossils, termed elements.

Fishes & Basal Tetrapods

Conodonts

- Three main groups have been established
- Protoconodonts such as Hertzina
- Paraconodonts like Furnishina
- Euconodonts more complex, with cones, bars and blades

Fishes & Basal Tetrapods



Descriptive morphology of the main types of conodont elements:

***(a) protoconodont Herzina* (×40);**

***(b) paraconodont Furnishina* (×40)**

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Fishes & Basal Tetrapods

Conodonts

- Conodonts are common in certain marine facies from the Cambrian to the Triassic.
- Paraconodonts: Mid Cambrian

Fishes & Basal Tetrapods

Conodont Element

- The three main morphotypes
- The basis of a crude single element or form taxonomy
- Coniform: simplest, with the base surmounted by a cone-like cusp

Fishes & Basal Tetrapods

Bars or Ramiform

- These elements consist of an elongate blade-like ridge with up to four processes developed posteriorly, anteriorly or laterally to the cusp

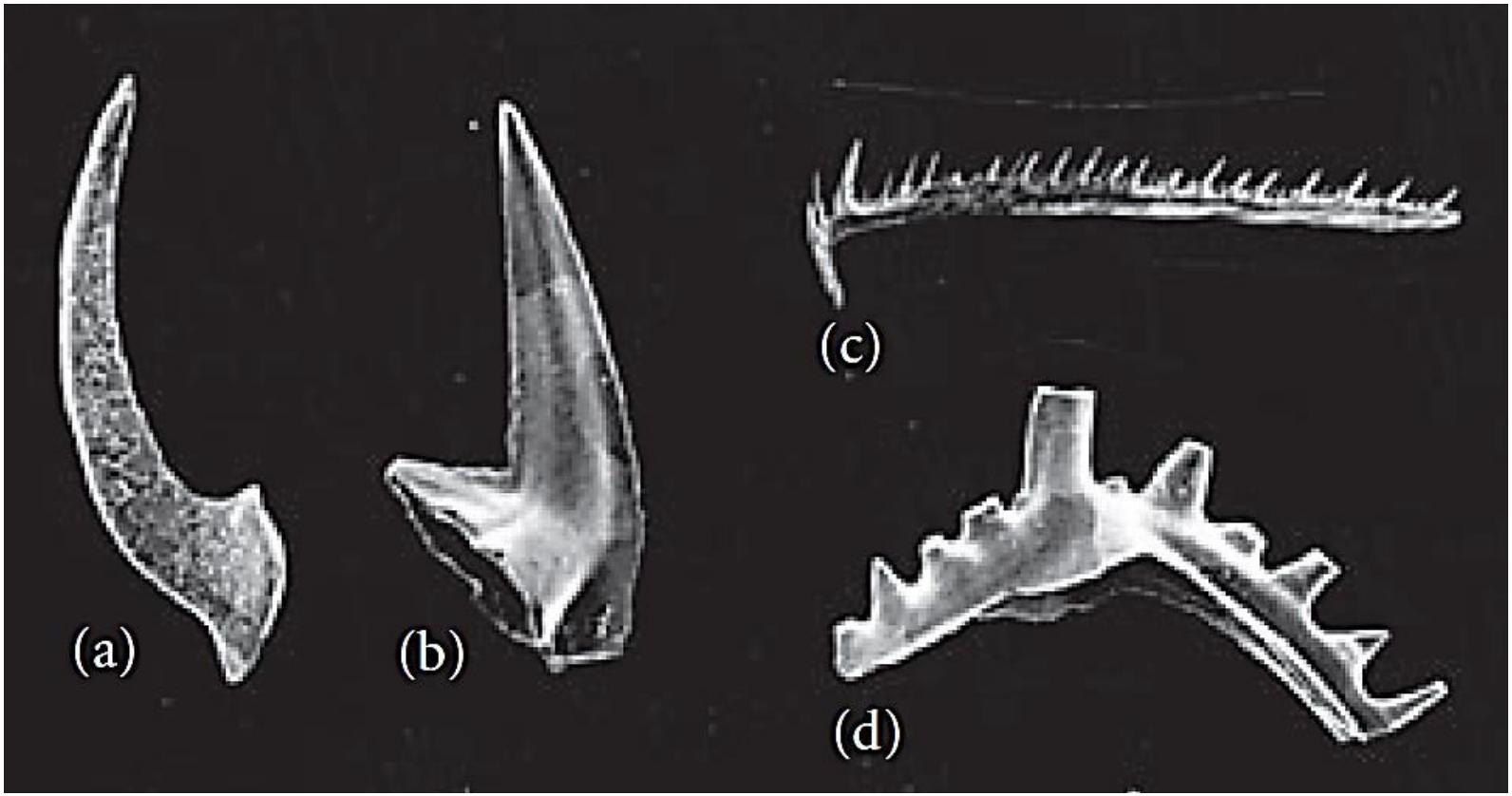
Fishes & Basal Tetrapods

Pectiniform elements

- Platforms have a wide range of shapes, with denticulate processes extending both anteriorly, posteriorly and/or laterally from the area of the basal cavity

END-1

Fishes & Basal Tetrapods



Conodont elements:

(a, b) coniform, lateral view;

(c, d) ramiform, lateral view;

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Fishes & Basal Tetrapods

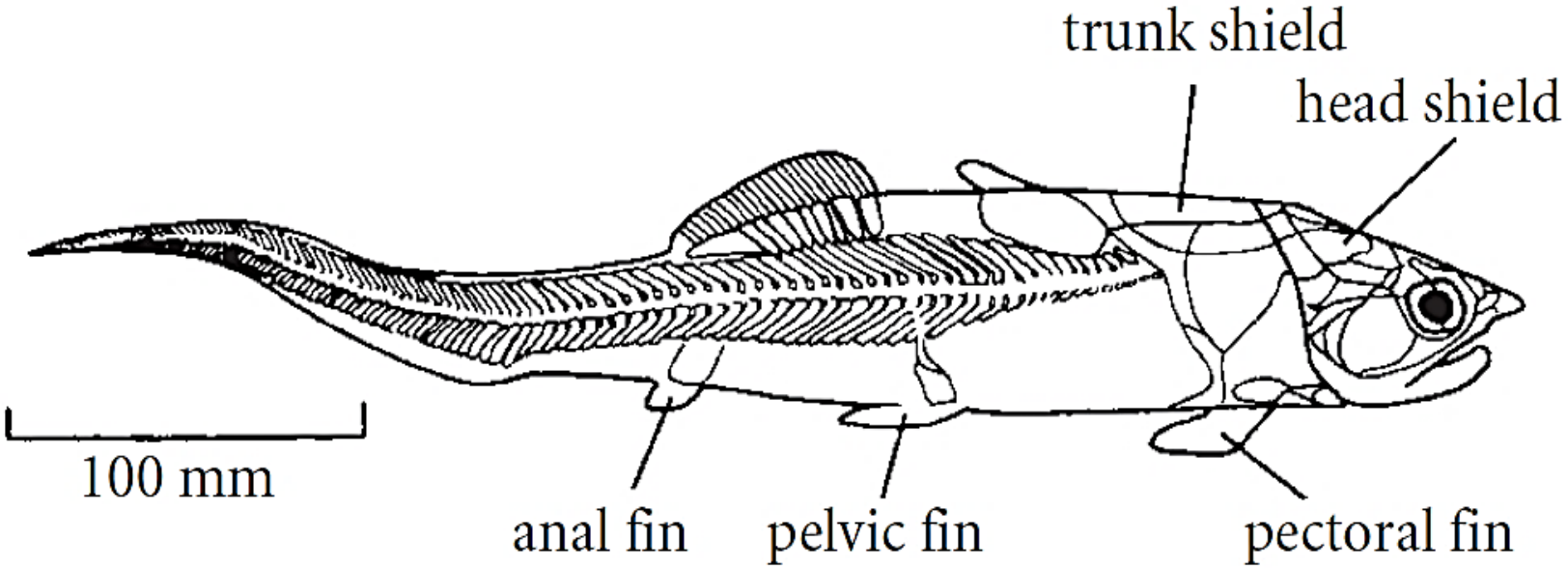
Jaws and Fish Evolution

Fishes & Basal Tetrapods

The First Jaws

- The basal vertebrates, lacked jaws, jaws probably evolved during the Ordovician
- Oldest jaw-bearing fishes were the placoderms, such as *Coccosteus*

Fishes & Basal Tetrapods



Jawed Fish of the Devonian
the placoderm *Coccosteus*

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Fishes & Basal Tetrapods

Chondrichthyans

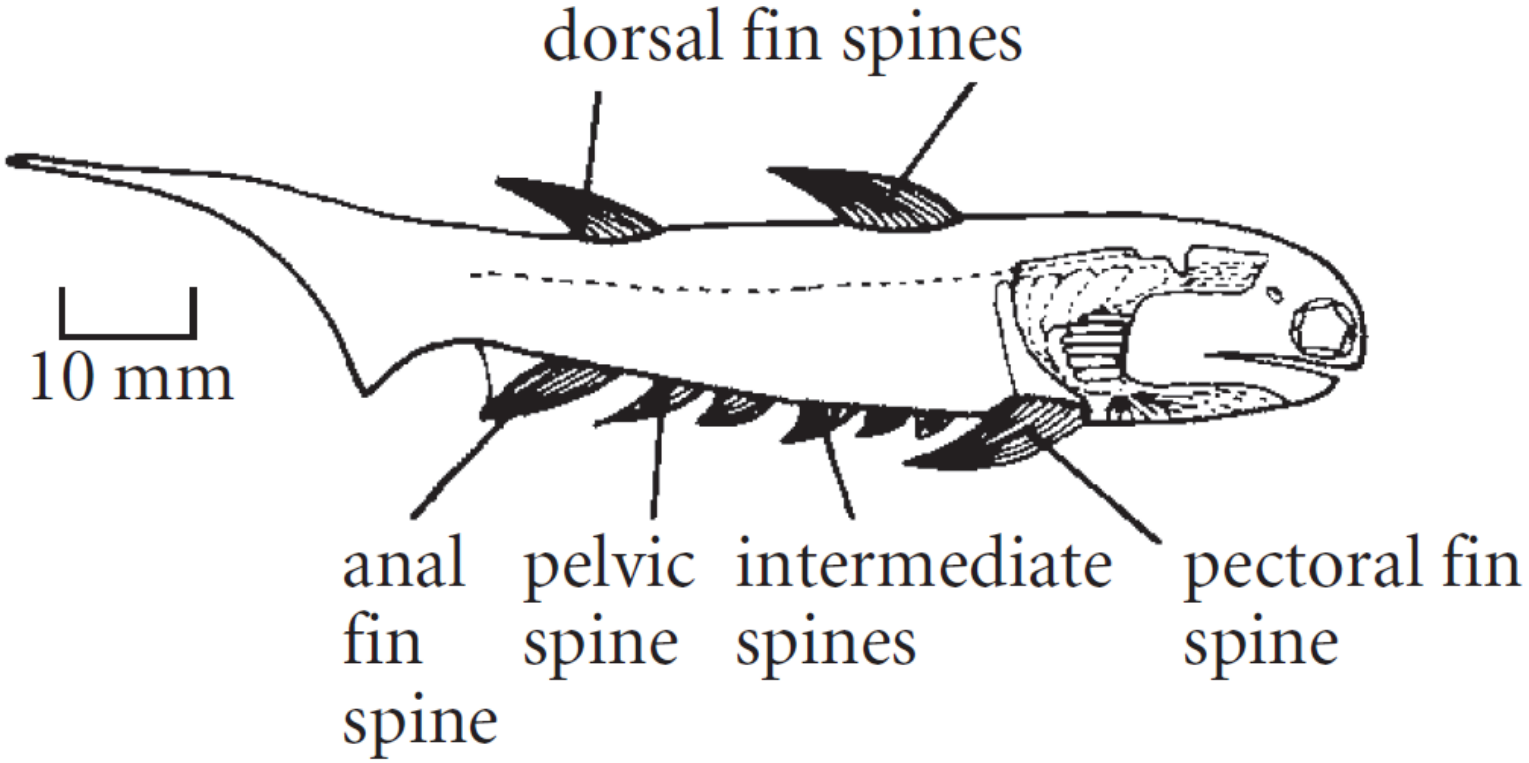
- The first shark like chondrichthyans or cartilaginous fishes, came on the scene during the Early Devonian

Fishes & Basal Tetrapods

Acanthodians

- Small fishes, mostly in the range 50–200 mm in length, and they bore numerous spines at the front of each fin and spaced rows on their undersides

Fishes & Basal Tetrapods



Jawed Fish of the Devonian
the acanthodian *Climatius*

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Fishes & Basal Tetrapods

Bony fishes

- The osteichthyans, or bony fishes, also appeared in the Devonian
- Two groups
- Ray fins and lobefins

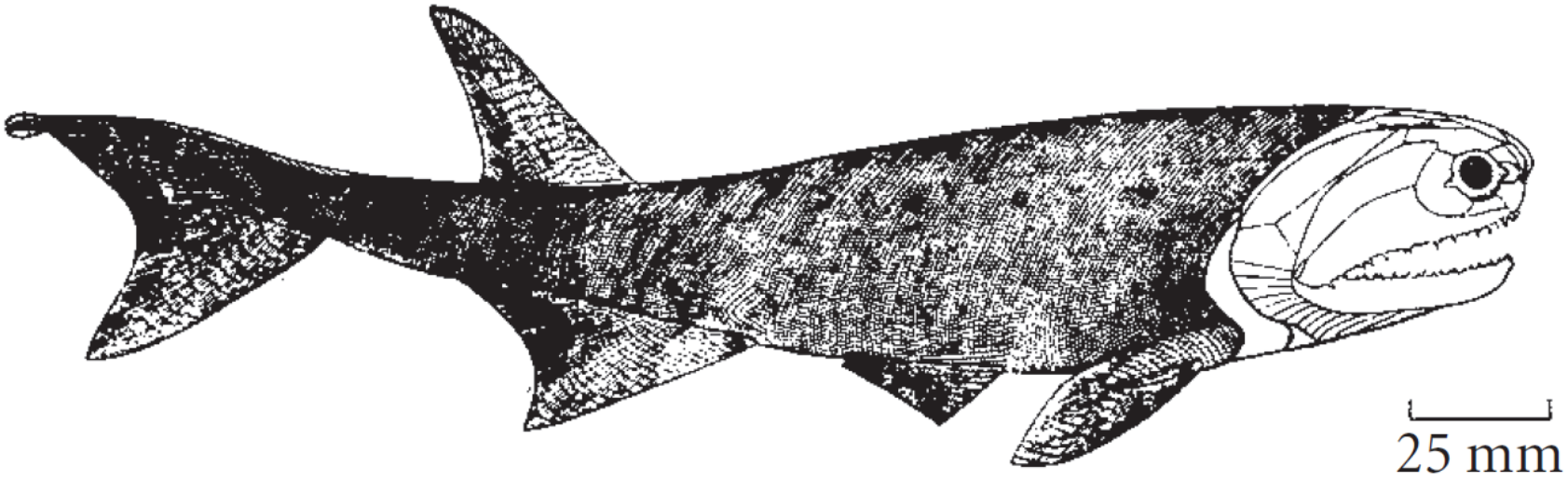
Fishes & Basal Tetrapods

END-2

Ray fins

- Those with ray-like fins, the actinopterygians
- Ancestors of most fishes today
- From carp to salmon
- Seahorse to tuna

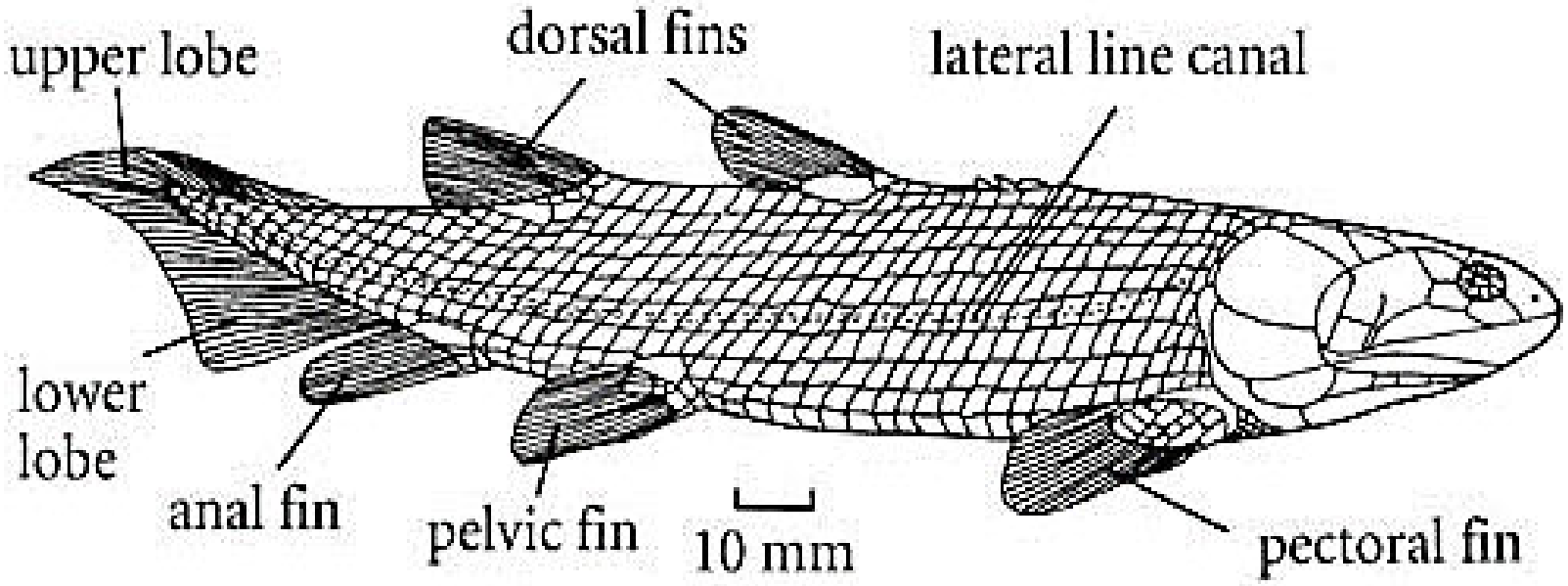
Fishes & Basal Tetrapods



Jawed Fish of the Devonian
the actinopterygian bony fish
Cheirolepis;

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Fishes & Basal Tetrapods



Jawed Fish of the Devonian
The lobefin Osteolepis

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Fishes & Basal Tetrapods

Tetrapods

Fishes & Basal Tetrapods

Fins to Limbs

- When a fish became a land animal, surely the key problem was breathing air?
- Not so.
- Early bony fishes almost certainly had both lungs and gills

Fishes & Basal Tetrapods

Main problem

- Support
- In water, an animal "weighs" virtually nothing,
- On land the body has to be held up from the ground
- The internal organs have to be supported in some way

Fishes & Basal Tetrapods

Other problems

- Reproductive, osmotic (water balance) and sensory systems had to adapt
- These changes did not happen all at once

Fishes & Basal Tetrapods

Origins

- Tetrapods arose from fishes during the Devonian
- Closest fishy relatives?
- The lobefins
- Complex bony and muscular pectoral (front) and pelvic (back) paired fins

Fishes & Basal Tetrapods

Origins

- The first good evidence of tetrapods comes from the Late Devonian
- Best-known forms are *Acanthostega* and *Ichthyostega,* which were 0.6 and 1 m long, respectively.

Fishes & Basal Tetrapods



Skull of the Late Devonian amphibian

Acanthostega, showing the streamlined shape, deeply-sculpted bones and small teeth, all inherited from its fish ancestor

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Fishes & Basal Tetrapods

The Amphibians:

- Half-way land animals
- 4000 species of mainly small animals that live in or close to water
- Still rely on the water for breeding and water balance.

Fishes & Basal Tetrapods

The Amphibians:

- The oldest fossil form, with reduced limbs, is Jurassic in age.
- All living amphibians appear to be closely related
- Forming a clade, the *Lissamphibia*

Fishes & Basal Tetrapods

The Amphibians:

- Subclass Lissamphibia
- Frogs, salamanders (newts) and caecilians (gymnophionans)
- Early Triassic to Recent

Fishes & Basal Tetrapods

END

The Amphibians:

- Subclass Reptiliomorpha
- Narrow-skulled, fish-eating amphibians
- High-skulled terrestrial amphibians
- Late Carboniferous to Late Permian

Fishes & Basal Tetrapods

Reign of the Reptiles

Fishes & Basal Tetrapods

Origin of the Reptiles

- Amphibians only made it halfway on to land
- They still produce swimming tadpoles
- Reptiles and their descendants: egg not have to be laid in water

Fishes & Basal Tetrapods

Origin of the Reptiles

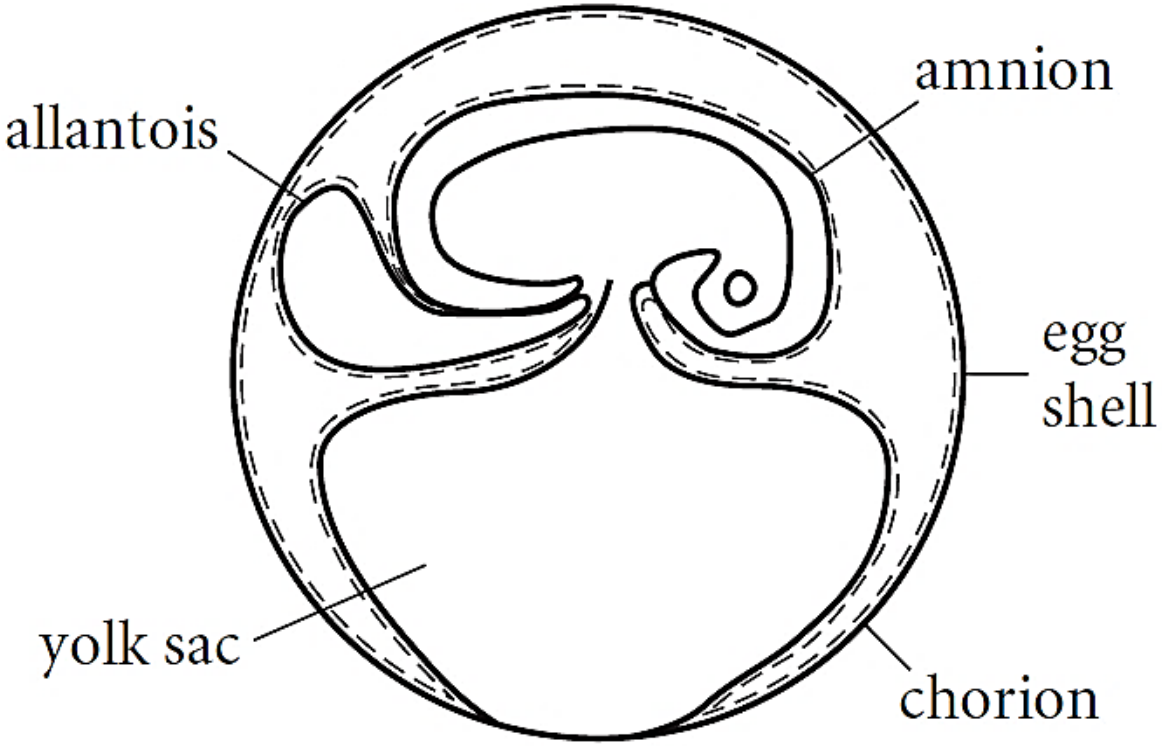
- The cleidoic ("closed") egg,
- Amniotic egg
- Tough semipermeable shell
- Clade Amniota (reptiles, birds, mammals)

Fishes & Basal Tetrapods

Membranes

- Enclose the embryo (the amnion)
- Collect waste (the allantois)
- line the eggshell
- (the chorion)
- Yolk, a yellow material rich in protein.

Fishes & Basal Tetrapods



The Cleidoic egg of amniotes in cross-section, showing the eggshell and extraembryonic membranes.

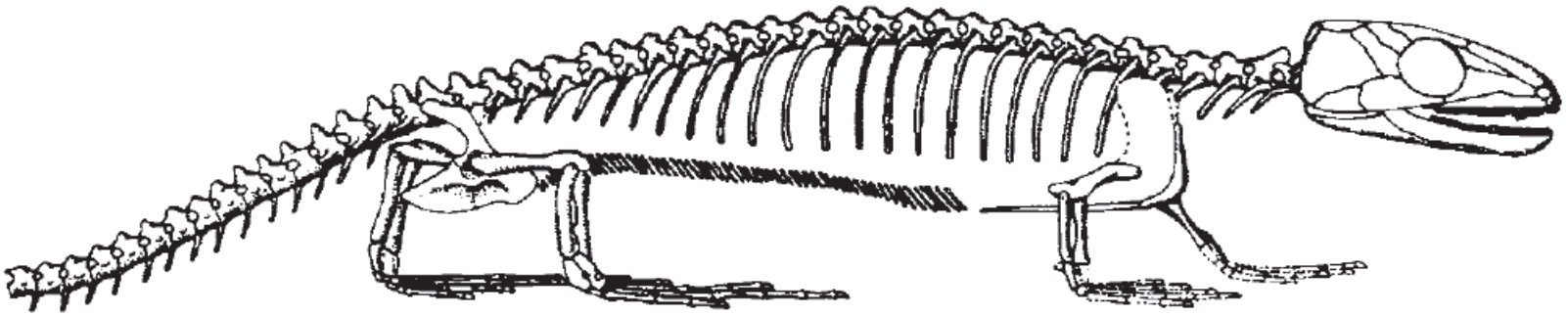
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Fishes & Basal Tetrapods

Oldest Amniote

- Oldest-known amniote, *Hylonomus* from the mid-Carboniferous of Canada.
- Only skeleton; no amniotic eggs are known from the Carboniferous.

Fishes & Basal Tetrapods



(a)

The earliest reptile, The mid-Carboniferous Reptile *Hylonomus*, skeleton and skull

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Fishes & Basal Tetrapods

Radiation

- Amniotes: Late Carboniferous
- Three main clades.
- Distinguished by the pattern of openings in the side of the skull
- Temporal openings behind the eye socket

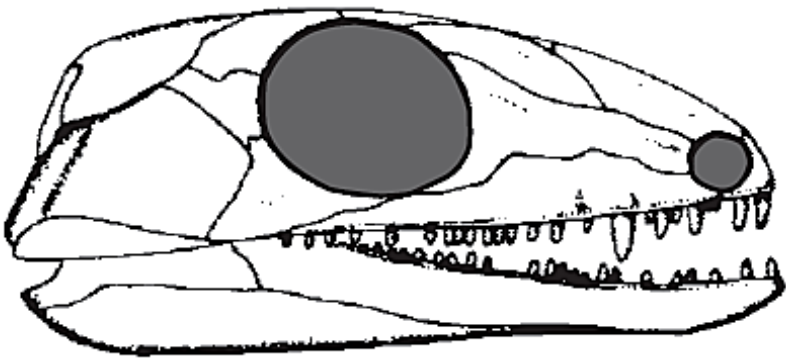
Fishes & Basal Tetrapods

END-2

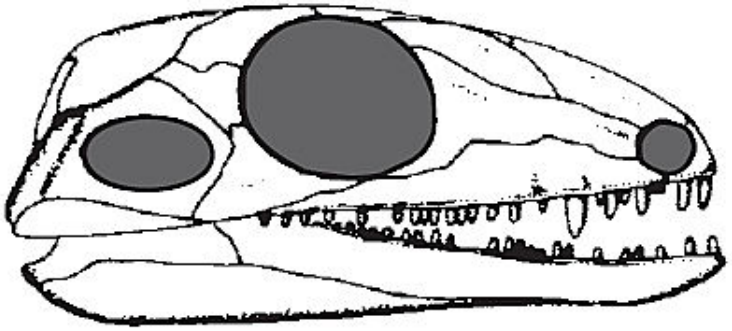
Radiation

- The anapsid ("no arch") skull pattern, no temporal openings.
- The synapsid ("same arch"), lower temporal opening,
- The diapsid ("two arch") pattern, two temporal openings.

Fishes & Basal Tetrapods



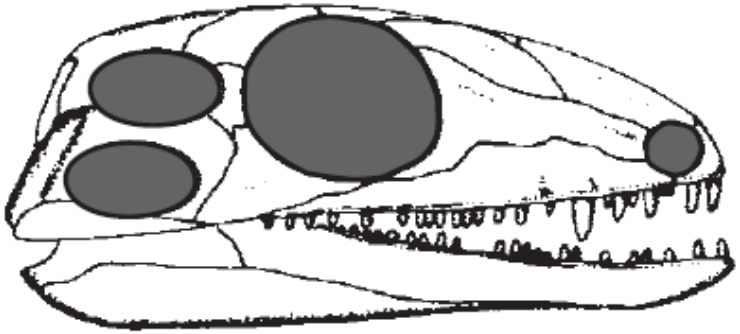
(c) Anapsid



(d) Synapsid

Early Reptile Evolution

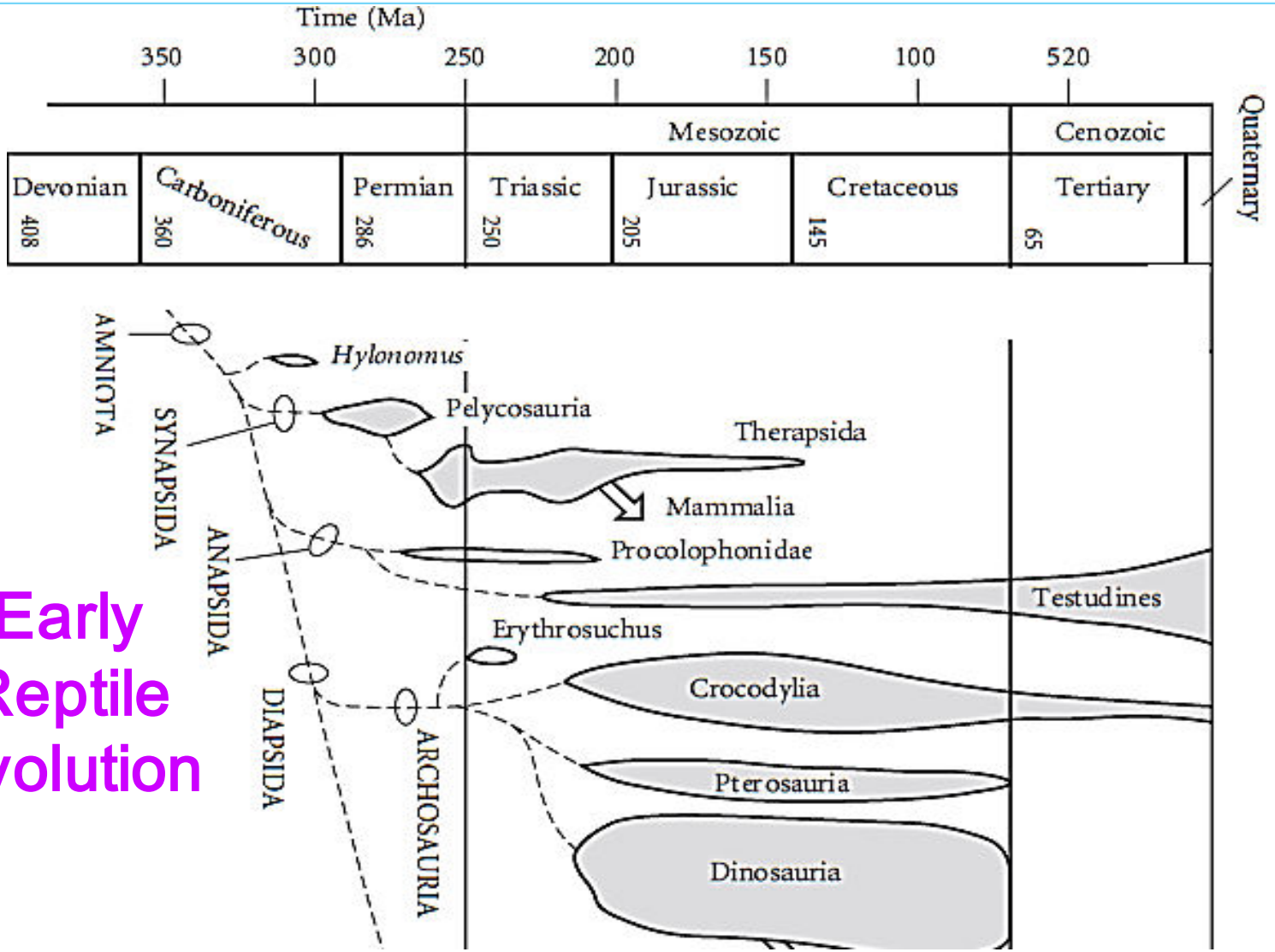
(c-e) The three major skull patterns seen in amniotes: anapsid, diapsid and synapsid. (Based on Carroll 1987.)



(e) Diapsid

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Fishes & Basal Tetrapods



Early Reptile Evolution

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Fishes & Basal Tetrapods

Reign of the Reptiles (Part 2)

Fishes & Basal Tetrapods

Three Clades

- The Anapsida: early forms such as *Hylonomus*, as well as some Permian and Triassic reptiles, and the turtles.
- The Synapsida: "mammal-like reptiles" and the mammals

Fishes & Basal Tetrapods

Three Clades

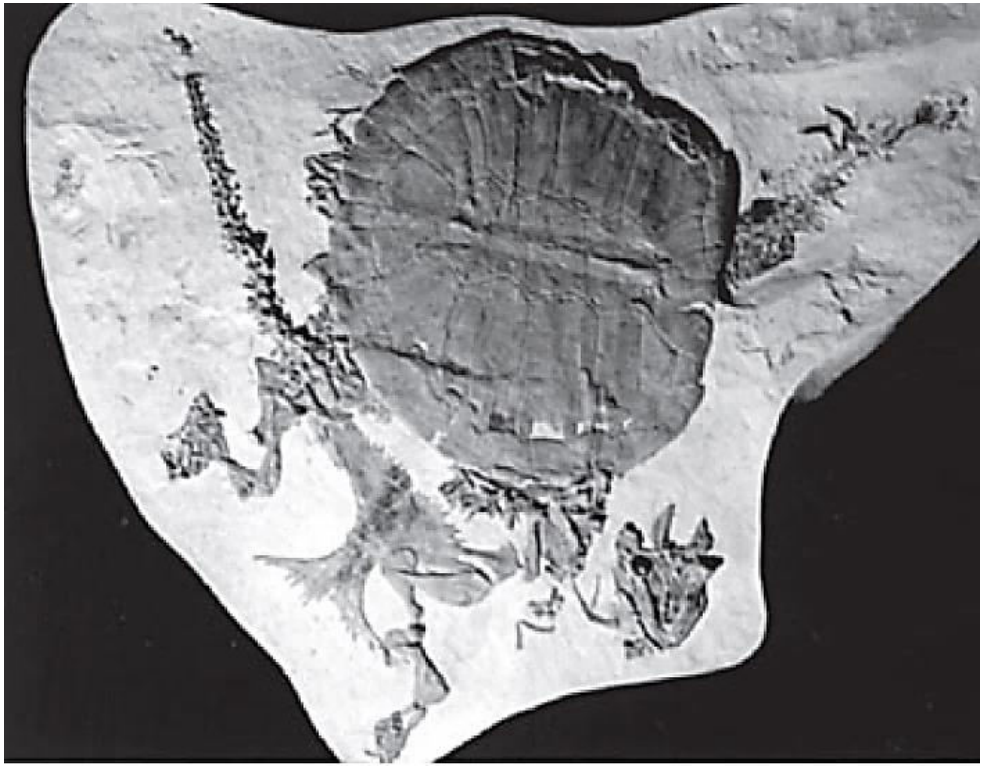
- The Diapsida : Early groups, as well as the lizards and snakes and the crocodiles, pterosaurs, dinosaurs and birds.
- All modern amniotes produce the cleidoic egg

Fishes & Basal Tetrapods

The Anapsida

- Turtles and relatives
- The oldest were small insect eaters.
- During the Permian and Triassic
- The turtles appeared first in the Late Triassic,
- *Proganochelys*

Fishes & Basal Tetrapods



A Fossilized Snapping Turtle, with the head (bottom right) and skeleton at Steinheim, Germany.

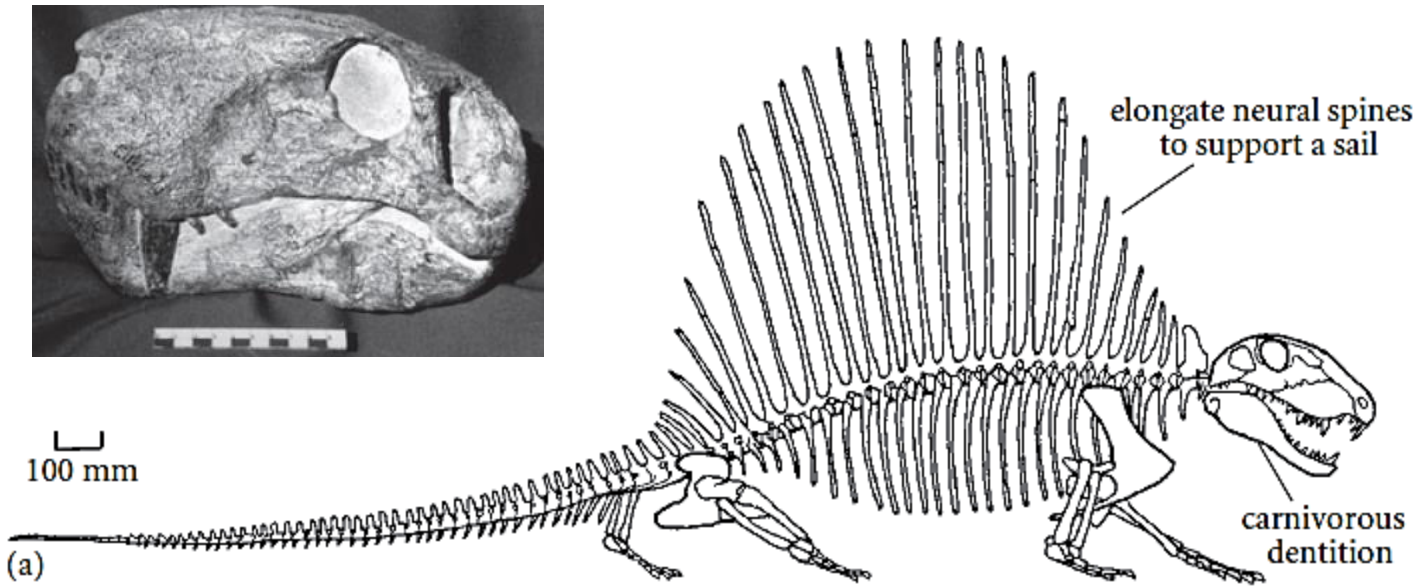
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Fishes & Basal Tetrapods

The Synapsids

- First synapsids, known from the Late Carboniferous and Early Permian
- Grouped loosely as "pelycosaurs"
- insectivores and carnivores
- With powerful skulls and sharp, flesh piercing teeth.

Fishes & Basal Tetrapods



Synapsids of the Permian:

(a) the carnivorous pelycosaur Dimetrodon; (Upper left) the carnivorous gorgonopsian *Lycaenops*

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Fishes & Basal Tetrapods

The Synapsids

- Synapsids radiated dramatically in the Late Permian as a new clade, the Therapsida.
- The carnivores were the *gorgonopsians*
- The cynodonts: The Early Triassic form *Thrinaxodon*

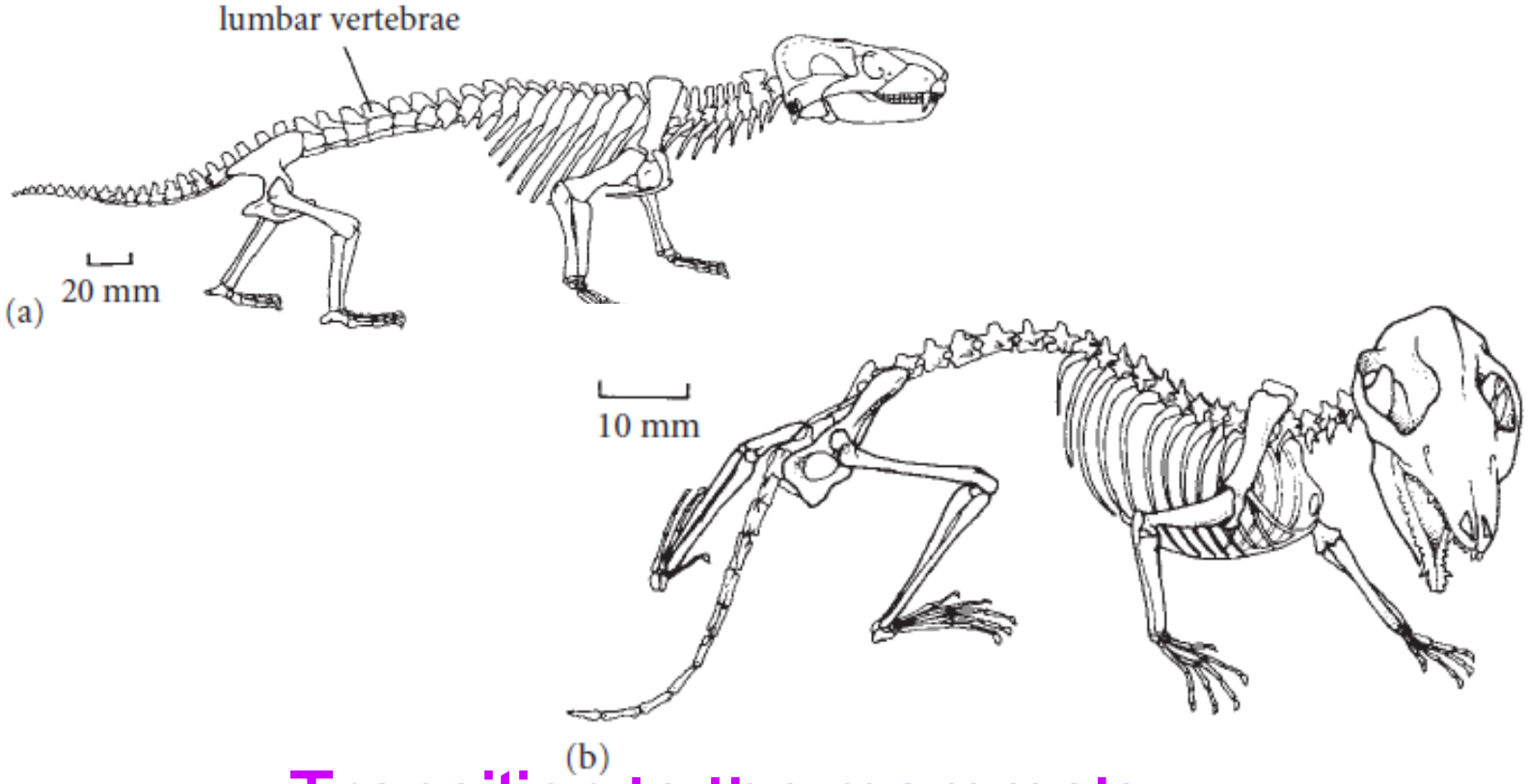
Fishes & Basal Tetrapods

Thrinaxodon

- Looked dog-like
- In the snout area: the roots of sensory whiskers
- Thrinaxodon had whiskers,
- Hair on other parts of its body
- Insulation and temperature control

END-1

Fishes & Basal Tetrapods



Transition to the mammals:

(a) The Early Triassic cynodont *Thrinaxodon*; (b) the Early Jurassic mammal *Megazostrodon*;

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Fishes & Basal Tetrapods

Dinosaurs and Mammals

Fishes & Basal Tetrapods

The Transition

- The transition from basal synapsid to mammal is marked by an extraordinary shift of the jaw joint into the middle ear.

Fishes & Basal Tetrapods

The Transition

- Reptiles typically have six bones in the lower jaw and the articular bone articulates with the quadrate in the skull

Fishes & Basal Tetrapods

The Transition

- In mammals, on the other hand, there is a single bone in the lower jaw, the dentary, which articulates with the squamosa

Fishes & Basal Tetrapods

The Transition

- The reptilian articular–quadrate jaw joint became reduced in Triassic cynodonts, and moved into the middle ear passage.

Fishes & Basal Tetrapods

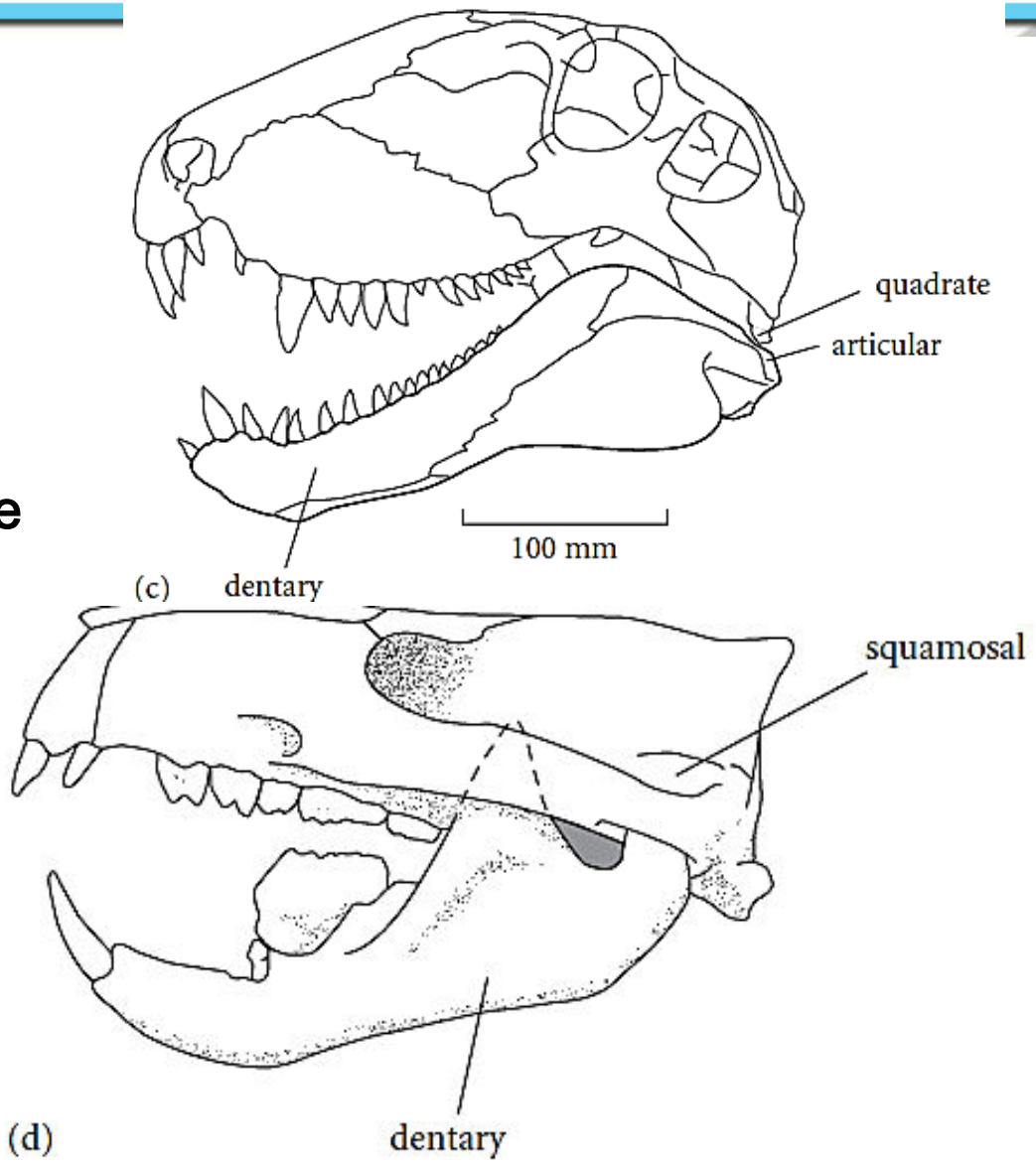
The Transition

- That is why we have three tiny ear bones, the hammer, anvil and stirrup, which transmit sound from the ear drum to the brain
- Reptiles have only one, the stirrup or stapes.

Fishes & Basal Tetrapods

Transition to the mammals:

skulls of an early synapsid (c) and a mammal (d) to show the reduction in elements in the lower jaw and switch of the jaw joint



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Fishes & Basal Tetrapods

A Misconception

- People think of the dinosaurs as precursors of the mammals.
- Dinosaurs famously ruled the Earth for 160 Myr of the Mesozoic
- Replaced by the mammals 65 Ma.

Fishes & Basal Tetrapods

Evolution of Mammals

- However, the mammals arose in the Late Triassic, about the same time as the first dinosaurs.

Fishes & Basal Tetrapods

END

Evolution of Mammals

- Both groups of mammals and Dinosaurs evolved side by side through the Late Triassic, Jurassic and Cretaceous

Dinosaurs and Mammals

Introduction

Dinosaurs and Mammals

Dinosaurs

- After the end-Permian mass extinction event 251 Ma, diapsid reptiles diversified in the Triassic.
- Dinosaurs were a hugely successful group for 160 myr of the Mesozoic.

Dinosaurs and Mammals

Dinosaurs

- Pterosaurs: key Mesozoic flyers
- Most important marine reptiles: plesiosaurs and ichthyosaurs
- Birds evolved from dinosaurs, and radiated during Tertiary

Dinosaurs and Mammals

The First Mammals

- Small insect eaters of the latest Triassic
- The mammals achieved great diversity and abundance only after the extinction of the dinosaurs.

Dinosaurs and Mammals

Radiation of the Mammals

- Molecular and paleontological evidence show that modern mammals radiated in the Late Jurassic and Early Cretaceous

Dinosaurs and Mammals

Radiation of the Mammals

- Basic splits were geographic – with major clades separated in South America, Africa, Australasia and the northern hemisphere.

Dinosaurs and Mammals

Emergence of Humans

- Humans arose 6–8 Ma, and fossil evidence points to repeated human migrations out of Africa.

Dinosaurs and Mammals

- Evolution of vertebrates forward into the Mesozoic and Cenozoic.
- The rise of the diapsids, and especially the dinosaurs and their descendants, the birds.

Dinosaurs and Mammals

END

- Modest evolution of mammals through the Mesozoic set the scene for their explosive radiation at the beginning of the Cenozoic, after the dinosaurs had gone

Dinosaurs and Mammals

Dinosaurs and Their Kin

Dinosaurs and Mammals

The Diapsids Take Over

- Abundance of the synapsids in the Carboniferous or Permian.
- Change during the Triassic, (end-Permian extinction event)
- Devastating effect on therapsid communities.

Dinosaurs and Mammals

The Diapsids Take Over

- Small and large meat eaters such as *Erythrosuchus*
- One of the first of the archosaurs
- A group include the dinosaurs, pterosaurs, crocodilians and birds.

Dinosaurs and Mammals



Archosaurs:
Skull of the Early Triassic archosaurs
Erythrosuchus

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Dinosaurs and Mammals

The Diapsids Take Over

- Triassic: diapsids diversified
- Archosaurs: large carnivores
- Proficient flyers: (pterosaurs)
- another extinction Late Triassic (about 220 Ma)
- Another Diapsid radiation

Dinosaurs and Mammals

The Diapsids

- New kinds of land tetrapods radiate
- The dinosaurs, pterosaurs, crocodilians and lizard ancestors, as well as the turtles
- Modern amphibians and true mammals

Dinosaurs and Mammals

The Pterosaurs

- Proficient flapping flyers with a lightweight body
- Narrow hatchet-shaped skull
- a long narrow wing
- Almost certainly endothermic
- Fed on fishes and insectivorous.

Dinosaurs and Mammals

Archosaurs:
the Late Jurassic
pterosaur
Rhamphorhynchus,
showing the elongated wing finger on each side, and the long tail with its terminal "sail" made from skin (×0.3)



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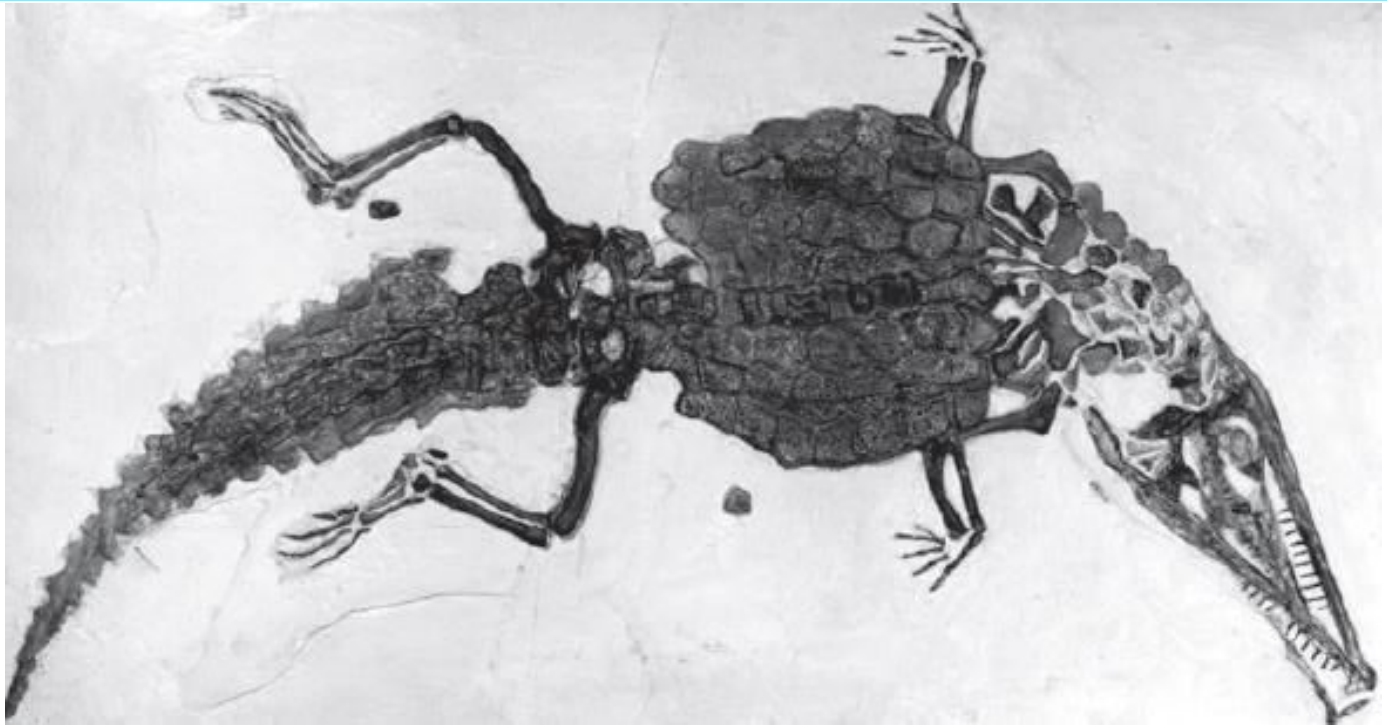
Dinosaurs and Mammals

Early Crocodilians

- Largely terrestrial in habits, walked on all fours and had an extensive armor of bony plates
- More diverse and abundant during the Jurassic and Cretaceous than now.

END-1

Dinosaurs and Mammals



Archosaurs:

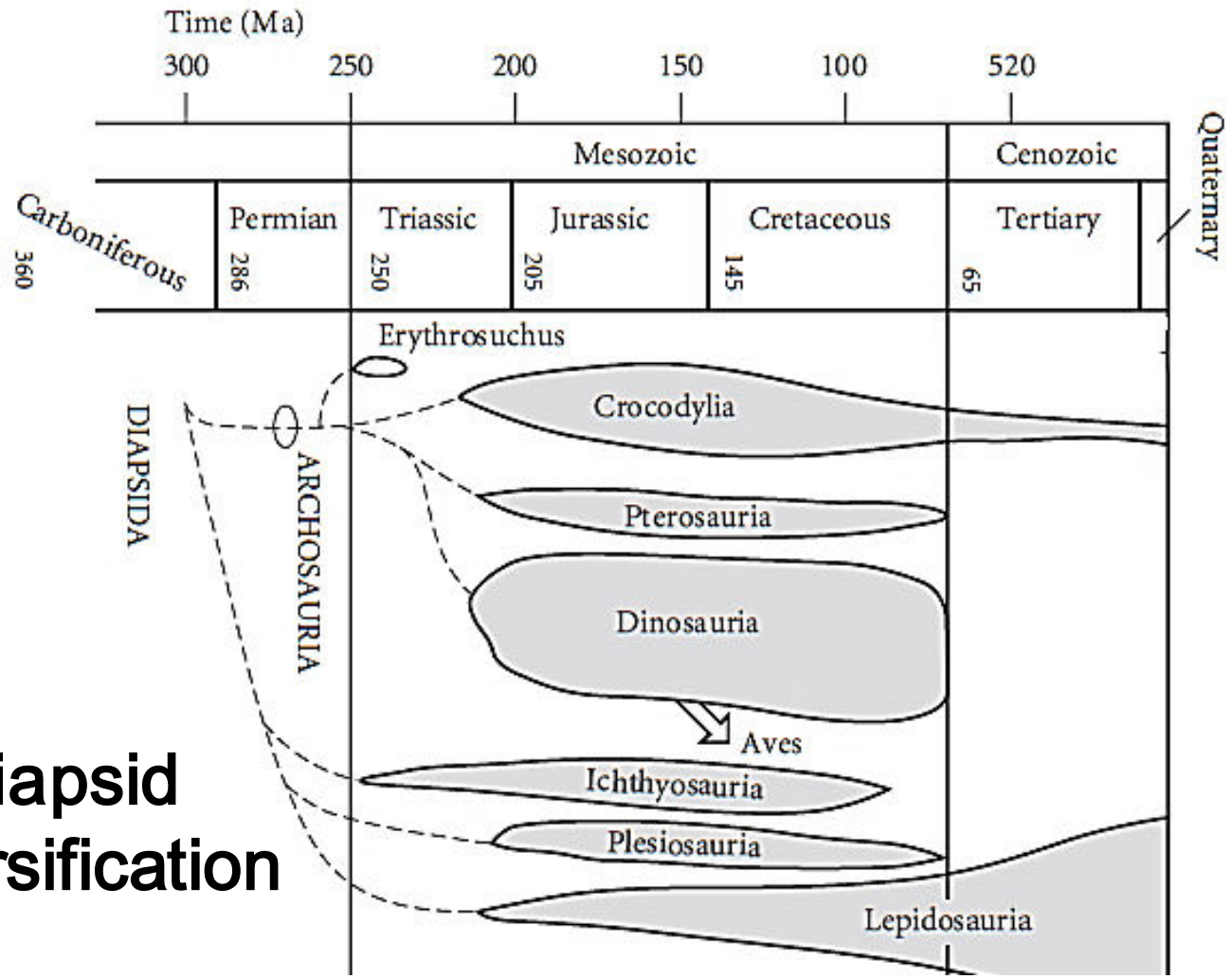
The Late Jurassic crocodilian Crocodilemus, showing the skeleton and armor covering (×0.2).

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Dinosaurs and Mammals

Dinosaurs and Their Kin (Part 2)

Dinosaurs and Mammals



Diapsid Diversification

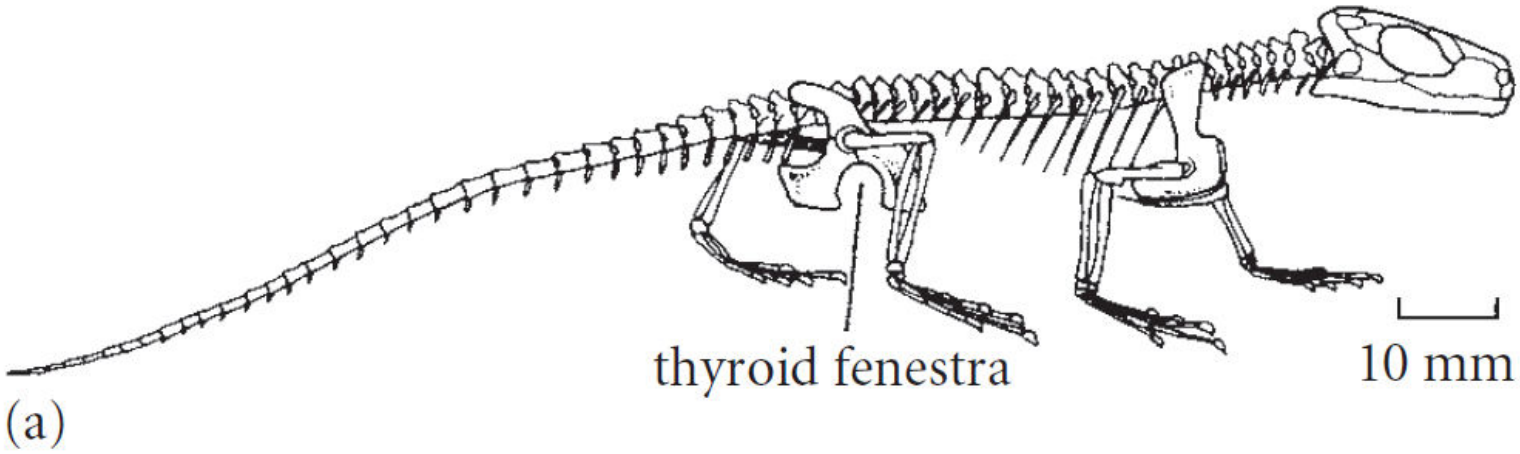
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Dinosaurs and Mammals

Lepidosaurs

- The second major diapsid clade
- Lizards and snakes
- Diversified in the Late Triassic
- Key forms: sphenodontids – snubnosed, lizard-sized animals

Dinosaurs and Mammals



Lepidosaurs:
The Late Triassic sphenodontid
Planocephalosaurus

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Dinosaurs and Mammals

Lepidosaurs

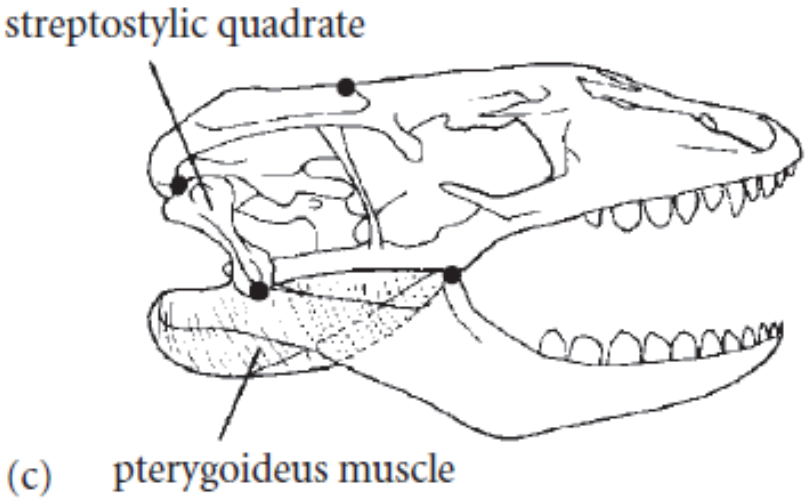
- The group dwindled after the Jurassic, except for a single living representative
- Sphenodon, the tuatara of New Zealand
- A famous “living fossil”.

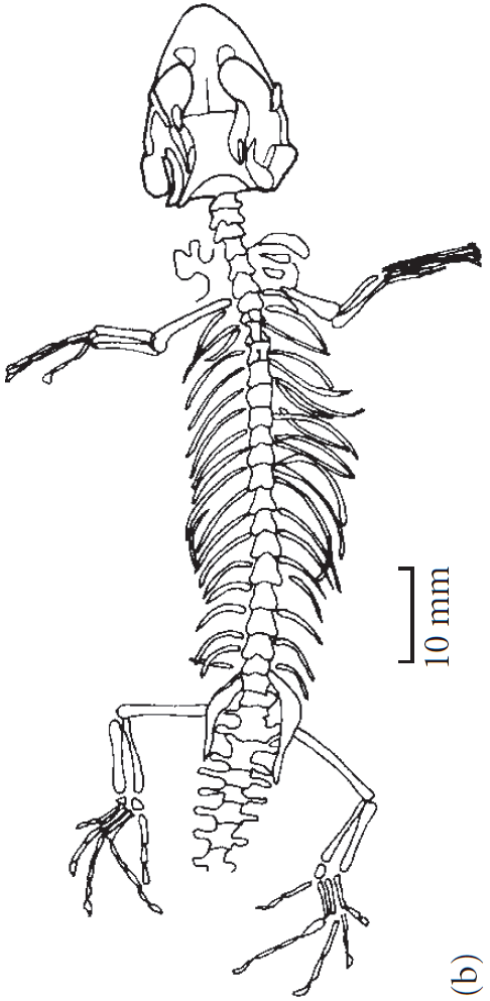
Dinosaurs and Mammals

Lepidosaurs

- First true lizards: Mid and Late Jurassic
- Characteristic mobility of the skull
- The quadrate is mobile, and the snout portion of the skull can tilt up and down

Dinosaurs and Mammals





Lepidosaurs:

(b) the Late Jurassic lizard *Ardeosaurus*; and (c) skull of a modern lizard

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Dinosaurs and Mammals

Loosening Skull

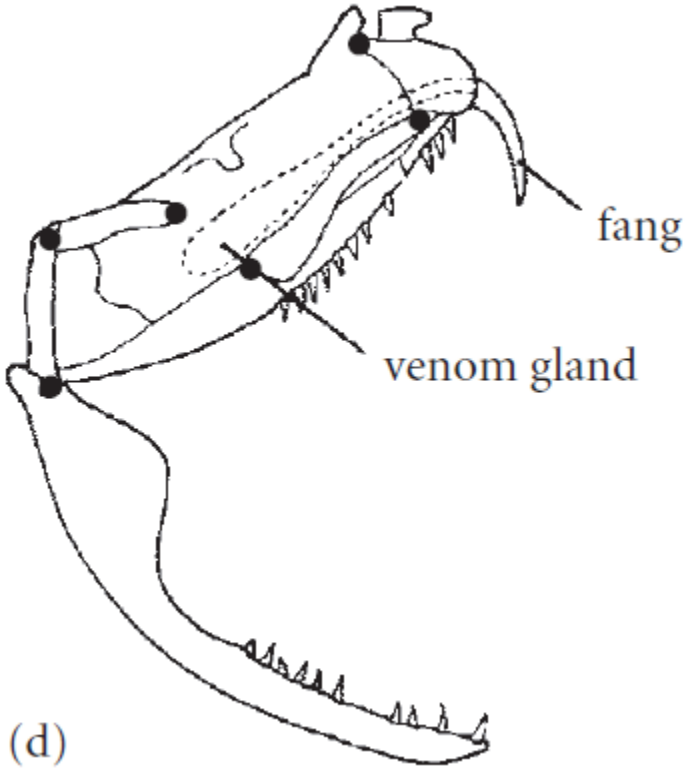
- Taken further in the snakes
- Known first in the Early Cretaceous
- Snakes have mobile skulls
- swallow prey animals several times the diameter of the head

END-1

Dinosaurs and Mammals

Lepidosaurs:

Skull of a Snake (d), showing the points of mobility that permit wide jaw opening.



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